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THE TAX IMPACT ON MULTINATIONAL CORPORATE FINANCIAL DECISIONS

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2022

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THE TAX IMPACT ON MULTINATIONAL CORPORATE FINANCIAL DECISIONS

Phd thesis presentend to the Graduate Program in Business Administration at Universidade Presbiteriana Mackenzie, as part of the requirements to obtain the doctorate degree in Business Administration.

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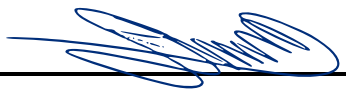
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Abstract

According to the trade-off theory, companies resort to third-party capitals - to the nevel at which the costs associated with bankruptcy risks - overcomes the benefit from tax interest deduction. From the 1990s onwards, the world market presents greater business flow and competition among countries starts to stimulate reduction of the corporate income tax rate. In addition, the world economical crisis - 2007/2008 - enlarges the tax competition process, tax deficits acceleration in developed countries. As a result, government and companies rise taxes to a first order theme. There is an incentive, even greater, for companies to organize their strategies in world platform. The result of this all is that production may take place in different continents, as well as the (in) tangible assets may be registered in units located in countries, other than the parents' own country. There is the possibility of opening subsidiaries in tax havens, as a part of their tax planning implementation. Given the above, this research proposes to analyze the tax influence on MNCs' investment and financing decisions. To this end, three essays are developed. The first one identifies objective aspects related to the papers analyzing this topic - journals, authors, keywords, etc. In addition, some knowledge gaps are pointed out - and a proposition of a future research agenda for researchers. The second one verifies the impact on MNCs' foreign direct investments decisions - in crisis and financial stability scenarios. Multinacional company's investment decision is highlighted for financing the domestic company. In these studies, tax proxies in literature are considered dua to their complexity - marginal benefit area, marginal tax rate, and last dollar marginal tax rate. Through structural equations econometric models, a positive relationship is identified between bankruptcy cost and multinational companies' value and their financial leverage level on, the other hand, interest rates and tax benefits present a negative relationship with these companies' indebtedness. Besides that, multinational companies are more indebted than more domestic ones - among other reasons - thanks to their negotiation power in obtaining lower interest rates. Finally, the third essay analyses the impact of MNCs' specific characteristics on their decisions of establishment subsidiaries in tax havens. Among the contributions of the study, there is the use of the non-conventional tax proxy - AREA - Which captures the short and long term investments. Another differential is the transfer pricing intensity (IPI) variable whose data are manually collected. The results indicate that MNCs make use of different instruments, - even combined - aiming at the conglomerate's total tax reduction. Therefore, despite the governments, and international entities efforts, the essay suggests that the application of a minimum rate - theme in these organizations' discussion focus - may not reach the desired effect in case there still remains jurisdictions with low-taxation options.

Keywords: Taxation, multinational, emerging markets, tax havens, financial crisis

Resumo

Segundo a teoria de *trade-off*, a empresa recorre a capitais de terceiros - até o nível em que os custos, associados aos riscos de falência - superem o benefício da dedução tributária dos juros. É a partir dos anos 90, que o mercado mundial apresenta maior fluxo de negócios e que a concorrência entre os países passa a estimular a redução das alíquotas de imposto de renda das pessoas jurídicas. Além disso, a crise econômica mundial - 2007/2008 - amplia o processo de competição tributária, a partir da aceleração de déficits orçamentários nos países desenvolvidos. Como resultado, governos e empresas ascendem os tributos a um tema de primeira ordem. Há um incentivo ainda maior para que as corporações organizem suas estratégias em plataformas mundiais. Como resultado, a produção pode ocorrer em diferentes continentes, assim como o os ativos (in) tangíveis podem ser registrados em unidades estabelecidas em países distintos das sedes das companhias. Assim, há a possibilidade de abertura de subsidiárias em paraísos fiscais, como parte da implementação de seu planejamento tributário. Diante do exposto, esta pesquisa propõe analisar a influência dos tributos nas decisões de investimento e financiamento de empresas multinacionais. Para tanto, são desenvolvidos três ensaios. O primeiro identifica aspectos objetivos relacionados aos artigos que analisam esse tema – periódicos, autores, palavras-chave etc. Além disso, são apontadas as principais lacunas do conhecimento, havendo uma proposição de agenda futura para outros pesquisadores. O segundo verifica o impacto dos tributos nas decisões de investimento estrangeiro direto das multinacionais – em cenários de crise e de estabilidade financeira. Destaca-se que a decisão de investimento da multinacional é a de financiamento da empresa doméstica. Nesse estudos, são consideradas proxies tributárias pouco utilizadas na literatura devido à sua complexidade – marginal benefit area, marginal tax rate e last-dollar marginal tax rate. Por meio de modelos de equações estruturais, identifica-se uma relação positiva entre os custos de falência e valor das multinacionais e seu nível alavancagem financeira. Ao contrário, taxas de juros e benefícios fiscais apresentam uma relação negativa com o endividamento dessas companhias. Além disso, empresas multinacionais são mais endividadas que as domésticas – entre outros motivos – graças ao seu poder de negociação na obtenção de taxas de juros menores. Por fim, o terceiro ensaio analisa o impacto de características específicas das multinacionais na sua decisão de estabelecer subsidiárias em paraísos fiscais. Entre as contribuições do estudo, está a utilização da proxy tributária não convencional – AREA – que captura investimentos de curto e longo prazos. Outro diferencial é a análise da variável de transfer pricing intensity, cujos dados são coletados manualmente. Os resultados indicam que as empresas multinacionais fazem uso de diferentes instrumentos - inclusive combinados - visando à redução de tributação total do conglomerado. Assim sendo, apesar dos esforços dos governos e entidades internacionais, o ensaio sugere que a aplicação de uma alíquota mínima - tema em pauta nas discussões dessas organizações - pode não atingir os efeitos desejados, caso ainda existam jurisdições com opções de baixa tributação.

Keywords: Tributação, multinacionais, mercados emergentes, paraisos fiscaís, crise financeira,

FIGURE LIST

Figure 1.1 - Macroeconomic indicators of OECD´s key-partners....16
Figure 2.1 - Total annual citations of the final sample....30
Figure 2.2 - Map of the influence of journals....31
Figure 2.3 - Map of the most influential sample authors....32
Figure 2.4 - Map of the most influential authors from the WoS database....34
Figure 2.5 - Map of the influence of articles from the WoS database....35
Figure 2.6 - Map of keywords co-occurrence....36
Figure 2.7 - Conceptual structure map - correspondence analysis method....37
Figure 2.8 - Map of co-authorship between countries of the articles´ samples....38
Figure 2.9 - Independent variable proxies (taxes)....43
Figure 2.10 - Dependent variable proxies (investment)....44
Figure 2.11 – Databases....44
Figure 2.12 - Econometric methods....45
Figure 4.1 - AREA....130

TABLES LIST

Table 2.1– Search filters 27
Table 2.2– Evolution of the final sample 28
Table 2.3 – Influence and co-citations of journals 30
Table 2.4 – Influence and citations of the sample authors 32
Table 2.5 – Influence and citations of the authors from the WoS database 33
Table 2.6 – Influence and citations of articles 34
Table 3.1 – Variables of econometric models 66
Table 3.2 – Descriptive statistics 74
Table 3.3 – Bankruptcy costs and leverage in financial crisis moments 76
Table 3.4 – Bankruptcy costs and leverage in nonfinancial crisis moments 78
Table 3.5 – Interest rates and leverage in financial crisis moments 79
Table 3.6 – Interest rates and leverage in nonfinancial crisis moments 80
Table 3.7 – Tax benefit and leverage in financial crisis moments 81
Table 3.8 – Tax benefit and leverage in nonfinancial crisis moments 82
Table 3.9 – Market value and leverage in financial crisis moments 84
Table 3.10 – Market value and leverage in nonfinancial crisis moments 85
Table 4.1 – Variables of econometric models 104
Table 4.2 – Descriptive statistics 108
Table 4.3 – Correlation Analysis 111
Table 4.4 – Direct corporate taxes and tax havens 112
Table 4.5 – Direct taxes of companies with intangible assets and tax havens 114
Table 4.6 – Direct taxes of companies with transfer pricing practices and tax havens 115
Table 4.7 – Intangible assets and tax havens 116
Table 4.8 – Transfer pricing and tax havens 117
Table 4.9 – Intangible assets with transfer pricing and tax havens 119
Table 4.10 – Model´s hit level classification 119
Table 4.11 – Result synthesis of the hypotheses 120
Table 4.12 – Data for calculating the AREA 131
Table 4.13 – Tax Haven jurisdiction 132

ABBREVIATIONS AND ACRONYMS LIST

2SLS	Two-Stage Least Squares
AAC	Average Annual Citations
AETR	Annual Effective Tax Rate
AMT	Alternative Minimum Tax
AREA	Marginal Benefit Area
BEA	Bureau of Economic Analysis
BEPS	Base Erosion and Profit Shifiting
CA	Correspondence Analysis
CFC	Controlled Foreign Corporation
CIF	Corrected Impact Factor
CIQ Pro	Capital IQ Pro
DCs	Domestic Companies
DITR	Difference of International Tax Rates
DOI	Digital Object Identifier
DTH	Dummy for Tax Haven
EATR	Effective Average Tax Rate
EMTR	Effective Marginal Tax Rate
ETR	Effective Tax Rate
EU	European Union
FDI	Foreign Direct Investment
GDP	Gross Domestic Product
HMRC	Her Majesty's Revenue and Customs
IFJCR	Impact Factor of Journal Citation Reports
IMF	International Monetary Fund
INTER	Nominal Interest Rate
IOB	Interest-over-Book
IRS	Internal Revenue Service
ITC	Investment Tax Credits
ITR	Indirect Tax Rate
IV	Instrumental Variables
JEL	Journal of Economic Literature
KKM	Kaufmann, Kraay and Mastruzzi index
LDMTR	Last Dollar Marginal Tax Rate
LM	Logit Model
LR	Likelihood Ratio
MB	Marginal Tax Benefit
MC	Marginal Cost Curve
MiDi	Microdatabase Direct Investment - Deutsche Bundesbank
ML	Maximum Likelihood
MNCs	Multinational Companies
MTB	Market-to-Book
MTR	Marginal Tax Rate
MTT	Market Timing Theory
NDTS	Non-Debt Tax Shields
NOLS	Net Operating Losses
OECD	Organization for Economic Cooperation and Development
OLS	Ordinary Least Squares
PLI	Property, Location and Internalization

POT	Pecking Order Theory
RD	Research and Development
ROA	Return on Assets
ROC	Receiver Operating Characteristic
SARB	South Africa Reserve Bank
SEM	Structural Equation Model
SFSO	Swiss Federal Statistical Office
STR	Statutory Tax Rate
TCRs	Thin-Capitalization Rules
TLS	Total Link Strength
TOT	Trade-off Theory
TPI	Transfer Price Intensity
UNCTAD	United Nations Conference on Trade and Development
VIF	Variance Inflation Factor
WIIW	Vienna Institute for International Economic Studies
WoS	Web of Science
WTRF	Withholding Tax Rate

SUMMARY

1. INTRODUCTION 12

1.1 Paper 1 13

1.2 Paper 2 14

1.3 Paper 3 17

References **18**

2. THE INFLUENCE OF TAXES ON FOREIGN DIRECT INVESTMENT: SYSTEMATIC LITERATURE REVIEW AND BIBLIOMETRIC ANALYSIS 21

Abstract 21

2.1 Introduction 22

2.2 Literature review 24

2.3 Methodology 26

2.4 Bibliometric analysis 28

2.5 Classifications and contributions 39

2.6 Discussion 45

2.7 Conclusion 48

References 50

3. HOW DEBT AND CRISES AFFECT MULTINATIONAL FINANCIAL POSITIONS 59

Abstract 59

3.1 Introduction 60

3.2 Literature review and research question development 61

3.3 Data description and sample selection 65

3.4 Methodology 66

3.5 Econometric models 68

3.6 Analysis of results 73

3.7 Conclusion 86

References **88**

4. TAX HAVENS AND TRANSFER PRICING: TAX EFFECTS FOR MULTINATIONAL COMPANIES OF EMERGING COUNTRIES 98

Abstract 98

4.1 Introduction 99

4.2 Literature review 101

4.3 Data description and sample selection 103

4.4 Methodology 103

4.5 Econometric models 105

4.6 Analysis of results 108

4.7 Conclusion 120

References 124

Appendix 1 129

Appendix 2 132

5. CONCLUSION 133

1. INTRODUCTION

In 1905, the German economist Georg Friedrich Knapp (Routh, 2003) formulates the thesis that currency and taxes are inseparable. All economic agents have assets and liabilities with the State. Thus, the currency becomes the unit of accounting. According to Knapp (1924), the acceptance of the currency stems from the fact that it may be used to pay taxes. Taxation is known as having an influence on financing and investment corporate decisions. According to the trade-off theory (TOT), the company, encouraged by the indebtedness tax advantage, resorts to third party capital to the level at which the costs, associated with bankruptcy risks, outweigh this advantage.

From the 1990s onwards – when the world market gains a greater business flow - competition among countries starts to stimulate the reduction of corporate income tax rates. This process is accelerated with the financial crisis of 2007, which begins with mortgage problems in the United States of America (USA), and impact – throughout 2008 – several countries and companies around the world. In 2009, the North American unemployment rate rises from 4.4% to 9.8%. This crisis is then considered as being the biggest recession since the Great Depression (1930). In 2020, the coronavirus pandemic affects the world economy even more strongly.

The North American Gross Domestic Product (GDP), in the second quarter, 2020, for instance, has a reduction of 32.9%, raising the country´s unemployment rate to 14.8%. Amidst the needs for resources to face the challenges of the pandemic, governments and world organizations rise taxes to a first order matter. Tax competition, in turn, is the consequence of the budget deficits acceleration in developed countries.

Facing this scenario, companies implement strategies on global platforms, producing in different continents. Thus, they begin to register their assets in several countries, not necessarily in the one of the organization's origin. Some of these jurisdictions operate with low or no taxation. The increase in intra-company transactions, in turn, makes transfer pricing a challenge for supranational entities and tax authorities globally. The Organization for Economic Cooperation and Development (OECD) estimates that these intra-group flows represent more than 70% of the total world trade (Merle *et al.*, 2019).

This topic has been receiving increasing public attention in recent years, particularly since the US presidential campaign in 2020. As a candidate, Joe Biden promises to impose a minimum tax rate of 21% on the profits of multinational corporations (MNCs). In 2021, already as elected president, he reduces his proposal to a minimum rate of 15%.

This project is supported by the 139 countries that make up the Inclusive Framework on Base Erosion and Profit Shifting (BEPS), the world's leading tax international discussion institution, hosted by the OECD (Poux, 2021).

The academy seeks to understand the influence of taxes on companies' decision-making. However, the complexity of tax systems challenges the depth of the studies on the subject. Thus, this research proposes to analyze the influence of taxes on investments and financing carried out by MNCs, in a global context. For that, three essays are presented. The first one examines the literature through a systematic review and bibiometric analysis. Its purpose is to provide a research agenda and framework to address the main gaps in current knowledge on the theme. The results obtained in this first essay drive the two subsequent studies of this research.

The second essay analyzes the validity of TOT for MNCs, in a pre and post financial crisis scenarios - which originated in USA in 2007. Macroeconomic indicators show that, even with a reduction in corporate tax rates, there is an increase in companies´ debts. This way, aspects related to the bankruptcy risk of these companies are investigated, as well as macroeconomic indicators that may shed light on the understanding of this fact, such as interest rates and GDP. For Almendros and Mira (2018), it is essential for the companies to calculate leverage costs in relation to their tax benefits.

Finally, the third essay investigates the investments of MNCs in countries with low or no taxation as a strategy to reduce the conglomerate's total taxes. MNCs seek to obtain benefits in these jurisdictions classified as tax havens. Their purpose is to maximize profit and minimize the economic group's taxes (Lin & Chang, 2010). This being so, the objective of this last essay is to verify whether MNCs effectively transfer profits to tax havens and through which tax-accounting mechanisms – e.g. transfer pricing, intangible assets etc.

1.1 Paper 1

The study aims at identifying general data from the papers that deal with the impact of taxes on the MNCs´ financing and investment - e.g.: main authors, journals, keywords, countries in which the educational institutions, to which the authors are linked, are located, etc. In addition, it also seeks to present knowledge gaps and propose an agenda for future research on the theme. To do so, initially, papers published in the Web of Science (WoS) database, related to the influence of taxes on foreign direct investment (FDI), are identified. In this study, MNC's FDI decision is considered as the domestic company´s financing.

The 41 papers in the final sample are those most cited in this database, whose identification is through the corrected impact factor (CIF). Next, there is a frequency count and investigation of the relationship of the general data of these studies, as well as the verification of specific citation metrics - total link strength (TLS), h-index, etc. Finally, there is the analysis of the conclusions and other contents of the papers, with identification of knowledge gaps on the research topic. Among the aspects verified, the databases, (in) dependent variables, methods considered in the econometric models, etc. stand out.

The results indicate that the tax is a determining factor for MNCs. However, the complexity of the legislations implies estimation problems and difficulties in defining and calculating the tax proxies. Furthermore, the occurrence of economic crises leads to problems in validating the resulting theories and hypotheses. Additionally, there is the influence of other determinants, such as the use of tax havens, verticalization and agglomeration. The papers in the final sample present opposite results for the same information bases and research problems, depending on how the tax independent variable is calculated.

The study contributes to the understanding of the criteria that influence the foreign direct investment by governments, world organizations, academy, companies and investors. To the best of the authors' knowledge, this is the first systematic review and bibliometric analysis identifying the gaps - and proposes an agenda for future studies - on taxation and MNCs´ financing and investment decisions.

1.2 Paper 2

In the 20th century, tax burdens are increased, as are restrictions on international mobility. High tariffs, strict capital controls, limited currency convertibility, visa and immigration strict laws reduce the scope for international tax evasion. This scenario is altered with the beginning of economic integration, in the 1980s.

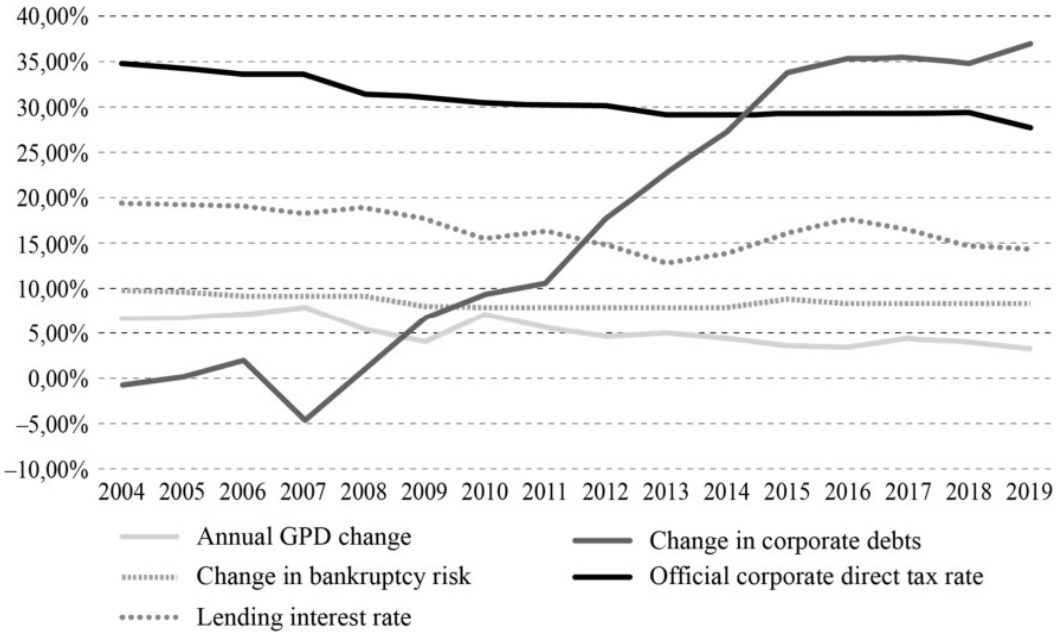
Trade liberalization, uncontrolled capital, global currency convertibility, regional integration schemes, new communication and transport technologies reduce transaction costs of transferring goods, services, capital and jobs (Genschel & Schwarz, 2011). According to the authors, from then on, reducing tax rates becomes a strategic solution by governments. Huizinga *et al.* (2008) observe a gradual decline in effective marginal rates, from about 38% in 1994 to 33% in 2008.

The economic crisis, originated in the USA in 2007, quickly spreads to other countries, given the interdependence of the global economy in trade and investments. The effects on developed and developing countries are different, considering their global economic integration and policy responses (Kahouli & Maktouf, 2015). According to the authors, it is crucial to examine how the crisis impacts FDI. For Desai *et al.* (2004), MNCs increase their investments, sales and assets more than local companies, during and after the crisis. They attribute the differences to the MNCs´ ability to finance inward investment to a greater extent than local companies. For Doytch and Eren (2012), although MNCs continue investing in efficiency projects and in search of the market, the main concerns about the post-crisis business environment remain.

According to De Mooij (2012), the deductibility of financial expenses has been creating a tax bias to the debt, favoring the use of leverage. This fact makes companies more vulnerable in times of financial crises. For the International Monetary Fund (IMF, 2009), given the potential negative impact caused by excess leverage, to understand why financing is still favored by taxes is difficult. Fratzscher (2012) studies the influence of increased global capital flows, in times of crisis and recovery. The author concludes that, for advanced economies, liquidity and risk shocks have a significant effect on developing countries´ investment flows.

Figure 1.1 shows the evolution of macroeconomic indicators of the Group of Twenty (G20) countries – between 2004 and 2019. During this period of study, there is an increase in the level of corporate debt (34.92%) and a reduction in the official direct tax rates - from 34.14% (2004) to 28.40% (2019). In addition, in subsequent moments of GDP retraction, the real interest rates of credit operations are observed as also decreasing - from 2.40% (2004) to 1.09% (2019). These rates exclude inflation, but incorporate the risk free rate and the other components of the bank spread. Finally, from the beginning of the systematic crisis - 2007 to 2011 - there is a more pronounced drop in tax rates (-3.17%) and in real interest rates (-1.06%), as well as an increase in corporate debts (15.37 %), which suggests a more accurate investigation into the validity of TOT, before and after this period.

Figure 1.1 - Macroeconomic indicators of OECD´s key-partners



Sources: World Bank, National Monetary Fund, Tax Foundation and Country Economy
Notes: Real interest rate = arithmetic average of bank interest rates for credit operations, adjusted by inflation and weighted by GDP; Annual GDP change = annual percentage change of total GDP in dollars; Official corporate direct tax rate = arithmetic average of the maximum official direct corporate tax rates; Change in corporate debts = annual percentage change in credit operations and corporate debt securities in relation to GDP; Change in bankruptcy risk = 1 / Moody´s sovereign rating, in which rating Aaa = 20 and rating Ca = 1. OECD´s key partners: Brazil, China, India, Indonesia and South Africa

For Bénassy-Quéré *et al.* (2007), the concept that capital migrates from countries with high to low tax rates is popular; however, not quite real. From the 1990s onwards, researchers begin to assert that investment decisions are indifferent to the existence of taxes (Wheeler & Mody, 1992; Ning & Reed, 1995; Porcano & Price, 1996). Thus, this study proposes to analyze the influence of taxation on the MNCs´ financial decisions, in emerging countries. The essay presents tax proxies little used in the literature, due to the complexity of their obtaining.

The marginal benefit area (AREA) variable projects the amount of debt that the firm may still contract to reach its optimal leverage point – TOT. In turn, the variable marginal tax rate (MTR) estimates the company´s future taxable income. Based on it, MNC takes indebtedness decisions at present, with the objective of maximizing the tax benefit arising from debt. The variable last dollar marginal tax rate (LDMTR) estimates the tax rate on companies - considering the contracting of new debts from their deleveraging. At last,

the non-debt tax shield (NDTS) variable considers tax benefits not arising from leverage, such as depreciation and amortization.

1.3 Paper 3

Among the knowledge gaps presented in the first essay of this research is the relevance of MNCs' investment in tax havens. According to Jones and Temouri (2016), that countries´ governments and the OECD fear the role that tax havens may play in the corporate tax base erosion is clear. The authors conclude that the use of taxes - via subsidiaries - will be an even more recurrent practice in the future. According to the Base Erosion and Profit Shifting Action Plan (OCED, 2013), a tax haven - in addition to present low taxation - tends to facilitate tax evasion for companies, by artificial means of segregating taxable income from activities that generate it. For Davies *et al.* (2018), large MNCs use tax havens to reduce their tax profit via transfer pricing.

Egger and Seidel (2013) confirm that differences in corporate taxes across economies encourage the vertical integration of producers of final goods as well as suppliers of intermediate goods. Such facts end up by increasing the commercial flow among companies. This occurs, because vertically integrated companies may transfer profits from a high to another low tax jurisdiction. Thus, most productive companies opt for verticalization, in detriment to an eventual outsourcing.

Devereux *et al.* (2018) emphasize the importance of using more complex tax proxies capturing long-term decisions, in research on the MNCs´ capital structure. However, the studies carried out in developed countries end up by using traditional tax rate proxies (Buttner *et al.*, 2018; Hopland *et al.*, 2018). Given the above, this study proposes to analyze the impact of specific characteristics of MNCs - from emerging countries - in their decision to open subsidiaries in tax havens.

To this end, the tax proxy of econometric models is calculated in an unconventional way, through the AREA - which captures short- and long-term investments. Another differential of this essay is the analysis of the transfer pricing intensity (TPI) between the MNCs and tax haven (DTH) variables, whose data are collected manually.

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2. THE INFLUENCE OF TAXES ON FOREIGN DIRECT INVESTMENT: SYSTEMATIC LITERATURE REVIEW AND BIBLIOMETRIC ANALYSIS

Abstract

In the last 30 years, foreign direct investment has increased tenfold and has become a top priority issue for governments and companies. The complexity of national and international tax legislations is a challenge for researchers who seek to understand the effect of taxes in corporate decisions. Thus, this study aims at identifying the main contributors, key areas, current dynamics and paths for future research in the field of the influence of taxes on financing and investment by multinational companies. For this purpose, a systematic literature review methodology combined with bibliometric analysis is adopted, in a sample of 41 studies obtained from the Web of Science. The results indicate that tax is a determinant for multinational companies. However, the complexity of the legislations implies problems in estimating and in the results of these studies. The theme may be concluded as being current and timely, both for the academy and for governments and corporations worldwide.

Keywords: Tax, foreign direct investment, capital structure, systematic literature review, bibliometric analysis

2.1 Introduction

In 1905, the German economist Georg Friedrich Knapp formulates the thesis that currency and taxes are inseparable. All economic agents have assets and liabilities with the State. Thus, the currency becomes the unit of accounting. According to Knapp (1924), the acceptance of the currency stems from the fact that it may be used to pay taxes. Taxation is known as having an influence on financing and investment corporate decisions. According to the trade-off theory (TOT), the company, encouraged by the indebtedness tax advantage, resorts to third party capital to the level at which the costs, associated with bankruptcy risks, outweigh this advantage. Barclay *et al.* (2013) synthesize the relevance of this discussion by highlighting the core question about the existence or not of an optimal capital structure for companies. They inquire: How could investment financing be divided between equity and third parties´ capital, to maximize the company's value?

From the 1990s onwards, the world market gains greater business flow and competition between countries stimulates the reduction of corporate income tax rates. Huizinga *et al.* (2008) observe a gradual decline in effective marginal rates, from around 38% in 1994 to 33% in 2008. In addition, the latest global financial crisis - which starts in 2007 in the United States - broadens tax competition as a consequence of the budget deficits acceleration in developed countries. In this scenario, companies implement strategies on global platforms, producing in different continents and registering their assets in several countries, which are not necessarily the organization´s origin´s. According to Damgaard *et al.* (2019), foreign direct investment (FDI) is an important driver of international economic integration. For the authors, FDI is also guided by the possibility of tax payment reduction, and part of the investments is channeled to tax havens. The International Monetary Fund (IMF) estimates a stock of 15 trillion dollars deposited in these countries in 2018.

Dunning (1977) presents studies on the performance of companies abroad, to clarify the modus operandi of international production and management. His theoretical approach - eclectic paradigm - points out the reasons why multinational companies (MNCs) opt for a global strategy rather than being limited to the national territory. According to the author, MNCs have advantages over domestic organizations, such as: lower production costs and better quality of goods - enabled by research and development, market power, advertising and branding. For Buettner *et al.* (2018), these companies may use the financial structure of foreign operations to shift the international allocation of taxable profits.In particular,

intercompany debt financing may be used to create interest payments, which are deducted in countries with high taxes (Fuest *et al.*, 2011; Dischinger *et al.*, 2014; Hansson & Olofsdotter, 2014; Devereux *et al.*, 2018). From a theoretical point of view, a MNC - which operates in a developed capital market - may minimize its global tax payments by managing loans in the economic group to which it belongs - from parent to subsidiary and among subsidiaries (Desai *et al.*, 2004; Bénassy-Quéré *et al.*, 2005; Azémar, 2010; Barrios *et al.*, 2012; Egger *et al.*, 2014; Hebous & Ruf, 2017). Governments seek FDI through tax incentives to MNCs. The effectiveness of these incentives depends on the taxation regime of companies´ residence country (Coeurdacier *et al.*, 2009). According to Egger and Seidel (2013), the raise in tax rate on profits increases the magnitude of intra company imports. FDI has an effect on economic growth, reducing unemployment, increasing government revenues from tax collection, reducing imports and / or increasing exports, contributing to technology transfer and developing local entrepreneurship (Tian, 2018).

For Buettner and Wamser (2013), the complexity of international taxation hinders accurately measure incentives to profit shifting and to FDI flow analysis. According to Herger *et al.* (2016), the role of taxation in a globalized economy, largely organized around MNCs, remains at the top of the political agenda. Ongoing debates focus on different issues, such as: tax base erosion; international tax competition; how to deal with double taxation; transfer pricing; capitalization restriction; tax havens; and inversion of headquarters to reduce corporate tax burden. In the tax and public finance literature there is an understanding that taxes affect the decision by multinationals to invest in a given country. Bénassy-Quéré *et al.* (2007), in turn, state that the concept that capital migrates from countries with high taxes to countries with low taxation is both widespread and popular; however, it may be misleading. In the 1990s, researchers begin to claim that investment decisions are indifferent to the existence of taxes (Wheeler & Mody, 1992; Ning & Reed, 1995; Porcano & Price, 1996). The researchers´ question is renewed in this scenario with fiscal crisis in many countries and income tax rate reduction: Do taxes influence the investment decision by MNCs?

In view of the above, the objective of this study is to map and analyze the existing bibliography on the MNCs´ taxation. It is intended, therefore, to identify: a) trend of publication on the topic; b) most influential journals, works and authors; c) intellectual structure of research in the area; and d) knowledge gaps offering opportunities for the evolution of this relevant, current and multidisciplinary theme. To this end, a systematic literature review and a bibliometric analysis are carried out on the relationship between taxes

and financing / investment decisions by MNCs. The period considered is twenty years (2000-2019). The systematic review aims at systematically researching and locating studies to carry out a critical evaluation of the literature and to point out gaps in the research (Tranfield *et al.*, 2003). The bibliometric method, in turn, uses data from publications to construct structural images of scientific fields, allowing the recognition of the current state of research and the identification of future approaches (Li *et al.*, 2017).

The other items in this study are organized as follows. The literature review is presented in section 2. In section 3, the methodology of the systematic review is detailed. The bibliometric analysis is itemized in section 4. In section 5, the classification and main contributions of the works are outlined. Section 6 highlights the results of the research, including the knowledge gaps. Finally, section 7 points out the main conclusions and limitations.

2.2 Literature review

For Hebous and Ruf (2017), since the seminal contributions by Modigliani and Miller (1958 and 1963) and Miller (1977), the influence of taxes on companies´ capital structure is discussed. The theoretical assumption of its non-existence - in the initial propositions of these authors - gives rise to theories considering the reality of market imperfections, analyzing their relevance in the definition of the financing sources for companies. Among the theories that stand out are TOT (Kraus & Litzenberger, 1973; Bradley *et al.*, 1984; Kane *et al.*, 1984) and the pecking order (Stiglitz, 1973; Myers & Majluf, 1984), among others.

When company has taxable profit, the use of more third party capital reduces the taxes due. TOT uses the debt tax advantage and the costs associated with bankruptcy risks as fundamental arguments for the existence of an optimal level of debt (DeAngelo & Masulis, 1980). According to Faulkender and Smith (2016), the empirical research has been evincing that the variation in the capital structure stems from the change in the estimated corporate income tax rates. For the authors, both the tax rates applied to MNCs and the taxable profit earned are being obtained incorrectly. According to Buettner and Ruf (2007), the rates - statutory and marginal - used in the research, have no predictive power. The authors argue that significant differences are found, depending on the rate applied in the studies.

The Organization for Economic Cooperation and Development (OECD) states, in 2017, that the latest financial crisis culminated in a complex and unstable international environment. As a result, there is a combination of budget deficits in developed countries and the continued intensification of a tax competition among governments, with the purpose of attracting companies and generating jobs. Countries hosting foreign companies tend to introduce specific rules that limit the possibility for subsidiaries to transfer profits abroad. From initiatives by the European Commission (1997), the OECD (1998, 2013) and supranational organizations, suggestions are being defined for MNCs´ tax planning opportunities reduction (Buettner *et al.*, 2018). These suggestions are intended to restrict tax base erosion and profit shifting among countries.

In addition, the OECD estimates that the extent of the problem is a challenge, with a loss of up to US$ 240 billion annually, in global corporate income tax revenues. According to Cen *et al.* (2017), one of the most used ways for profit shifting from subsidiaries in tax havens is through the customer-supplier relationship. For Zucman (2014), the reduction in corporate income tax rates is not natural. Profit shifting is the main driver of this decline, over the fiscal competition for productive capital. This phenomenon is not a necessary by-product of globalization and technological change, but the result of policies by countries with a high tax burden. Globalization does not imply that corporate income tax rates are fated to decrease. Countries implement measures to restrict profit shifting, through capitalization rules and by limiting the possibilities of manipulating transfer pricing (Buettner & Wamser, 2013; Schindler & Schjelderup, 2016; Hopland *et al.*, 2018). The literature also examines the effectiveness of transfer pricing regulations (Buettner *et al.*, 2018).

According to the property, location and internalization (PLI) approach - also known as an eclectic paradigm - companies use these three potential sources of competitive advantage to decide on their global performance. For Dunning (2000), MNCs have the property or privileged access to certain assets that are not available to local competitors, showing the ability to coordinate them across borders. The location advantage is related to the characteristics of the country – e.g., labor supply and incentives from the local government. Finally, internalization allows the control of production processes, reducing the dependence on intermediaries (Egger & Seibel, 2013; Herger *et al.*, 2016). MNCs exploit these resources, rather than selling or renting them to other companies.

Long *et al.* (2015) provide empirical evidence that, as studying the Chinese market, the presence of FDI positively affects the institutional quality of host countries. Subsidiaries

located in regions with a higher level of FDI tend to face a lower level of tax burden, less tax bureaucracy and better legal protection. According to Becker *et al.* (2012), public policymakers should not only be concerned with the FDI flow, but also with its quality, that is, the results that the investment provides to the host country. The studies by Fedderke and Romm (2006) about South Africa suggest that FDI is impacted by taxation, net return rate, wages and infrastructure. For Bénassy-Quéré *et al.* (2007), besides the tax rates, other aspects also interfere in companies´ location decision and the consequent increase in FDI - geography, market size, agglomeration effects and public infrastructure.

The studies by Becker and Riedel (2012) suggest that the activities of parent and subsidiary are positively correlated and that, a 10% increase in taxation in the invested country reduces the affiliate's share capital by 5.6%. According to them, from the well-being point of view, this cross-border tax effect on stock of capital generates negative fiscal externality in companies´ taxation. Empirical results by Ang (2008), in Malaysia, indicate that the Gross Domestic Product (GDP), financial development, investment in infrastructure, trade openness and macroeconomic uncertainty are positively related to FDI. In contrast, the increase in corporate income tax, exchange rate appreciation and social uncertainty have a negative impact on FDI. Arena and Roper (2010) demonstrate in their work that, in addition to the corporate rate, FDI is impacted by personal taxes and withholding taxation on dividends and interest.

2.3 Methodology

According to Fink (2019), the literature review is a systematic, explicit and reproducible project to identify, evaluate and interpret the existing body of documents. The analysis of this material should be rigorous, independent and follow a methodological approach. In addition, it needs to be explicit in the presentation of the procedures adopted, comprehensive in scope, including all relevant material. Therefore, it should be reproducible by others who follow the same path in the study of the topic. The literature review seeks to: i) summarize the existing research, recognize patterns, themes and issues, and ii) identify the conceptual content of the field, contributing to the theory. Thus, the purpose of this section is to present the work structure, providing a methodological organization on how the research is designed. The strict criteria used in the systematic review are linked to the objective of basing the analyses on the best quality evidence. The independent variable “taxes” is studied for its influence on MNCs´ financing and investment - dependent variable. The systematic review of the literature, based on the methodology adapted from Tranfield *et*

al. (2003), is detailed in steps from 1 to 6, while the bibliometric analysis is described in step 7. Step 8 combines both methods:

Step 1 - The Web of Science (WoS) database is chosen for the quality of its organization, data availability and selection of papers with relevance in the scientific community.

Step 2 - Filters are used to select the initial sample, with the objective of delimiting the research - see Table 2.1.

Table 2.1– Search filters

Description	Data
Search date	September, 25, 2019
Keywords	Tax* and FDI or Foreign Direct Investment and Capital Structure
Language	English
Material type	Article and review
Concentration areas	Economics or business or business finance or management.
Search period	From 2000 to 2019
Initial sample	**211**

Note: The asterisk (*) is used to allow searching for keywords by their root. As a result, it is possible to obtain them in their various suffix variations.

Step 3 - From the 211 papers found, 37 are excluded for not receiving, at least, one citation in the WoS database itself. Thus, the resulting sample consists of 174 articles, from January 1st, 2000 to September 25th, 2019 - see Table 2.2.

Step 4 - The articles are obtained from the academic bases: Science Direct, Ebsco, ProQuest, JSTOR and Google Scholar. From them, the following information is obtained - complementarily to those already extracted from WoS - to capture the general data of the papers: title, name, affiliation institution and authors´ country of origin, journal name, volume, start and end page number, year of publication, origin country, keywords, Digital Object Identifier (DOI), Journal of Economic Literature (JEL) and number of the article citations in the WoS database.

Step 5 - A reading of the abstract, introduction and conclusion of each paper is performed, to check whether they are in accordance with the defined theme, excluding the non-pertinent articles. Thus, 55 are excluded for referring to domestic companies´ capital structure, 24 for not considering multinational investment and financing as the dependent variable, 23 for not having corporate taxes as an independent variable, 19 for not presenting an empirical test, 9 for being financial institutions or real estate companies, 3 for dealing exclusively with taxes

of individuals. At the end of this process, the final sample consists of 41 articles - see Table 2.2.

Table 2.2– Evolution of the final sample

Description	Quantity of articles
Initial sample	**211**
(-) Articles with less than one citation in WoS	37
(=) Intermediate sample	**174**
(-) Domestic companies	55
(-) Dependent variable	24
(-) Independent variable	23
(-) With no empirical test	19
(-) Financial institutions or Real estate companies	9
(-) Individual tax	3
(=) Final sample	**41**

Step 6 – Referred articles are classified and reviewed in order to identify their objectives, contributions and results.

Step 7 – The 41 articles of the final sample are submitted to the bibliometric analysis, using the VOSviewer and RStudio softwares.

Step 8 – Both results of bibliometric analysis and systematic review are analyzed together, with the identification of information, standards and content gaps, in order to indicate future directions of knowledge.

2.4 Bibliometric analysis

By means of the WoS database, the bibliometric analysis is employed to construct structural images. The method allows presenting the influence of articles, journals and authors as measuring the co-citations among them by using the total link strength (TLS). Co-citation is defined as the frequency with which two documents are cited together by other documents. The more co-citations two documents receive, the greater their TLS will be - see equation 1 (Van Eck & Waltman, 2010):

$$TLS_{ab} = \frac{C_a}{W_a W_b} \quad (1)$$

Where:
C_{ab} = number of co-citations by the authors of A and B
W = total number of citations by authors of A and B

According to Castriotta *et al.* (2019), bibliometric analysis is the most appropriate method for studying the conceptual structure of a research area. The method allows the development of studies that recognize the current state of research and identifies opportunities for future studies, through different approaches presented – academy´s interest, analysis of the influence of journals, authors, articles, keywords and contribution by country.

2.4.1 Academy´s interest

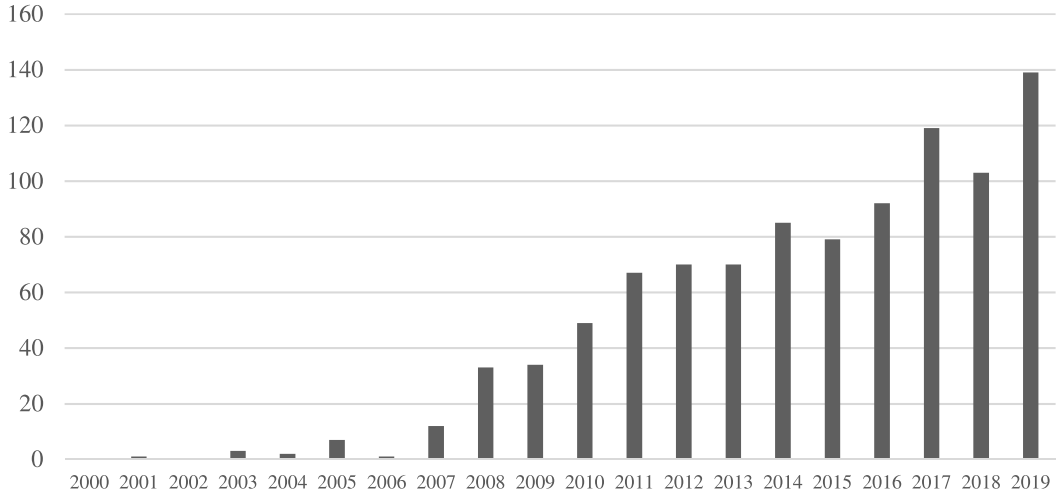
The WoS database calculates the h-index (16) of the final sample. The h-index is a measure of productivity and of impact of the theme, made available by the databases. Productivity is quantified by the number of papers, while the impact is measured by the number of citations which the researchers' publications received. In the construction of the indicator, the number of articles in the sample (41) is weighted with their respective citations, discounting the disproportionate weight of the articles with few or many citations – outliers. The sample period is 20 years (2000-2019). With these data, it is possible to calculate the academic relevance of the topic by using the M-index (Banks, 2006) - see equation 2. According to Banks, the theme is promising when "M" is in the range between 0.50 and 2.00, presenting a window of opportunity for future research.

$$M = \frac{H}{NY} = \frac{16}{20} = 0.80 \quad (2)$$

Where:
H = h-index of sample articles
NY = number of years since the sample was first published

The result of M = 0.80 indicates that the academy is likely to have an interest in the topic. Figure 2.1 ratifies the relevance of this matter, considering the evolution of the total annual citations of the final sample.

Figure 2.1 - Total annual citations of the final sample



2.4.2 Analysis of the influence of journals

Through VOSviewer, the 10 most influential journals of the sample of this study are identified, as well as the total co-citations received - see Table 2.3. These journals are important sources of research and reveal a potential target journal for future publications on the theme. The number of papers published by the journal - International Tax and Public Finance, in 16% or 7 articles in the sample - should be noted as not necessarily representing a greater impact for academy. The influence of journals can be seen in Figure 2.2.

Table 2.3 – Influence and co-citations of journals

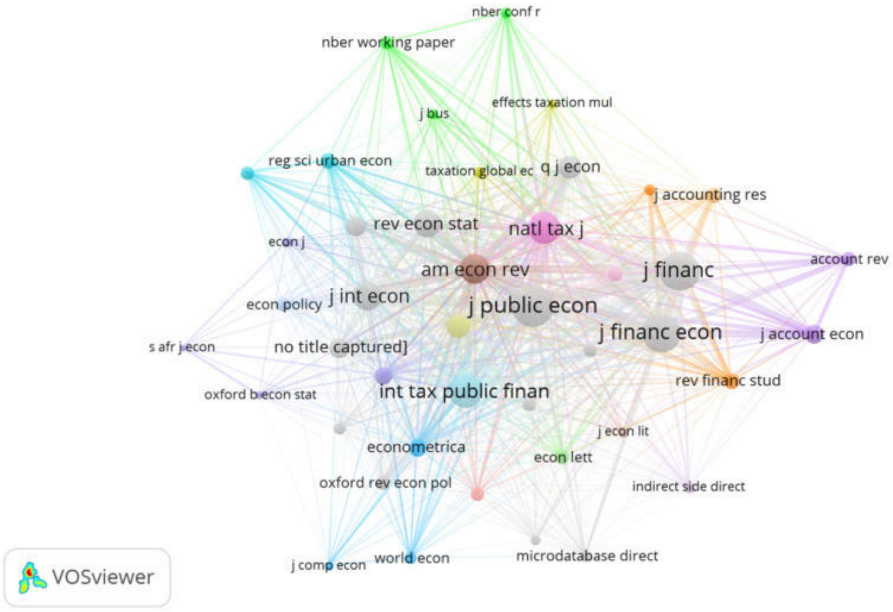
Journal title	Publisher	TLS	Co-citations
Journal of Public Economics	Elsevier BV	3989	103
Journal of Finance Economics	Elsevier BV	3674	88
Journal of Finance	American Finance Association	3392	86
International Tax and Public Finance	Kluwer Academic Publishers	2742	67
National Tax Journal	National Tax Journal	2308	62
Journal of International Economics	Elsevier BV	2286	56
American Economic Association	American Economic Association	2193	53
Review of Economics and Statistics	MIT Press	1812	44
Accounting Review	American Accounting Association	1350	28
European Economic Review	Elsevier BV	1277	30

Note: TLS = total link strength

In Figure 2.2, the nodes of the networks represent the journals, the size of the nodes is the dimension of influence of journals and the edges (lines) between the nodes represent

the strength of co-citation of publications. Co-citation occurs when two authors or two sources appear together in the reference list of a single publication (Tunger & Eulerich, 2018). According to Small (1973), co-citation is more dynamic, and thus, better accompanies changes in the field of knowledge. The main collaborating journals in the field are also identified in the co-citation network map, being Journal of Public Economics (j public econ), Journal of Finance Economics (j financ econ) and Journal of Finance (j financ).

Figure 2.2 - Map of the influence of journals



2.4.3 Analysis of the influence of the sample authors

Table 2.4 lists the 10 most influential authors in the sample, their citations and the total of articles published, according to the WoS database. Wamser, Egger, Buettner and Overesch are the most important researchers in the studies of the relationship between taxes and MNCs´ financing and investment. The identification of Wamser as the author with the greatest impact is only possible through the bibliometric tools - VOSviewer and RStudio - used in this study, since the researcher is not the main author in any of the works in the sample; however, he participates as a co-author in 6 papers with Buettner, Egger and Overesch.

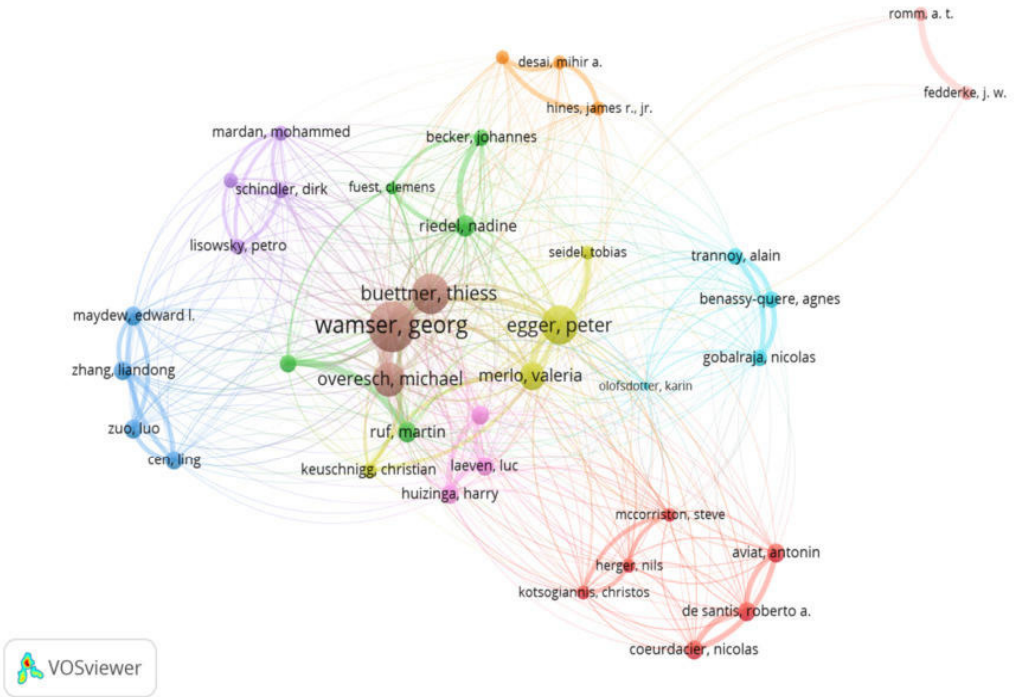
Table 2.4 – Influence and citations of the sample authors

Author	TLS	Citations	Articles
Wamser Georg	234.96	67	6
Egger Peter	196.97	42	4
Buetter Thiess	172.77	86	5
Overesch Michael	145.30	28	4
Merlo Valeria	110.08	17	2
Zhang Liandong	104.00	21	2
Bénassy-Quéré Agnes	97.67	117	2
Riedel Nadine	97.08	63	4
Ruf Martin	75.00	50	2
Olofsdotter Karin	74.92	11	2

Note: TLS = total link strength

Figure 2.3 shows the map of the influence- TLS - of the sample researchers, in which the relationship clusters between the authors are identified. The most influential are: 1) Wamser, Buetter and Overesch; 2) Egger, Merlo and Seidel; 3) Zhang, Cen, Maydew and Zuo; 4) Bénassy-Quéré, Olofsdotter; Gobalraja and Trannoy; and 5) Riede, Ruf and Fuest.

Figure 2.3 - Map of the most influential sample authors



2.4.4 Analysis of the influence of authors and articles from the WoS database

In the 41 articles in the final sample, 758 authors are cited, from whom Table 2.5 shows the 10 most influential ones. These researchers are a relevant source for the study of the relationship of taxes in the investment decision. Graham, for example, is an acknowledged researcher on the influence of taxes as a determinant of companies' capital structure. However, since his studies do not specifically investigate MNCs, he is not listed in Table 2.4. It happens that, given the relevance of his research on the development of tax variables, his name stands out as the second most influential author in the complete WoS database. Also noteworthy is the example of authors present in both Tables 2.4 and 2.5. Devereux is present in Table 2.6 with a recent work on the topic, being the most influential author in the complete database, as mentioned in Table 2.5. His researches are also references in the development of tax variables - see Table 2.10.

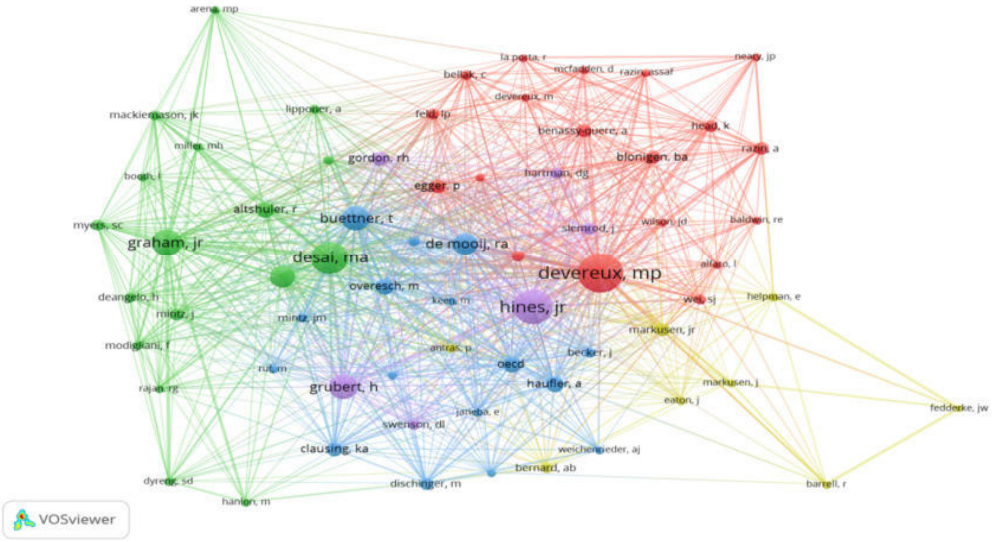
Table 2.5 – Influence and citations of the authors from the WoS database

Author	TLS	Citations
Devereux, Michael	1968	51
Graham, John	1605	35
Desai, Mihir	1493	38
Hines, James	1364	41
Buettner, Thiess	940	25
Grubert, Harry	920	27
De Mooij, Ruud	918	20
Huizinga, Harry	742	20
Overesch, Michael	596	13
Haufler, Andreas	573	13

Note: TLS = total link strength

Figure 2.4 presents the map of the influence network - TLS - of the authors from the WoS database, confirming the impact of these researchers. The relationship clusters between the authors are shown, with the most influential ones being: 1) Devereux, Bénassy Quéré and Egger; 2) Graham, Desai and Huizinga; 3) Hines, Grubert and Swenson; and 4) De Mooij, Buettner, Overesch and Haufler.

Figure 2.4 - Map of the most influential authors from the WoS database



2.4.5 Analysis of the influence of articles

The 41 articles in the final sample mention 1235 references, 215 from which are cited at least twice, while other 98 are cited at least three times. The popularity of an article is measured by its number of citations, while its prestige is measured by its influence - TLS. Table 2.6 shows the 10 most relevant articles on the theme. As analyzing them, it appears that, in addition to confirming the importance of direct taxes, they present complementary paths - relevance of indirect taxes (Desai *et al.*, 2004); impact of loans among subsidiaries (Huizinga *et al.*, 2008); dimension of rates - marginal, average and statutory (De Mooij & Ederveen, 2003; Devereux & Griffith, 2003); agglomeration effect (Devereux & Griffith, 1998; Bénassy-Quéré *et al.*, 2005; Demekas *et al.*, 2007) and tax havens (Hines & Rice, 1994).

Table 2.6 – Influence and citations of articles

Reference	Title	TLS	Citations
Desai *et al.* (2004)	Foreign direct investment in a world of multiple taxes	526	12
Huizinga *et al.* (2008)	Capital structure and international debt shifting	491	13
De Mooij e Ederveen (2003)	Taxation and foreign direct investment: a synthesis of empirical research	471	11
Devereux e Griffith (1998)	Taxes and the location of production: evidence from a panel of US multinationals	468	11
Hines (1999)	Lessons from behavioral responses to international taxation	425	11

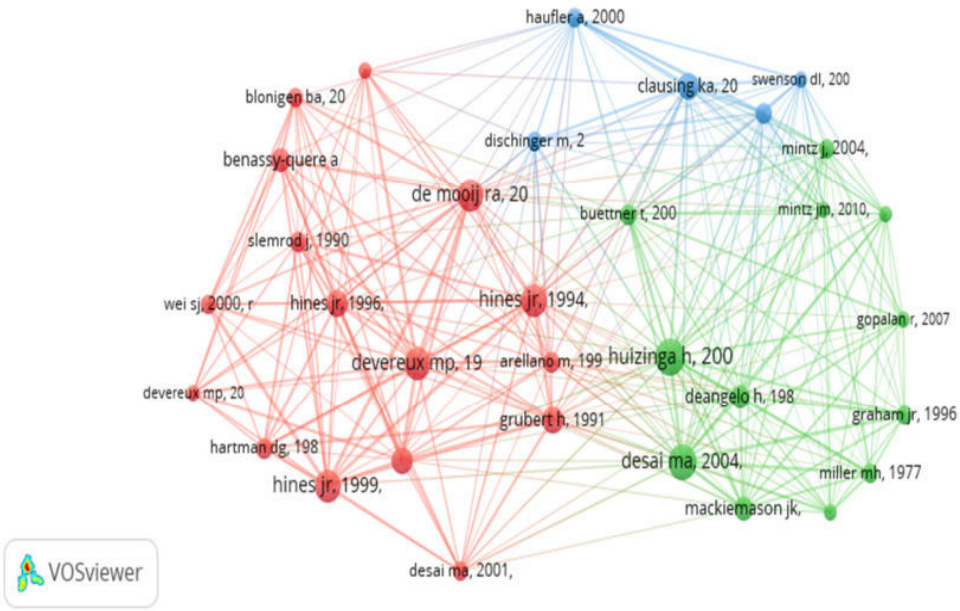
Table 2.6 – Influence and citations of articles (continuation)

Reference	Title	TLS	Citations
Hines e Rice (1994)	Fiscal paradise: Foreign tax havens and American business	365	11
Bénassy-Quéré *et al.* (2005)	How does FDI react to corporate taxation?	341	7
DeAngelo e Masulis (1980)	Optimal capital structure under corporate and personal taxation	335	7
Devereux e Griffith (2003)	Evaluating tax policy for location decisions	330	8
Clausing (2003)	Tax-motivated transfer pricing and US intrafirm trade prices	300	8

Note: TLS = total link strength

Figure 2.5 shows the map of the influence network - TLS - of the 98 papers with at least three citations. Relationship clusters among articles are identified: 1) Desai *et al.* (2004), Huizinga *et al.* (2008), DeAngelo and Masulis (1980) and Graham (1996); 2) De Mooij and Ederveen (2003), Devereux and Griffith (1998), Hines (1999), Hines and Rice (1994) and Bénassy-Quéré *et al.* (2005); and 3) Clausing (2003), Haufler and Schjelderup (2000), Grubert, H. (2003).

Figure 2.5 - Map of the influence of articles from the WoS database



2.4.6 Analysis of keywords

According to Strozzi *et al.* (2017), the keywords used by the authors indicate the content of articles and may mean the trend of research in the area - see Table 2.7. RStudio provides the option of obtaining the results through the "keywords plus", which often appear in work references, but not in their titles. The tool makes use of the database's algorithms and improves the search power of these references - see Table 2.8.

Table 2.7 – Keywords of the authors

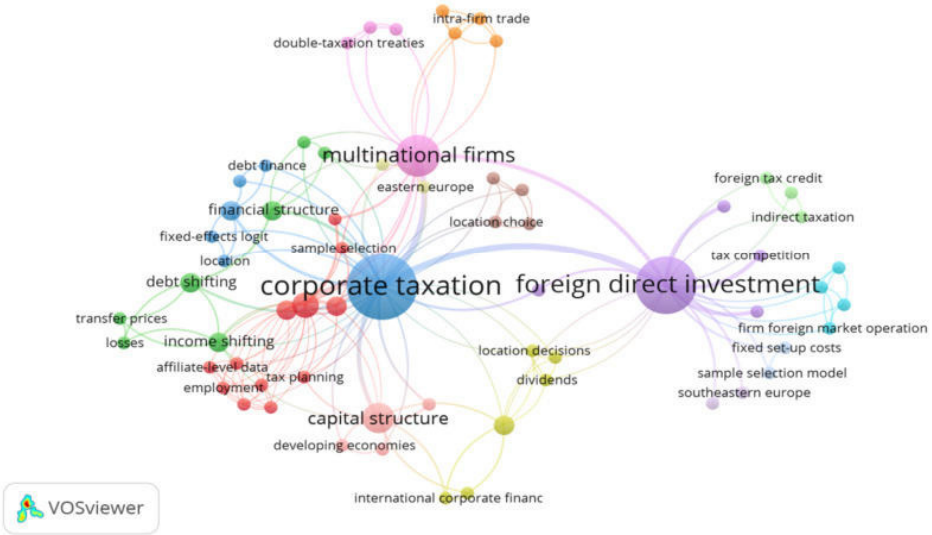
Keywords	No. articles
Corporate taxation	15
Foreign direct investment	12
FDI	7
Multinational firms	6
Capital structure	4
Profit shifting	4
Corporate tax rates	3
Corporate taxes	2
Debt shifting	2
Financial structure	2

Table 2.8 – Keywords of the references

Keywords	No. articles
Foreign direct investment	15
Location	12
Firms	9
Taxation	9
Capital structure	7
Income	7
Debt	6
FDI	5
Panel	5
Taxes	5

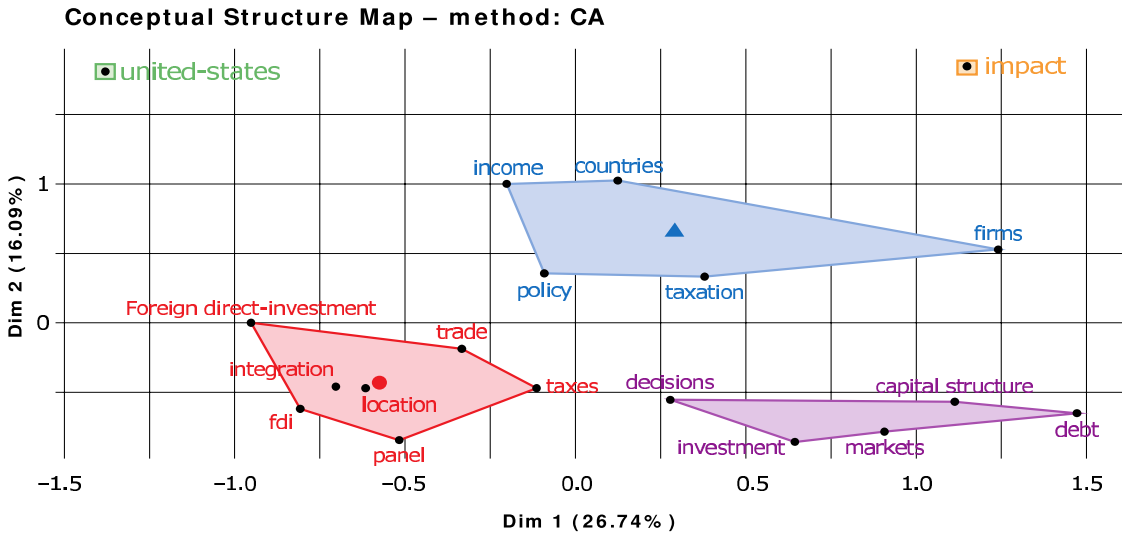
Figure 2.6 shows the VOSViewer´s co-occurrence map, with the most used keywords in the sample articles, as well as the relationship among them. Such a fact allows to observing the most addressed themes by the authors. The strength of the lines connecting the nodes illustrates the intensity of the relationships of the keywords mentioned in the sample (Carvalho *et al.*, 2013).

Figure 2.6 - Map of keywords co-occurrence



The analysis of the co-occurrence of keywords also allows to revealing the conceptual framework of the articles. Its objective is to identify relationships and trends emerging from the research on the influence of taxes on MNCs´ financing and investment. This verification is done through correspondence analysis (CA) that uses dimensionality reduction techniques to draw a conceptual structure map of the field and to cluster common concepts (Fisher, 1940) - see Figure 2.7. Cluster 1 - in lilac - identifies the relevance of research on location, verticalization and agglomeration decisions (Hubert & Pain, 2002; Hansson & Olofsdotter, 2013; Cen *et al.*, 2017). Cluster 2 - in red - highlights tax decisions influenced by issues of fiscal policy and the macroeconomic ones, in host countries (Coeurdacier *et al.*, 2009; Fuest *et al.*, 2011; Becker & Riedel, 2012). Cluster 3 - in green - highlights the importance of domestic capital markets and the financing and investment decisions by economic groups (Desai *et al.*, 2006; Huizinga *et al.*, 2008; Egger *et al.*, 2014). Finally, Figure 2.7 highlights the words strongly related to the 3 clusters, which are "the United States" and "impact".

Figure 2.7 - Conceptual structure map - correspondence analysis method



2.4.7 Analysis of contribution by country

The theme “influence of taxes on the financing and investment by MNCs” is analyzed by the articles whose most of samples are made up of companies from Germany, the United States and England, in that order of relevance. Together, authors represent more than 55% of the sample of the 41 papers analyzed in this study. This fact is due to the

economic relevance of these countries that concentrate 32% of the world GDP (IMF, 2019), as well as due to the differentiated quality of the databases made available by their governments. The total TLS of these countries appears prominently in the academy - 49% of the sample - see Table 2.9. This concentration, on the one hand, contributes robustly to the literature. However, on the other hand, it skews the results on the topic. Tax laws and macroeconomic conditions in these three countries - highly developed - tend not to represent most of the variables for other countries.

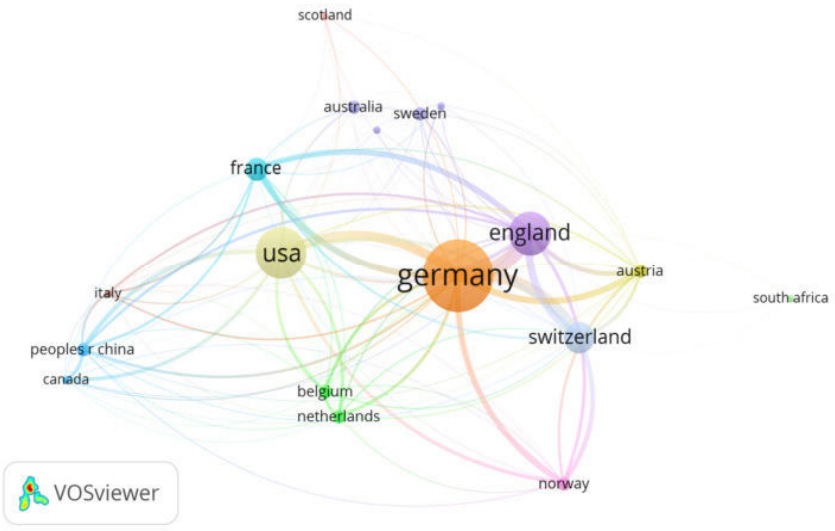
Table 2.9 – Contribution by countries

Country	TLS	Articles	Citations
Germany	2084	17	255
England	1487	9	125
The United States	1119	11	368
Switzerland	1092	6	69
France	619	4	154
Norway	583	2	13
Austria	557	2	34
China	467	2	16
Belgium	319	2	165
Netherlands	319	2	165

Note: TLS = total link strength

The map of network of countries that most collaborate and contribute, jointly, to the topic under research is shown in Figure 2.8. The outstanding presence of Germany as the most influential country on the theme is confirmed, as well as the peripheral presence or absence in emerging countries.

Figure 2.8 - Map of co-authorship between countries of the articles´ samples



2.5 Classifications and contributions

Table 2.10 presents the complete list of the 41 papers in the final sample, with their classifications and main contributions. Studies are in decreasing order of Corrected Impact Factor - CIF - see first column - whose formula is presented in equation 3. For Jabbour (2013), classification is one of the most relevant aspects of systematization, as it identifies the characteristics of the articles under analysis. Thus, the second column indicates the reference of these articles - authors and year. The third column presents the main characteristics of their samples - country, number of observations, period of analysis and database. The fourth and fifth columns present the main dependent (DV) and independent variables (IV), respectively. The sixth column shows the econometric methods used. Finally, the seventh column presents their main contributions.

$$CIF = AAC \times (1+IFJCR) \quad (3)$$

Where:
CIF = Corrected Impact Factor
AAC = Average Annual Citations
IFJCR = Impact Factor of Journal Citation Reports

Table 2.10 – Classifications and contributions of articles

N.	Reference	Sample	DV	IV	Method	Contributions
1	Huizinga *et al.*, 2008	32 c. Europe 90,599 1994-2003 Amadeus	Indebted ness	EMTR; DITR	OLS	The international debt transfer reflects the tax regimes of all countries where the MNC operates, and not just the bilateral differences in tax rates in relation to the parent.
2	Cen *et al.*, 2017	USA 42,565 1994-2009 EDGAR	FDI subsidiar y dummy	STR; MTR	Logistic regression	Companies keeping close relationships with customers and suppliers are able to identify and implement tax avoidance strategies through supply chains.
3	Brülhart *et al.*, 2012	Switzerland 26,233 1999-2002 SFSO	New invest- ments	EATR	Poisson regression	MNCs´ investments react negatively to tax charges, but this discouraging effect is weaker in concentrated sectors.
4	Desai *et al.*, 2006	USA 128,368 1992-1997 BEA	Equity	DITR; ITR	Static panel data	American companies´ FDI responds to income tax rate variation in accordance with the level of capital control applied by subsidiary´s country.
5	Bénassy-Quéré *et al.*, 2007	USA 1,346 1994-2003 BEA	FDI stock	STR; EATR	OLS and static panel data	Jointly with the tax rates, geography, market size, agglomeration effects and public infrastructure participate in companies´ location decision: all on equal terms.

Table 2.10 – Classifications and contributions of articles (continuation)

N.	Reference	Sample	DV	IV	Method	Contributions
6	Ang, 2008	Malaysia n.i. 1960-2005 n.i.	FDI flow	STR	2SLS	FDI inflows react negatively to a corporate tax rate increase. Currently, Malaysia's rate is higher than of its neighbors. The country needs to consider reducing the tax to promote international competitiveness.
7	Coeurdacier *et al.*, 2009	Europe 83,034 1985-2004 Thomson Reuters's	Mergers and acquisitions	ETR	Poisson regression	Policymakers can help attract capital by reducing corporate tax rates and the degree of market regulation, improving the country's financial system.
8	Barrios *et al.*, 2012	33c. Europe 26,648 1999-2003 Amadeus	FDI subsidiary dummy	EATR; DITR; WTRF	Logistic regression	Additional taxation in the origin country has an independent and strongly negative effect on the likelihood of the subsidiary's location in potential host countries.
9	Fedderke e Romm, 2006	South Africa n.i. 1956-2003 SARB	FDI flow	STR	Simultaneous equations	Corporate taxation increases exclude FDI, with an elasticity of 2.65 to the average value of the STR sample.
10	Demekas *et al.*, 2007	24 c.World 1995-2003 OECD	FDI stock and flow	STR	Dynamic panel data	The host country's high corporate tax burden discourages non-privatization-related FDI.
11	Desai *et al.*, 2004	USA 42,856 1982, 1989 and 1994 BEA	Total assets	DITR; ITR	OLS	Direct and indirect taxes have comparable and independent effects on MNCs´ asset allocation.
12	Egger *et al.*, 2014	Germany 227,558 1996-2007 MiDi	Indebtedness	STR	OLS	MNCs´ internal debt is not only used to avoid taxes, but also to compensate for differences in other fundamentals in affiliates - weak institutional environment and underdeveloped capital market.
13	Bénassy-Quéré *et al.*, 2005	11 c.OECD 1,307 1984-2000 Eurostat	FDI flow	DITR; STR; EATR; AETR	Static panel data	There is an asymmetry in the impact of tax differentials on FDI: although lower rates in recipient countries are unable to significantly attract foreign investment, higher taxes tend to discourage new FDI inflows.
14	Buettner and Wamser, 2013	Germany 157,155 1996-2005 MiDi	Indebtedness	STR	Static panel data	Domestic debt is influenced by the differences between the host country´s income tax rates and the multinational group´s lowest rate. The effects are small and domestic debt appears to be an insignificant vehicle for German companies.
15	Buettner and Ruf, 2007	Germany 37,695 1996-2003 MiDi	Fixed assets	STR; EATR; EMTR	Logistic regression	FDI is influenced by taxes: while STR has a stronger predictive power than EATR, EMTR, in turn, is not predictive.
16	Becker *et al.*, 2012	22 c.Europe 100,697 2000-2006 Amadeus	Fixed assets	STR; EMTR	Static and dynamic panel data	The amount of FDI is influenced by corporate taxes. The effects of the quality of this FDI are the result of tax reduction, to the extent that the investment contributes to the corporate tax base.
17	Nagano, 2013	Asia and Oceania 22,314 1999-2009 Thomson Reuters's	FDI flow	STR	Simultaneous equations	Corporate tax rate, host country's population size, and per capita income are determinant for FDI inflow, either through M&A or greenfield.
18	Arena and Roper, 2010	23 c.World 8,248 1995-2004 Thomson Reuters's	FDI subsidiary dummy	ETR; WTRF	Logistic regression	MNCs are more likely to issue debt in subsidiaries with high corporate taxes, when the parent country adopts an exemption system on repatriated profits. FDI is also influenced by the incidence of withholding taxes on the interest remittance and on dividends.

Table 2.10 – Classifications and contributions of articles (continuation)

N.	Reference	Sample	DV	IV	Method	Contributions
19	Becker and Riedel, 2012	25 c.Europe 23,438 1995-2006 Amadeus	Fixed assets	EMTR	Dynamic panel data	Tax increases to parent company have a significantly negative and robust effect on the subsidiary's stock of capital. MNCs "export" the tax burden from the parent company to its affiliates.
20	Dischinger *et al.*, 2014	29 c.Europe 83,800 1995-2007 Amadeus	Subsidiary 's taxable profit	DITR	OLS	Differences between corporate tax rates - parent and subsidiaries - impact profit shifting. In other words, hosting multinational headquarters helps to protect countries with high taxes against large FDI outflows.
21	Altshuler and Grubert, 2001	USA 23,200 1996 American Treasury	FDI subsidiary dummy	EATR	Logistic regression	There is no evidence that FDI is influenced by low taxation, with the exception of MNCs having high deferred tax stocks - even so, with little significant effect.
22	Hansson and Olofsdotter, 2013	EU15 1,899 1996-2004 Eurostat	FDI flow	EATR; EMTR	OLS with Heckman' s estimation	FDI is impacted by corporate taxes, however, it is sensitive to the inclusion of agglomeration economies.
23	Egger *et al.*, 2009	OECD 2,361 1991-2002 UNCTAD	Bilateral FDI stocks	EMTR; EATR	OLS	Bilateral tax rates directly affect FDI and unilateral ones affect it indirectly - and positively - through its negative influence on competing domestic companies.
24	Fuest *et al.*, 2011	Germany 167,367 1996-2007 MiDi	Indebted ness	STR; DITR	Static panel data	The effect of the host country's corporate tax rate on the debt ratio of affiliates in developing economies is positive and greater than the same effect for affiliates in developed economies.
25	Hebous and Ruf, 2017	Germany 208,573 1999-2012 MiDi	Indebted ness	STR	OLS	The estimated marginal effect of STR suggests that an increase in the corporate tax rate of 1% leads to an increase of the investee's debt by 0.35%.
26	Buettner *et al.*, 2016	Germany 42,850 1996-2004 MiDi	Indebted ness	STR	OLS	The tax incentive for using internal loans is effectively removed for restricted subsidiaries and responds positively - albeit with a weak effect - to subsidiaries in unrestricted environments.
27	Overesch and Wamser, 2010	Germany 384,721 1996-2005 MiDi	Fixed assets	STR; ETR; EATR;	Poisson regression	From a single country point of view, tax policies may be used as an active tool to attract subsidiaries and capital.
28	Devereux *et al.*, 2018	United Kingdom 93,259 2001-2009 HMRC	Indebted ness	EATR; MTR	Dynamic panel data	MTR, based on tax returns, has greater explanatory power for MNCs´ leverage than that based on financial statements. Errors in measuring tax incentives, using financial statements, may lead to an underestimation of tax effects on MNCs´ capital structure.
29	Wolff, 2007	European Union 2,057 1994-2003 Eurostat	Equity	STR	OLS	Origin country´s high taxes increase the likelihood that MNCs will reinvest profits abroad, while market size factors play a less important role.
30	Hopland *et al.*, 2018	Norway 128 1998-2005 Compustat	Indebted ness and remittan-ces (royalties; licenses; rents)	STR	OLS	In response to changes in tax rates, the remittance of income may be greater than traditional studies suggest. Especially for internal debt transfers, where the flexibility to send *ex post* income is limited.

Table 2.10 – Classifications and contributions of articles (continuation)

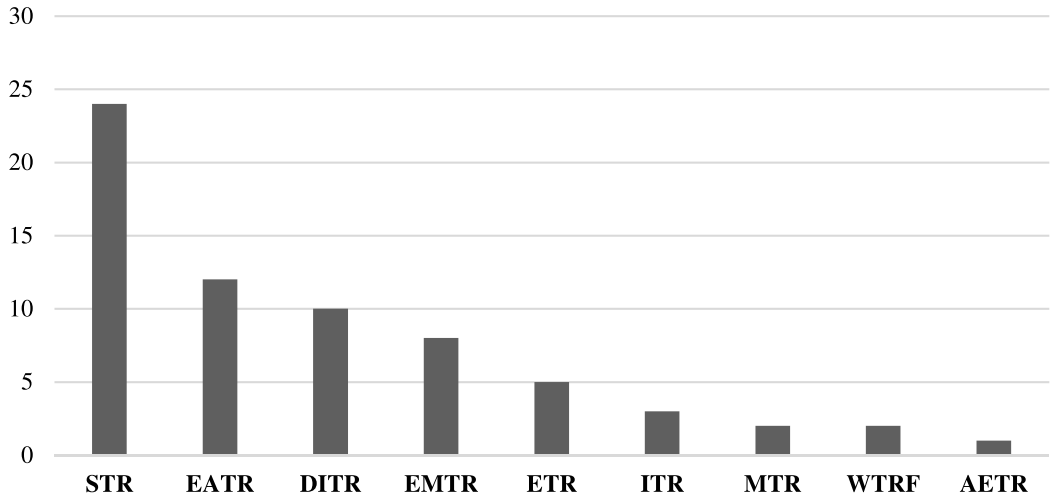
N.	Reference	Sample	DV	IV	Method	Contributions
31	Wijeweera *et al.*, 2007	USA n.i. 1982-2000 OECD	FDI stock	STR; ETR ETMR; EATR	OLS	The corporate income tax rate in the USA exerts a significantly negative impact on the American FDI inflow. The results are more robust for STR than for ETR.
32	Kersan-Škabić, 2015	Southeast Europe 287,000 2000-2011 WIIW	FDI stock	STR; DITR	Dynamic panel data	The corporate tax rate is not as important as other market-related determinants for FDI attractiveness. The low level of taxes alone is not sufficient to attract foreign capital.
33	Buettner *et al.*, 2018	Germany 94,187 1997-2006 MiDi	Fixed assets	STR	Static panel data	A stricter thin-capitalization rule has significant adverse effects on FDI in countries with high taxes. Moreover, in countries imposing thin-capitalization rules, the sensitivity to tax rate on FDI increases. Transfer pricing, for its part, has no significant effect on FDI.
34	Herger *et al.*, 2016	32 c. World 11,248 1999-2010 KPMG	Mergers and acquisitions	STR; EATR EMTR; ITR	Poisson regression	Though confirming that taxes have a negative impact on FDI, the results suggest a more subtle interpretation, as the tax effects will also depend on the nature of FDI strategies, which, in turn, depend on the different factors that determine these decisions.
35	Hubert and Pain, 2002	Germany 880 1981-1996 OECD	FDI stock	ETR	Dynamic panel data	Results show significant effects of host country's fiscal instruments on the location of intra-EU direct investment, with agglomeration effect importance.
36	Egger and Seidel, 2013	USA 1,383 1999-2006 OECD	Business volume-parent and subsidiary	DITR	Poisson regression	Differences in corporate taxes between economies stimulate vertical integration of producers of final and intermediate goods, generating more intra-company trade. Vertically integrated MNCs may shift profits from a highly taxed jurisdiction, making this type of organization more attractive compared to outsourcing.
37	Goodspeed *et al.*, 2011	OECD 4,448 1984-2002 UNCTAD	Equity	STR	OLS	FDI is sensitive to taxation in developed host countries, but not in developing countries. This is an important result contradicting the literature and may stem from weak tax management in developing countries.
38	Azemar, 2010	USA 216 1992,1994,1996,1998 and 2000 IRS	Total assets	EATR	OLS	Evidence indicates that taxes influence American MNCs´ foreign investment. In addition, once profit is generated, they play an important role in decisions about where to report them and whether to reinvest them or repatriate them.
39	Hansson and Olofsdotter, 2014	27 c. Europe 5,527 1997-2007 Eurostat	FDI stock	DITR	OLS with Heckman's estimation	Corporate taxation negatively affects FDI decisions, for new member countries of the European Union (EU), while for existing members - EU15 - labor taxes are a more important determinant of FDI flows.
40	Buettner *et al.*, 2011	Germany 78,020 1996-2006 MiDi	Indebtedness	STR	OLS	Tax sensitivity of capital structure is more impacted by tax shields unrelated to debt: depreciation and carry forward losses reduce it.
41	Egger and Merlo, 2011	Germany 63,496 1996-2005 MiDi	FDI stock	STR	Poisson regression	Higher statutory tax rates in a potential host country significantly reduce the probability for a German MNC to establish its first affiliate. Further, higher STRs reduce the activity at the intensive margin, that is, they tend to reduce the number of affiliates - in addition to the first.

Notes: n.i. = not informed; c = countries.

Databases: BEA = Bureau of Economic Analysis (USA); EDGAR = Securities and Exchange Commission's Electronic Data Gathering, Analysis, and Retrieval; IRS (USA) = Internal Revenue Service; HMRC = Her Majesty's Revenue and Customs; SARB = South Africa Reserve Bank; MiDi = Microdatabase Direct Investment - Deutsche Bundesbank; SFSO = Swiss Federal Statistical Office; UNCTAD = United Nations Conference on Trade and Development; WIIW = Vienna Institute for International Economic Studies. Independent variables: AETR = Annual effective tax rate (Mendoza *et al.*, 1994); DITR = Difference of international tax rates (Hines & Rice,1994); EATR = Effective average tax rate (Overesch, 2005); EMTR = Effective marginal tax rate (Devereux *et al.*, 2002); ETR = Effective tax rate paid (Devereux & Griffith, 2003); ITR = Indirect tax rate (McKenzie *et al.*, 1997); MTR = Marginal tax rate (Graham & Mills, 2008); STR = Statutory tax rate; WTRF = Withholding tax rate (Huizinga & Voget, 2009).

In Figure 2.9, the frequency of the different tax proxies used by the authors is presented as an explanatory variable - taxes - in order to better capture the influence on MNCs´ financing and investment decisions. This aspect is quite controversial in the literature and is far from a consensus. Some researchers state that an inadequate tax proxy may result in estimation errors (Buettner & Ruf, 2007; Wijeweera *et al.*, 2007). At this point, the systematic review result is interesting because, despite the intense debate about the appropriate measure to capture the influence of taxes, most articles (36%) make use of the simplest one among them. These studies only consider the statutory tax rate (STR). Yet, the second most used proxy - EATR - is also easy to obtain and is analyzed in 18% of the papers. On the other hand, proxies requiring deepening of calculations and of domestic and international tax legislations - MTR and ITR - have low adherence.

Figure 2.9 - Independent variable proxies (taxes)



As for Figure 2.10, there is a diversification of the proxies used for the dependent variable - MNCs´ investment, with emphasis on the most frequent ones in the articles - indebtedness (23%), FDI stock (18%) and FDI flow (18%).

Figure 2.10 - Dependent variable proxies (investment)

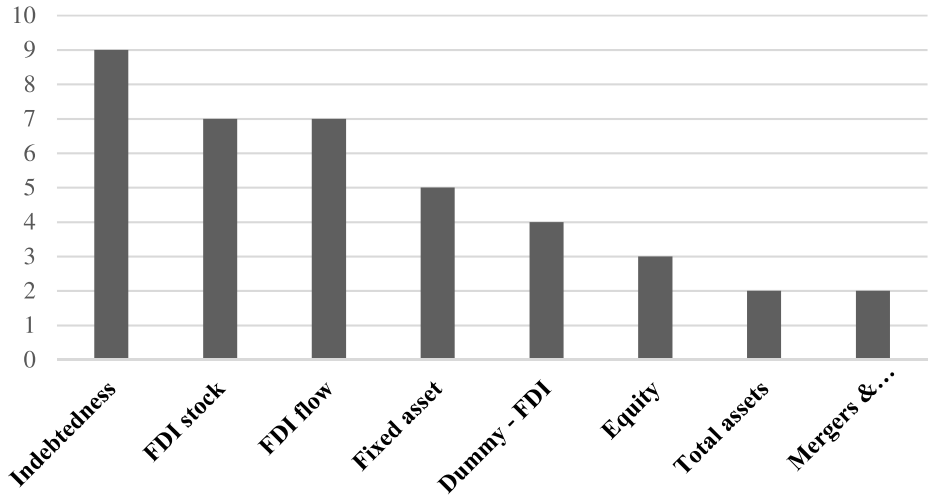
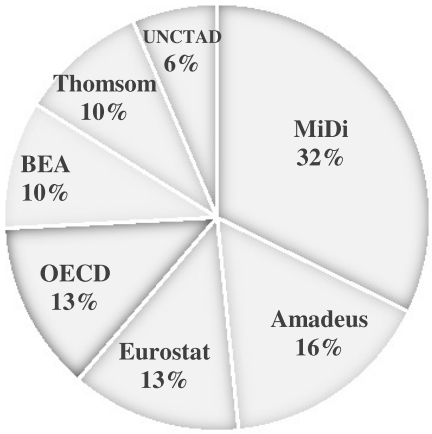


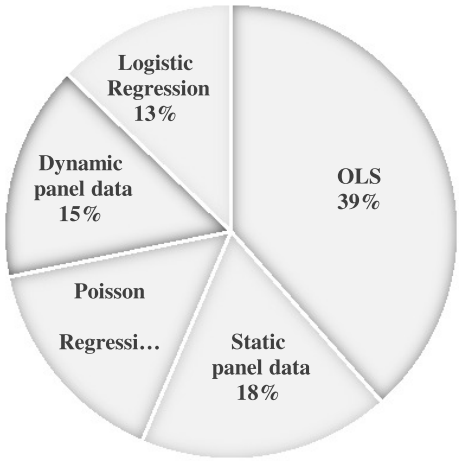
Figure 2.11, in turn, shows the databases frequency used in the samples of the 41 papers in this study. The MiDi database - made available by the German government - is the most used by researchers (32%). This tool's distinctive position results from the Bundesbank's attention to data - since 1976 - on foreign direct investment by local companies. The other highlighted databases are: Amadeus - Moody´s group - which provides data from European companies (16%); Eurostat, which is the European Union's main statistical database (13%); and OECD (13%).

Figure 2.11 – Databases



The 41 articles in the final sample apply econometric models in their empirical studies. Figure 2.12 shows the methods used by the authors, with emphasis on the Least Squares Method - Ordinary Least Squares (OLS) - which is a widely spread way of estimation in econometrics. It is present in 37% of the articles in this study, followed by static panel data (16%); Poisson regression (14%); dynamic panel data (14%) and logistic regression (12%).

Figure 2.12 - Econometric methods



2.6 Discussion

Research on the influence of taxes on the financing and investment by MNCs has been evolving and following the changes in the world economy, with impacts on public and corporate finances - economic turbulences (Ang, 2008); fiscal crises and tax competition among countries (Overesch & Wamser, 2010); verticalization of production chains (Egger & Seibel, 2013) and business strategies (Herger *et al.*, 2016); international organisms' concerns about fiscal erosion (Cen *et al.*, 2017) and country responses through restriction to thin-capitalization and transfer pricing rules (Buetter *et al.*, 2018).

Academy demonstrates being aware of the demands of governments and companies in seeking fiscal, financial and economic environment improvement. The analysis of the 41 articles in this study aims at identifying: a) trends in publication, b) journals, papers and most influential authors, c) research intellectual structure, and d) gaps offering opportunities for the theme evolution. As a result, the following main findings are highlighted.

Research on the impact of taxes on investment decision may be inferred as having increased at a growing pace in the last 20 years, arousing current interest by academy. This

fact is evidenced by the number of citations and results of the h-index calculation of the WoS database. The most relevant papers in the sample suggest as paths for research: the study of economic groups (Huizinga *et al.*, 2008) and their verticalization processes (Cen *et al.*, 2017), in agglomeration environments (Bénassy-Quéré *et al.*, 2007) and observing the activity sector (Brülhart *et al.*, 2012). The most influential works on the topic - sample and WoS complete database - in turn, suggest research on the influence of indirect taxes (Desai *et al.*, 2004) and estimation of studies, with attention to tax proxies (Devereux & Griffith, 1998; De Mooij & Ederveen, 2003).

The analysis of the keywords conceptual structure allows to graphically visualizing the categories of the variables and, thus, verifying the degree of interaction among them. With this method, it is possible to dimension 3 interest groups by the academy for research - a) verticalization and agglomeration; b) tax regulation in the countries of investees; and c) economic group.

The authorial analysis indicates that Wamser, Egger, Buettner and Overesch are the most important researchers in the sample. Their works study the themes of loans to affiliates and tax restrictions; capital markets of investees´ countries; proper selection of tax proxies, bilateral taxes and verticalization. The most influential authors in the field - WoS complete database - are Devereux and Graham who contribute, mainly, with the construction of tax proxies.

The journal with the largest number of publications on the subject is the International Tax and Public Finance, representing 16% or 7 articles in the sample. Considering the importance that this journal assigns to the theme, its recent publications are studied to identify opportunities for publication. The covered topics deal with tax rules on internal indebtedness and profit shifting in host countries, with emphasis on thin-capitalization and transfer pricing (Buetter *et al.*, 2016; Herger *et al.*, 2016; Buetter *et al.*, 2018). The most influential journals are: Journal of Public Economics, Journal of Finance Economics and Journal of Finance. The Journal of Public Economics publishes the article on the significant differences observed in the estimates of the taxes-investment relation, depending on the tax proxy applied in the studies (Hebous & Ruf, 2017).

The data samples in the 41 articles are predominantly from companies in Germany, the United States and England. On the other hand, noteworthy is that the only two papers focusing on developing countries - Malaysia and South Africa - are among the 10 most influential in the sample, which may suggest the interest by academy for studies in peripheral countries.

The econometric method of differences-in-differences (DID) is not used by researchers of the sample. However, its use mitigates the identification problems - associated to tax effects estimation - present in studies adopting regression models with panel data (St. Clair & Cook, 2015). The possibility of assessing the influence of taxes, based on relevant changes in tax systems, seems promising for future publications. The United States, for example, make in 2017 - in force in 2018 - the biggest restructuring of their tax rules, since 1996. The effect of financial crises on MNCs´ capital structure and investment is poorly studied, offering opportunities for research.

The analysis of the empirical studies of the 41 works of the sample in this systematic review identifies 30 papers which consider corporate tax as an important determinant in MNCs´ decision. As for the results of the other studies, most researchers conclude that tax is determinant in the financing and investment decisions by MNCs. Three of them define taxes as a weak determinant (Bénassy-Quéré *et al.*, 2005; Buettner & Wamser, 2013; Herger *et al.*, 2016). Some four others point to taxes at the same level as other determinants - tax shields; market size; agglomeration effects; public infrastructure (Bénassy-Quéré *et al.*, 2007).

Finally, there are still four others who vary between considering impact on certain conditions or irrelevance of taxes, such as: a) FDI is not influenced by taxes, with the exception of MNCs having deferred tax high stocks - even so, with little significant effect (Altshuler & Grubert, 2001); b) FDI is sensitive to taxation in developed host countries, but not in developing countries (Goodspeed *et al.*, 2011); c) corporate taxation negatively affects FDI decisions, for new member countries of the European Union (EU), however, for existing members - EU15 - labor taxes are a more important determinant (Hansson & Olofsdotter, 2014); d) taxes do not influence subsidiaries in countries with thin-capitalization restrictions, but they positively influence - weak effect - subsidiaries with unrestricted environments (Buettner *et al.*, 2011).

The main issue of this research is about the influence of corporate taxes in the decision of financing and investment by MNCs, in a world scenario with continued reductions in tax rates, provided by countries seeking for new resources, amid low growth economies. Most part of the sample studies confirms the relationship expected by the literature. However, the authors of this study have many concerns about the effectiveness of this relationship by the authors - tax proxies; multiple taxes; complexity of tax legislation; tax havens; effect of the financial crisis; increasing indebtedness with decreasing corporate rates - challenging researchers to new studies.

2.7 Conclusion

The relationship between taxes and investment decisions by MNCs has been arousing the interest by researchers, especially in recent years, given the strong growth of international business - globalization - and the challenges of world economies - low growth. The literature review of any topic tends to be complex, due to the volume of published works and the variations that the subject builds over time. The systematic review, combined with bibliometric analysis, aims at identifying the main contributors, the key areas, the current dynamics, and suggests guidelines for future research on the theme. From the application point of view, this study contributes to the understanding of the research on the influence of taxes on MNCs´ financing and investment decisions, the paths trailed, the current stage and the direction to where the research is evolving. It also highlights the gaps in the current body of knowledge and its future perspectives.

The work classifies and uses bibliometrics in the articles extracted from the systematic literature review. The 41 selected articles are cited in the WoS database. Citations are influenced by scientific factors - such as the magnitude of the study - and by non-scientific factors - such as the author's literature knowledge. Subsequently, the papers are reviewed and their main contributions are identified. These elements of research and analyses allow to deepening the study of the topic.

Among the knowledge gaps identified, the estimation problems of the models stand out, due to the difficulty in calculating the tax proxies. The papers present opposite results for the same information bases and research problems, depending on how the tax independent variable is modulated. The complexity of tax laws and the specifications of each country pose a challenge to researchers. Identifying the direct corporate tax rate is not enough to analyze its impact on investments. There are a multitude of rules that may accompany the calculation of taxable profit, such as offsets, exemptions, tax benefits, destination of results, exclusions and additions to the accounting calculation. Therefore, there is an opportunity for deeper studies on the topic, which include all the tax rules of the countries in which the MNC develops business. In addition, the sample articles are verified as not analyzing the effect of the financial crisis on the MNCs´ financing-investment relationship. The study of bankruptcy costs, in the midst of an increase in corporate debts, has a potential for contribution.

At last, also found is that the studies focus on the relationship between direct taxes and FDI, with little analysis on the impact of indirect taxes. These taxes become even more

relevant, in a scenario with business global chains. Multinational companies have several production options, including verticalization. Thus, the analysis, combined of (in) direct taxes, allows a better understanding of the theme.

As for the limitations of this study, the results presented refer to the set of articles from the journals that meet the established selection criteria and the database used. In addition, the analysis of articles, exclusively with empirical tests, may have excluded the examination of papers with mathematical modeling, research and essays also having relevant contributions.

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3. HOW DEBT AND CRISES AFFECT MULTINATIONAL FINANCIAL POSITIONS

Abstract

This study is among the first ones to investigate whether multinational companies are more resilient to economic crises. This is done by analyzing the role of tax effects and capital structure - in periods of crises and financial stability - on the financial position of emerging market companies with special tax proxies. The sample includes 977 public companies from the key-partners of the OECD (Brazil, China, India, Indonesia, and South Africa). Using structural equations for pre- (2004–2007) and postcrisis (2008–2011) periods, as well as for the long-run period (2004–2019), we find a positive and statistically significant impact of multinational companies´ leverage level on bankruptcy costs and firm value and negative and statistically significant relations among leverage, interest rates and tax benefits during crisis periods. Analysing the entire period (2004 to 2019), we find an increase in the leverage level of multinational companies and a decrease in that of domestic companies. Compared with domestic companies, multinational companies have higher financial negotiation power and can achieve lower interest rates than domestic companies. These results are crucial for understanding the financial behaviour of multinational companies in emerging countries in periods of crisis and financial stability.

Keywords: Tax benefits, financial crisis, leverage, multinational, structural equations

3.1 Introduction

Financial crises impact companies' financial position, mostly their bankruptcy costs, interest rates, tax benefits and market value. The subprime crisis, which began in late 2007 in the United States, also reduced the investment level of firms. In the face of uncertainties, there was a reduction in demand by customers, and a shock in the supply of credit (Jucá & Fishlow, 2021). Nevertheless, according to Desai *et al.* (2008), before and after economic crises, multinational companies (MNCs) increase their investments, sales and assets more than domestic companies (DCs). After the subprime crisis, MNCs increased their leverage level, taking advantage of the reduced interest rates in global capital markets (Duran & Stephen, 2020). Currently, there is a consensus among G7 finance ministers to set a minimum global taxation level. The Organization for Economic Cooperation and Development (OECD)´s country members seek a final agreement ensuring that foreign subsidiaries also pay fair rates where they are located, not only in the country of the MNC's headquarters (OECD, 2021).

In fact, since the beginning of the subprime crisis, there has been a more pronounced drop in tax and real interest rates, as well as an increase in corporate debt. According to Trade-off Theory (TOT), companies´ optimal capital structure seeks the balance between the tax benefit and the bankruptcy risks arising from debts. Therefore, it is equally relevant to verify whether this debt increase - after crisis - also causes a risk raise of companies' bankruptcy, as expected.

Some studies have aimed to better understand the leverage level of companies from the perspective of tax issues. However, according to Wijeweera *et al.* (2007), an inadequate tax proxy may result in estimation errors. Sorbe *et al.* (2017) studied the financing decisions made by MNCs, using the statutory rate as a proxy for tax. Nicodano and Regis (2019) and Dhawan *et al.* (2020) analysed the relations among bankruptcy costs, taxes, and leverage, considering the effective payment of taxes as a tax proxy. Nevertheless, the use of adequate proxies to capture the influence of taxes on MNCs´ decisions remains a challenge, given the complexity of tax legislation.

This essay - as mentioned at the Introduction of this thesis (item 1.2) - has the objective of studying the impact of MNC´s financing decisions on their financial position. Thus, it examines whether MNCs are more resilient than DCs - in times of crisis and economic stability - on bankruptcy costs, interest rates, taxes, and market value. The hypotheses arising from this analysis are as follows: i) leverage positively affects MNCs´

bankruptcy costs; ii) leverage negatively affects MNCs´ interest rates; iii) leverage negatively affects MNCs´ tax benefit; and iv) leverage positively affects MNCs´ market value.

This verification is performed using a structural equation model (SEM) and ordinary least squares (OLS) model. The final sample consists of 977 publicly traded nonfinancial MNCs and DCs from five OECD`s key partner countries: Brazil, China, India, Indonesia, and South Africa. The data were obtained from the Capital IQ database, totalling 15,632 observations, from 2004 to 2019. An analysis of the complete sample period of 16 years was conducted to study the financial position of multinational companies. Moreover, the impact of the crisis on these companies is analysed before (2004–2007) and after (2008–2011) the last systemic crisis. Finally, the challenging estimated tax proxies are the marginal benefit area, marginal tax rate, last-dollar marginal tax rate, and nondebt tax shields.

Econometric tests confirmed the hypotheses of this study. MNCs' leverage has a positive influence on their bankruptcy costs and value. Additionally, we found that the greater the MNCs´ leverage is, the lower their nominal interest rates and tax benefits are. Such relations occur in crisis scenarios. This indicates the existence of differences between the financing policies by MNCs and DCs. Unlike DCs, MNCs increase their leverage level after the systematic crisis, mainly because of their ability to transfer resources among their foreign subsidiaries.

The remainder of this paper is organized as follows. The following section summarizes the relevant literature review and the development of the research question. Section 3 describes the data, Section 4 presents the methodology of this study, Section 5 discusses the empirical results, and Section 6 concludes and draws inferences from the findings.

3.2 Literature review and research question development

The seminal paper of Modigliani and Miller (1958) described that, in a perfect world, a firm's capital structure is irrelevant to the company´s value. Some years later (1963), the authors recognized that companies' value is increased by debt tax benefits, indicating that companies should seek a greater degree of leverage. Since then, a great number of academicians worldwide have attempted to identify the determinants of the best debt-to-equity ratio, with theories such as trade-off theory (TOT) (Kraus & Litzenberger, 1973), agency (Jensen & Meckling, 1976), signalling (Ross, 1977), personal taxes (Miller, 1977), debt overhang (Myers, 1977), informational asymmetry (Myers & Majluf, 1984),

pecking order theory (POT) (Myers, 1984) and market timing theory (MTT) (Baker & Wurgler, 2002). These theories have been tested through regressions studying the variation in capital structure policies based on slope coefficients estimated for factors such as company size, taxes, tangibility, profitability, growth opportunities, age, liquidity, operating cash flow, depreciation, bankruptcy costs, agency costs, and dividend distribution policy (Van Binsbergen *et al.*, 2010).

Van Binsbergen *et al.* (2010) analyse the definition of companies´ target capital structure based on the previous definitions of debt costs versus tax benefits. Among the verified costs, in addition to interest, are those of financial distress, agency conflicts between managers and investors, and other nontax benefits reflected in the ideal debt options. In this diagnosis, there may be divergences in managing real costs and perceived costs. A company with a low level of leverage has a high potential for tax benefits. However, this may stem from the high cost of raising debt or from managerial bias.

According to Myers (1984), tax benefits encourage third parties´ capital usage by companies, as pointed out by TOT, as their optimal leverage level results from choosing how much debt financing and how much equity financing to use after balancing the tax benefits and the debt costs. However, this capital source is limited by the increase in debt costs resulting from an increase in the company's probability of default. Thus, companies tend to lose value and increase bankruptcy costs by moving away from their optimal capital structure. Bankruptcy costs arise from the likelihood of financial distress and the inherent costs of a company's insolvency (Miller, 1977).

At the end of the 20th century, the world economy was affected by three financial crises: the collapse of the exchange rate mechanism in Western Europe (1992), the devaluation of the Mexican peso (1994/1995), and the East Asian financial crisis in 1997 and 1998 (Ballester *et al.*, 2019). The 21st century, in turn, began with the financial crisis of 2007, which originated in the default of real estate credit operations, carried out by the subprime segment in the United States, impacting other economies as of 2008. Global stock markets slumped nearly 40% from their previous high, investment banks collapsed, and world economic activity fell dramatically (Vieito *et al.*, 2013).

In 2009, the American unemployment rate increased from 4.4% to 9.8%. This crisis has led to the second largest recession since the Great Depression (1929), being surpassed only by the COVID-19 pandemic. In the second quarter of 2020, American GDP retracted 32.9%, raising the unemployment rate to 14.8% (Siegel & Van Dam, 2020). Fratzscher

(2012) and Kahouli and Maktouf (2014) examine how liquidity shocks and the risk of financial crises impact foreign direct investment (FDI).

According to Harris and Raviv (1991), leverage increases with nondebt tax benefits, such as company size, tangibility, and growth opportunities. However, it decreases with the volatility and costs of bankruptcy. Luciano and Nicodano (2014) argue that MNCs provide guarantees among their subsidiaries, aiming to reduce the probability of bankruptcy and lower the costs of this process, which allows an increase in their optimal leverage level. In a crisis scenario, Deesomsak *et al.* (2004) demonstrated that companies´ specific conditions and the countries in which they are located affect their decisions on capital structure.

MNCs use their network of subsidiaries to take on debts and capture business opportunities, changing their strategies and obtaining lower rates (Song, 2021). They are more concerned with the effects of future inflation on debt costs than the immediate risk of default. Higher monitoring costs and institutional risk factors associated with internationalization intensify bankruptcy risks and impact the lower leverage of MNCs (Gvimah *et al.*, 2021). Thus, the government's monetary policy instrument during financial crises encourages them to contract more debt by reducing risk-free interest rates. The hypotheses arising from these arguments are as follows.

H1a - Leverage positively affects MNCs´ bankruptcy costs in financial crisis moments.

H1b - Leverage positively affects MNCs´ bankruptcy costs in nonfinancial crisis moments.

Companies tend to issue more debt when current interest rates are lower than their historical levels (Barry *et al.*, 2008). Research in international environments has assessed the importance of global macroeconomic and institutional factors as determinants of leverage. Some authors have suggested that MNCs use more external debt than local companies to mitigate political/systemic risks and exchange rate variations (Reeb *et al.*, 1998; Erel *et al.*, 2020). However, Kwok and Reeb (2000) found a negative relationship between MNCs´ leverage and the investee's local market risk. They used the parent´s capital markets to overcome imperfections in investee markets (Desai *et al.*, 2005; Wang *et al.*, 2020). Regarding the maturity of debts, Doukas and Pantzalis (2003) observed that MNCs use more short-term and less long-term debt than DCs.

Recently, faced with a scenario of stagnation in world growth, legislators have wondered the correctness of their expansionist global monetary policies by reducing interest

rates and facilitating access to credit for companies. An undesired effect is the excess corporate debt in an environment of a financial crisis (IMF, 2015). These arguments support the following hypotheses:

H2a - Leverage negatively affects MNCs´ interest rates in financial crisis moments.

H2b - Leverage negatively affects MNCs´ interest rates in nonfinancial crisis moments.

As determined by Bénassy-Quéré *et al.* (2005), higher taxes tend to reduce MNCs´ investments. Richardson and Lanis (2007) found a negative relationship between taxes and debt. MacKinlay (2015) stated that changes in corporate taxes do not change companies' propensity to issue debt. MNCs are more likely to issue debt at subsidiaries with high corporate tax rates when the parent´s country adopts an exemption system for repatriated profits (Desai *et al.*, 2006; Foley *et al.* 2007; Arena & Roper, 2010). In addition, MNCs´ internal debt is verified as being used to avoid taxes and compensate for differences in other fundamentals if their affiliates are located in countries with a weak institutional environment and an underdeveloped capital market (Egger *et al.*, 2014).

Given this, the complex structures of MNCs appear to impose challenges in investigating the effects of taxation on their capital structures. According to Keen *et al.* (2010), during financial crises, distortions in tax laws cause companies to contract more debt than necessary. Therefore, correcting these distortions is important to reduce their exposure to future crises and the vulnerability of capital inflows in these countries. Becker and Riedel (2012) suggested that parent and subsidiary activities are positively correlated. In addition, they mention that a 10% increase in taxation in the invested country reduces the affiliate´s share capital by 5.2%, negatively impacting the international capital flow. The empirical evidence supports that the tax benefit of interest is a weaker determinant of debt utilization than was previously thought.

According to Jin (2021), companies prefer to consider other ways to avoid taxes—such as tax shelters and non-debt tax shields (NDTSs) rather than reduce corporate debt. He states that these tax avoidance methods are less sensitive in periods of crisis than in periods of stability. These facts point to the following hypotheses:

H3a - Leverage negatively affects MNCs´ tax benefit in financial crisis moments.

H3b - Leverage negatively affects MNCs´ tax benefit in nonfinancial crisis moments.

According to TOT, companies reach their maximum value when they obtain the optimal point of their capital structure. For Jensen (1986), leverage works as a tool for monitoring and reducing agency problems, with positive impacts on the company's value. The study by Mishra and Gobeli (1998) is consistent with the theory of internalization, in which multi-nationality influences the company's valuation as long as it maintains a high investment in technology. Information asymmetry impacts a company's value, and its adverse effect is influenced by leverage (Fosu *et al.*, 2016). Recent studies on MNCs and DCs have found a positive relationship between leverage and firm value both in times of crisis and economic stability (Luo & Wang, 2018; Botta, 2020; Huynh *et al.*, 2020). The hypotheses arising from these arguments are as follows.

H4a – Leverage positively affects MNCs´ market value in financial crisis moments.

H4b – Leverage positively affects MNCs´ market value in moments of nonfinancial crises.

3.3 Data description and sample selection

The initial sample considered in this study consists of 1,249 publicly traded companies from OECD's key partner countries: Brazil, China, India, Indonesia, and South Africa. Company data are collected from Capital IQ, while country variables are obtained from the World Bank and include interest rates, inflation and populations (World Bank, 2021a) and the Kaufmann Kraay and Mastruzzi indexes (World Bank, 2021b). The sample analyses 16 years, from 2004 to 2019, aiming to study multinational companies´ financial positions. Additionally, the impacts of crisis on these companies was analysed before (2004–2007) and after (2008–2011) the systemic crisis that began in late 2007 in the United States.

From this initial sample, some companies were excluded for the following reasons: (i) insufficient data for analysis (20), (ii) negative equity during the sample period (78), and (iii) belonging to the real estate or financial sector (174); since real estate and financial companies have specific regulations, they also have differentiated capital structures in relation to the other sectors. The final sample consists of 977 publicly traded companies and 15,632 observations. Among 977 companies, 349 are MNCs and 628 are DCs. MNCs are companies whose sum from their foreign branches´ incomes is greater than 5% of the economic group´s total income (Erel *et al.*, 2020).

3.4 Methodology

In this section, the econometric models and their variables are described for each hypothesis. Moreover, there is a brief outline of the instrumental variables (IVs) related to SEM.

3.4.1 Variable descriptions

Table 3.1 describes the variables used in this study and some authors who have already used them in their studies.

Table 3.1 – Variables of econometric models

Initials	Name	Formula	Components	References
Dependent variables				
IOB	Interest-over-book	IOB = IEX/TA	IEX = Interest expenses TA = Total assets	Van Binsbergen *et al.* (2010); Clark (2015)
INTER	Nominal interest rate	INTER = IEX/TD	IEX = Interest expenses TD = Total debt (short and long terms)	Karpavičius & Yu (2017)
LDMTR	Last-dollar marginal tax rate	LDMTR = $(EBT_0 - \Delta IEX)$ $\Delta IEX = (TD_1 - TD_0)$ * RATE $TD_1 = \Delta 2\%$*TA	EBT = Earnings before taxes IEX = Interest expenses TD = Total debt (short and long terms) RATE = Interest rate TA = Total assets	Matheson (2006)
MTB	Market-to-book value	MTB = nl (MV/BV)	nl = Nepierian logarithm MV = Market value of equity BV = Book value of equity	Vo *et al.* (2021)
Instrumental variables				
AREA	Marginal benefit area	$A = \int_0^{\infty} f_{it}(x_{it})dx_{it}$	A= Integral of the x function x = Marginal benefit on an additional dollar (US$) of income	Van Binsbergen *et al.* (2010)
MTR	Marginal tax rate	$\Delta TI_{it} = \mu_{it} + \varepsilon_{it}$	ΔTI_{it} = First difference in pre-tax income of company (i) at year (t) μ_{it} = Sample ΔTI mean $\mathcal{E}_{it}$ = Random variable normally distributed with zero mean and variance equal to sample ΔTI_{it} variance	Graham (1996); Byoun (2008)
IOB	Interest-over-book	IOB = IEX/TA	IEX = Interest expenses TA = Total assets	Van Binsbergen *et al.* (2010); Clark (2015)

Table 3.1 – Variables of econometric models (continuation)

Initials	Name	Formula	Components	References
NDTS	Non-debt tax shield	NDTS = DPA/TA	DPA = Depreciation and amortization TA = Total assets	Graham (1996); Jadiyappa *et al.* (2020)
Independent variables				
MNC	Multinational company	MNC = TIR/TGR	TIR = total international revenue TGR = total global revenue	Tallman & Li (1996); Bausch & Krist (2007)
LEV	Leverage	LEV = TD/TA	TD = Total debt (short and long terms) TA = Total assets	Matheson (2006); Van Binsbergen *et al.* (2010)
CRISIS	Global financial crisis	Dummy for crisis	0 if pre-crisis (from 2004 to 2007) 1 if post-crisis (from 2008 to 2011)	Duran & Stephen (2020); Jucá & Fishlow (2021)
Companie´s control variables				
SIZE	Total asset logarithm	SIZE = nl TA	nl = Nepierian logarithm TA = Total assets	Korteweg (2010); Vo *et al.* (2021)
COL	Collateral	COL = [(INV + NFA)/TA]	INV = Inventory NFA = Net fixed assets TA = Total assets	Graham & Tucker (2006); Van Binsbergen *et al.* (2010)
MTB	Market-to-book value	MTB = nl (MV/BV)	nl = Nepierian logarithm MV = Market value of equity BV = Book value of equity	Vo *et al.* (2021)
INTA	Intangible assets	INTA = NIA/TA	NIA = Net intangible assets TA = Total assets	Van Binsbergen *et al.* (2010); Graham *et al.* (2015)
CF	Cash flow	CF = (EBD –DIV) / TA	EBD = Earnings before extraordinary items and depreciation DIV = Dividend distribution TA = Total assets	Almeida *et al.* (2004)
DPAY	Dividend payout	DPAY = DIV/TA	DIV = Dividend distribution TA = Total assets	Doidge & Dyck (2015); Chang *et al.* (2020)
ZSCORE	Altman´s Z-Score	Z = (1.2*X1) + (1.4*X2) + (3.3*X3) + (0.6*X4) + (1*X5)	X1 = Working capital/Total assets X2 = Retained earnings/Total assets X3 = EBIT/Total assets X4 = Market value of equity/Book value of demanded liabilities X5 = Sales/Total assets	Altman (1968)

Table 3.1 – Variables of econometric models (continuation)

Initials	Name	Formula	Components	References
Countries´ control variables				
INF	Inflation	INF = [(CPI_1 – CPI_0)/CPI_0]	CPI_1 = Consumer price index in year 1 CPI_0 = Consumer price index in year 0	Li & Liu (2005); Olalere *et al.* (2021)
POP	Population	POP = nl POP	nl = Nepierian logarithm POP = Population in million	Li & Liu (2005); Dharmapala & Hines Jr (2009)
KKM	Kaufmann Kraay and Mastruzzi index	KKM = It varies between -2.5 and 2.5. The higher the regulatory environment index, the better	The index is derived from the average six-dimensional estimate - control of corruption, government effectiveness, political stability and absence of violence/terrorism, regulatory quality, rule of law and voice and accountability. The estimate gives the country score, for each dimension, in units of a standard normal distribution.	Kaufmann *et al.* (2011); Jucá & Fishlow (2020)

3.5 Econometric models

Korteweg (2010) identified challenges in the regression approach for studying companies´ leverage. Profitability and size have biased effects on the optimal capital structure. Van Binsbergen *et al.* (2010) noted that estimating the marginal cost curve (MC) by using the ordinary least squares (OLS) method tends to cause endogeneity problems since leverage is determined simultaneously with the marginal costs and benefits. The OLS method cannot distinguish whether a given variation results from changes in the marginal cost curves or in the benefit curves.

To address this issue, a structural equation models (SEM) with a two-stage least squares (2SLS) linear probability model is adopted using Stata software. The instrumental variables and other variables are listed in Table 3.1. According to Bauldry (2014), interest in Stata software for SEM increased as a set of commands are introduced to adjust the models. These commands allow the use of popular estimators, such as maximum likelihood (ML), direct maximum likelihood estimator, adjusted variance for missing data, and asymptotic distribution-free estimator. SEM have been frequently used over the past 40 years to explore causal relationships between pairs of variables in social, behavioural, and economic research. As described by Tsai *et al.* (2006), researchers find more robust responses using the SEM, especially for large samples.

According to Larcker and Rusticus (2010), endogeneity is an econometric challenge in accounting studies. Researchers are aware of this problem and use IV methods to mitigate inconsistencies in parameter estimates. According to them, simultaneity is an important source of endogeneity. This arises when the dependent variable jointly determines one or more explanatory variables. If simultaneity exists, then the structural error term is, by definition, correlated with the explanatory variable. For Chenhall and Moers (2007), IV can solve this problem because it is correlated with the explanatory variable and uncorrelated with the omitted variables.

For Roberts and Whited (2013) apart the controlled experiments, there is no way to assure that endogeneity problems are eliminated to ensure adequate inferences. Properly addressing endogeneity - its causes and consequences - is based on the strength of the arguments supporting how the proposed identification strategy addresses the issue.

The identification strategy proposed at this work, as in Van Binsbergen et al. (2010), is constructed by simulating a marginal tax benefit function, for each observation of the company´s year, allowing the use of panel data. Since the benefit function varies, the cost curve must remain constant with the inclusion of supposedly comprehensive control variables, to capture the costs. Thus, depending on the company´s characteristics, the remaining variation in marginal benefits will permit the curve to change location. The area between the cost of debt and the tax benefit curves estimates the net benefit of debt financing.

Besides, industry and country fixed effects are used to control for time-varying characteristics across sector and macroeconomic types (Schmal *et al.*, 2021). This study does not directly control taxation at the investor level since investors' corporate source revenue taxation is uniform among companies in a given year and/or sector. The fixed effects of the models must capture its effects.

3.5.1 Leverage positively affects multinational companies´ bankruptcy costs

In this section, Hypothesis 1 is tested through an analysis of the influence of MNCs' leverage on bankruptcy costs during financial crisis periods and nonfinancial crisis moments. Van Binsbergen *et al.* (2010) simulated the function of the marginal tax benefit (MB) for each firm-year through the marginal tax rate (MTR), according to Graham´s model (2000). The variation in MB allows the estimation of the cost curve, which must remain fixed. Estimating the model with firm-year observations, in which a substantial rebalancing of the capital structure occurs, reduces the effect of fixed adjustment costs.

The model assumes that companies with no financial restrictions or distress make (almost) optimal debt choices. In the regression methodology using the panel data considered in this study, similar to Van Binsbergen *et al.*'s (2010) study, the AREA under the MB curve is used as the identification instrument. It measures the total potential tax benefits. To measure the changes, the first equation calculates the total potential debt tax benefit ($AREA_{it}$) for each company in each year, which is equal to the area under the MB curve (see Equation 1). Appendix 1 presents an example of variable AREA calculus.

$$AREA_{it} = \int_{0}^{\infty} f_{it}(x_{it})dx_{it} \quad (1)$$

SEM tests *H1a - Leverage positively affects MNCs´ bankruptcy costs in financial crisis moments,* and *H1b - Leverage positively affects MNCs´ bankruptcy costs in nonfinancial crisis moments.* To test H1b, the interaction variable is MNC_{it}*LEV_{it}. In Equation 2b, IV (Z) is the area. This demonstrates the effects of companies´ leverage on bankruptcy costs.

$$IOB_{it} = \beta_0 + \beta_1(MNC_{it}*LEV_{it}*CRISIS_{kt}) + \beta_2 \text{Company controls}_{it} + \beta_3 \text{Country controls}_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (2a)$$

$$MNC_{it}*LEV_{it}*CRISIS_{kt} = \beta_0 + \beta_1 Z_{it} + \beta_2 \text{Independent variables} + \beta_3 \text{Company controls}_{it} + \beta_4 \text{Country controls}_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (2b)$$

Where, dependent variable = IOB; independent variables = MNC, LEV, and CRISIS; company controls = SIZE, COL, MTB, INTA, CF, DPAY, and ZSCORE; Country controls = INF, POP, and KKM; Z = instrumental variable AREA; i = company; j = industry; k = country; t = year; λ_{jt} = industry-year effect; θ_{kt} = country-year effect; γ_{it} = company´s specific effects; and μ_{ijkt} = residuals.

3.5.2 Leverage negatively affects multinational companies´ interest rates

In this section, research Hypothesis 2, which analyses the influence of MNCs' leverage on their interest rates during financial crisis periods and nonfinancial crisis moments, is tested. According to trade-off theory, the net tax savings per dollar of corporate debt are proportional to nominal interest rates. Therefore, everything else being equal, the size of the tax incentives affecting the use of corporate debt should be high when nominal interest rates are high and vice versa (Gordon & Lee, 2007). However, the authors highlight

an interaction between taxes and interest rates; that is, it is imperative to study the effects of both factors in decision-making regarding companies' leverage.

MacKinlay (2015) argued that corporate leverage is related to the cost of debt. Due to the tax deductibility of interest, the incentive to operate with greater leverage is a popular political issue. Some legislators advocate lowering the corporate tax rate to induce companies to incur less debt. They further argue that lower debt levels reduce bankruptcy risk and aggregate risk in the economy. For the author, the impact of reduced taxes does not change companies´ general propensity for leverage. Instead, the companies migrate from public to private debt, with this change occurring more frequently in market environments whose cost of debt is higher.

The corporate marginal tax rate (MTR) is an important instrument for obtaining the cost of capital and consequent budget decision-making (Graham & Mills, 2008). However, academic research on MTR calculations is rare because of the complexity of interpreting tax codes (Graham, 1996). MTR is an IV that explores the relationship between taxation and corporate leverage. The MTR is defined as the tax rate paid on an additional $1.00 dollar of income.

The calculation involves two essential characteristics: 1) considering net operating losses (NOLS), investment tax credits (ITCs), and alternative minimum tax (AMT), according to the federal tax code, and 2) taking into consideration managers´ debt policy about their tax rate expectation (Graham, 1996). Applying the carry-forward loss rule when calculating the MTR requires a forecast of future revenue based on the manager's expectation. The model proposed by Shevlin (1987) may be used to generate a proxy for this expectation. It is based on the assumption that income follows a pseudorandom path before taxes (see Equation 3).

$$\Delta TI_{it} = \mu_{it} + \varepsilon_{it} \quad (3)$$

Where, ΔTI_{it} = first difference in pre-tax income of company (i) in year (t); μ_{it} is the sample mean of ΔTI, and ε_{it} is a normally distributed random variable with mean zero and variance equal to variance of $\Delta TIit$ to the number of years being considered.

SEM tests *H2a - Leverage negatively affects MNCs´ interest rates in financial crisis moments* and *H2b - Leverage negatively affects MNCs´ interest rates in nonfinancial crisis moments*. To test H2b, the interaction variable is MNC_{it}*LEV_{it}. In Equation 4b, IV (Z) is the marginal tax rate. This demonstrates the effects of companies´ leverage on their interest rates.

$$INTER_{it} = \beta_0 + \beta_1(MNC_{it}*LEV_{it}*CRISIS_{kt}) + \beta_2 Company\ controls_{it} + \beta_3 Country\ controls_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (4a)$$

$$MNC_{it}*LEV_{it}*CRISIS_{kt} = \beta_0 + \beta_1 Z_{it} + \beta_2 Independent\ variables + \beta_3 Company\ controls_{it} + \beta_4 Country\ controls_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (4b)$$

Where, dependent variable = INTER; independent variables = MNC, LEV, and CRISIS; company controls = SIZE, COL, MTB, INTA, CF, DPAY, and ZSCORE; Country controls = INF, POP, and KKM; Z = instrumental variable MTR; i = company; j = industry; k = country; t = year; λ_{jt} = industry-year effect; θ_{kt} = country-year effect; γ_{it} = company´s specific effects; μ_{ijkt} = residuals.

3.5.3 Leverage negatively affects multinational companies´ tax benefit

This section, Hypothesis 3, investigates the influence of MNCs' leverage on tax benefits during the financial crisis period and in nonfinancial crisis moments. Matheson (2006) used the last dollar marginal tax rate (LDMTR) to study corporate debt levels. MTR proxies derived from financial statements appear to provide relatively reliable estimates of the MTR that firms face in the current year. The coefficients of these variables only slightly underestimate the effects of taxation (Plesko, 2003). The LDMTR is calculated as the ratio between the company´s tax liability change, resulting from an increase in its deductions by 2% of total assets and that amount (Matheson, 2006). The instrument must be correlated with the cost benefit curve (leverage) and not to fiscal benefits (taxes). Van Binsbergen *et al.* (2010) reported that interest overbook, calculated as the ratio of interest expense to total assets (IOB), is exogenous for taxes.

The SEM tests *H3a - Leverage negatively affects MNCs´ tax benefit in financial crisis moments* and *H3b - Leverage negatively affects MNCs´ tax benefit in nonfinancial crisis moments.* To test H3b, the interaction variable is $MNC_{it}*LEV_{it}$. In Equation 5b, IV (Z) is the interest-over-book. This demonstrates the effects of companies´ leverage on their tax benefits.

$$LDMTR_{it} = \beta_0 + \beta_1(MNC_{it}*LEV_{it}*CRISIS_{kt}) + \beta_2 Company\ controls_{it} + \beta_3 Country\ controls_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (5a)$$

$$MNC_{it}*LEV_{it}*CRISIS_{kt} = \beta_0 + \beta_1 Z_{it} + \beta_2 Independent\ variables + \beta_3 Company\ controls_{it} + \beta_4 Country\ controls_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (5b)$$

Where, dependent variable= LDMTR; independent variables = MNC, LEV, and CRISIS; company controls = SIZE, COL, MTB, INTA, CF, DPAY and ZSCORE; Country controls = INF, POP, and KKM; Z = instrumental variable IOB; i = company; j = industry; k = country; t = year; λ_{jt} = industry-year effect; θ_{kt} = country-year effect; γ_{it} = company´s specific effects; μ_{ijkt} = residuals.

3.5.4. Leverage positively affects multinational companies´ market value

Hypothesis 4 analyses the influence of MNCs' leverage on market value during financial crisis and nonfinancial crisis periods. Jadiyappa *et al.* (2020) addressed the bias of the variable omitted in the coefficient estimate and the potential for endogeneity problems. In this study, the two-stage least squares (2SLS) estimation of IV is used. The main problem with this approach is finding exogenous IVs that are not directly related to the firm's value. The authors use the non-debt tax shield (NDTS) as the depreciation and amortization ratio to total assets. NDTS is applied as an instrumental variable since it is correlated with the cost benefit curve (leverage) and is not related to tax benefits (market-to-book).

SEM tests *H4a – Leverage positively affects MNCs´ market value at financial crisis moments* and *H4b – Leverage positively affects MNCs´ market value at nonfinancial crisis moments.* To test H4b, the interaction variable is MNC_{it}*LEV_{it}. In Equation 6b, IV (Z) is the non-debt tax shield. This demonstrates the effects of companies´ leverage on their market value.

$$MTB_{it} = \beta_0 + \beta_1(MNC_{it}*LEV_{it}*CRISIS_{kt}) + \beta_2 Company\ controls_{it} + \beta_3 Country\ controls_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (6a)$$

$$MNC_{it}*LEV_{it}*CRISIS_{kt} = \beta_0 + \beta_1 Z_{it} + \beta_2 Independent\ variables + \beta_3 Company\ controls_{it} + \beta_4 Country\ controls_{kt} + \lambda_{jt} + \theta_{kt} + \gamma_{it} + \mu_{ijkt} \quad (6b)$$

Where, dependent variable = MTB; independent variables = MNC, LEV, and CRISIS; company controls = SIZE, COL, INTA, CF, DPAY and ZSCORE; Country controls = INF, POP, and KKM; Z = instrumental variable NDTS; i= company; j = industry; k = country; t = year; λ_{jt} = industry-year effect; θ_{kt} = country-year effect; γ_{it}= company´s specific effects; μ_{ijkt} = residuals.

3.6 Analysis of results

The verification of each hypothesis is done by its descriptive statistics and econometric outputs.

3.6.1. Descriptive statistics

Table 3.2 describes the main characteristics of the models' dependent, instrumental, independent, and control variables. A proxy for bankruptcy costs (IOB) reports a mean of 0.02, in line with the results obtained by Clark (2015) for a sample with American companies. Companies had a leverage (LEV) mean of 0.25, the same number found by Vo *et al.* (2021). They considered a sample of 37,190 companies listed in 81 countries. The marginal tax rate (MTR) variable projects future taxable income. Companies use it as a tool for debt decisions. MTR presents a mean of 0.23 and a median of 0.25. Chen *et al.* (2021) found in American companies - before the tax reform (1988 to 2016) - 0.30 and 0.32, respectively.

Moreover, inflation (INF) presents a negative minimum value as a result of the deflation in China in 2009 (-0.728%). The Altman´s Z-Score (ZSCORE) has an average of 3.57. This means that companies are in the safety-risk bankruptcy probability area (Z > 2.99). Concerning intangibles, companies present low levels of it (INTA = 0.06), if compared to their collateral assets (COL = 0.48). In turn, market to book (MTB) shows a minimum negative (-2.00). Since the variable is calculated via nepierian logarithm, the ratios between 0 and 1 show negative values. As for the leverage (LEV), 415 firms-year do not disclose debt values in their financial statements, which explains the minimum of zero (0.00) for the variable. Normality is verified through the skewness and kurtosis coefficients of the Jargue-Bera test.

Table 3.2 – Descriptive statistics

This table presents the descriptive statistics for the dependent, independent, and control variables: IOB is the interest overbook, calculated as the ratio of interest expenses to total assets. INTER is the real interest rate, calculated as the ratio of interest expenses to total debt. LDMTR is the last dollar marginal tax rate calculated as the ratio between the change in a company´s tax liability resulting from the deleveraging company and an increase in its debt by 2% of total assets and that amount. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. MTR is the marginal tax rate, which is the ratio between the change in a company's tax liability and the change in its net profit as a result of the change in its deductions. NDTS is the nondebt tax shield calculated as the ratio of depreciation and amortization to total assets. CRISIS is the global financial crisis calculated through a proxy, where 0 = if pre-crisis and 1 = post-crisis. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. Zscore is the Z-score obtained by Altman. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices.

Variables	Observations	Mean	Median	Standard Deviation	Min.	Max.
IOB	15,209	0.02	0.01	0.02	0.00	0.14
INTER	15,209	0.07	0.06	0.05	0.00	0.36
LDMTR	15,632	0.18	0.25	0.14	0.00	0.51
LEV	15,626	0.25	0.24	0.18	0.00	0.74
AREA	15,632	2.84	3.23	1.93	0.00	6.63
MTR	15,632	0.23	0.25	0.11	0.00	0.51
NDTS	14,812	0.03	0.03	0.02	0.00	0.18
CRISIS	15,632	0.75	1.00	0.43	0.00	1.00
MNC	14,976	0.11	0.00	0.21	0.00	0.99
SIZE	15,632	6.36	6.27	1.58	0.11	12.88
COL	15,632	0.48	0.48	0.20	0.00	0.91
MTB	15,632	0.76	0.72	0.87	-2.00	3.63
INTA	15,632	0.06	0.03	0.09	0.00	0.63
CF	15,632	0.05	0.04	0.07	-0.29	0.30
DPAY	15,632	0.02	0.17	0.02	0.00	0.19
ZSCORE	15,624	3.55	2.49	3.28	0.02	18.81
INF	15,632	0.03	0.03	0.02	-0.01	0.13
POP	15,632	13.74	14.12	0.80	10.76	14.18

3.6.2 Strutural equation model (SEM) and ordinary least squares (OLS) results

Tables 3.3 to 3.10 present the SEM and OLS results, respectively. SEM is verified by the likelihood ratio (LR) through two tests: i) the first is a repetition of the model χ 2 test, reported at the bottom of the SEM output. The saturated model perfectly fits the covariances. For the null hypothesis, the equation variance is zero (p value > 0.0001); ii) the second hypothesis compares the baseline with the saturated model. The baseline model includes the mean and variances of all observed variables and the covariances of all observed exogenous variables. The null hypothesis—the baseline model fitting as well as the saturated model—is rejected at the 5% significance level (Kline, 2015). The OLS assumption tests applied are Breusch–Pagan, Cameron-Trivedi, Shapiro–Francia, Reset, Variance Inflation Factor (VIF), and White.

The analyses are performed by considering the SEM. This model addresses the simultaneity existing in capital structure decisions and taxes through IVs (Larcker & Rusticus, 2010). However, it is important to highlight that the OLS results are consistent with the SEM results. Moreover, Tables 3.3 to 3.10 report only the first-stage instruments' SEM results (Equations 2a, 4a, 5a, and 6a). This means that the results of the second equations (2b, 4b, 5b, and 6b), which include the instrumental variables, are not reported.

Additionally, a robustness test is performed using the Stata command of three-stage estimation for systems of simultaneous equations, and as OLS with fixed effects, namely,

industry-year, country-year, and company. The results of the simultaneous equations - obtained by Stata´s "reg3"- are also not reported, as they are surpassed in statistical significance by the SEM ones.

3.6.3 Leverage positively affects MNCs´ bankruptcy costs in financial crisi moments (H1a)

Table 3.3 presents the SEM and OLS results for H1a. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC × LEV × CRISIS) was statistically significant. It has a positive relationship with bankruptcy costs (0.085). This means that for every 1% increase in multinational companies´ leverage during the crisis, there is an increase of 0.085% in the company's bankruptcy costs. These results corroborate the H1a hypothesis *that leverage positively affects MNCs´ bankruptcy costs in financial crisis moments* and agree with other empirical studies (Kuo & Wang, 2005; Akhtar & Oliver, 2009). For the control variables, the negative coefficient of size (-0.002) and positive coefficient of dividend payment (0.030) stand out. In the case of size, this fact indicates that larger companies have lower bankruptcy costs, confirming the bankruptcy cost theory (Titman & Wessels, 1988). According to Brav *et al.* (2005), companies distributing dividends have higher debt costs, as they direct fewer resources to pay interest.

Table 3.3 – Bankruptcy costs and leverage in financial crisis moments

The dependent variable IOB is interest overbook, calculated as the ratio of interest expenses to total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. The independent variable (MNC*LEV*CRISIS) is the interaction variable among the MNC, LEV, and CRISIS. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. CRISIS is the global financial crisis calculated by a proxy, where 0 = pre-crisis and 1 = post-crisis. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. Zscore is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

Interest-over-book (IOB)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV*CRISIS	0.085***	8.29	0.014***	3.42
SIZE	-0.002***	-10.43	-0.002***	-13.96
COL	0.014***	11.56	0.008***	7.53
MTB	-0.000	-0.40	0.003***	8.21
INTA	0.027***	9.21	0.017***	4.75
CF	-0.015***	-4.45	0.014***	3.48
DPAY	0.030***	2.97	0.064***	5.73
INF	-0.002	-0.23	-0.028***	-3.12
POP	-0.003***	-6.42	-0.012***	-14.65
KKM	0.015***	8.12	-0.015***	-5.76
ZSCORE	-	-	-0.002***	-18.23
Cov(e.1,e.2)	-0.000***	-7.86	-	-
# Obs	7,235		7,235	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.2249	

Notes: significance levels of 1% (***) and 5% (**).

3.6.4 Leverage positively affects MNCs´ bankruptcy costs in moments of nonfinancial crises (H1b)

Table 3.4 presents the SEM and OLS results for the H1b hypothesis. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC*LEV) is statistically significant, showing a positive relationship with bankruptcy costs (0.049), with a 0.049% increase in the company's bankruptcy costs for every 1% increase in the multinational leverage. These results corroborate hypothesis H1b: *Leverage positively affects MNCs´ bankruptcy costs in nonfinancial crisis moments*, confirming the results found during the crisis period (H1a) and confirming the theory of bankruptcy costs.

Among the control variables, the negative coefficient for cash flow (-0.029) and the positive coefficient for intangibles (0.019) are highlighted. Dunning (1977) stated, through the theory of ownership, location, and internalization (OLI), that multinational companies have more intangible assets, such as technology, patents, and recognized brands. According to Lemmon *et al.* (2008), they allow these companies to borrow less, making less use of their tax benefits.

Table 3.4 – Bankruptcy costs and leverage in nonfinancial crisis moments

The dependent variable IOB is interest overbook, calculated as the ratio of interest expenses to total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. The independent variable (MNC*LEV) is the interaction variable between the MNC and LEV. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets.CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. ZSCORE is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equations error terms.

Interest-over-book (IOB)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV	0.049***	20.27	0.021***	9.73
SIZE	-0.001***	-6.80	-0.002***	-14.91
COL	0.013***	19.63	0.008***	11.84
MTB	0.000***	2.91	0.003***	13.42
INTA	0.019***	13.25	0.008***	5.31
CF	-0.029***	-13.29	0.001	0.44
DPAY	0.050***	8.11	0.067***	9.51
INF	-0.000	-0.02	-0.014**	-1.97
POP	-0.005***	-15.51	-0.008***	-22.30
KKM	0.001	1.25	-0.006***	-5.82
ZSCORE	-	-	-0.002***	-32.48
Cov(e.1,e.2)	-0.000***	-14.60	-	-
# Obs	14,590		14,590	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.2507	

Notes: significance levels of 1% (***) and 5% (**).

3.6.5 Leverage negatively affects MNCs´ interest rates in financial crisis moments (H2a)

Table 3.5 presents the SEM and OLS results for the H2a hypothesis. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC × LEV × CRISIS) was statistically significant. It has a negative relationship with interest paid (-0.233). This means that, for every 1% variation in the MNC's leverage during the crisis, there is a 0.233% reduction in the total interest paid by the company, arising from its leverage. This negative relationship results from the ability of the parent and subsidiaries to offer better interest conditions in intercompany transactions (Chowdhry & Nanda, 1994; Desai *et al.*, 2005). These results confirm the H2a hypothesis –

leverage negatively affects MNCs´ interest rates in financial crisis moments, in line with similar studies (Collin-Dufresne & Goldstein, 2001; Bartholdy & Mateus, 2011; Daskalakis *et al.*, 2017).

For the control variables, the positive cash flow coefficient (0.050) stands out. According to agency theory, companies with higher cash flow levels are more leveraged. Leverage exerts pressure on cash flow, minimizing the possibility that managers make investments with negative discounted cash flow (Jensen & Meckling, 1976).

Table 3.5 – Interest rates and leverage in financial crisis moments

The dependent variable INTER is the real interest rate, calculated as the ratio of interest expenses to total debt. The independent variable (MNC*LEV*CRISIS) is the interaction variable among the MNC, LEV, and CRISIS. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. CRISIS is the global financial crisis calculated by a proxy, where 0 = pre-crisis and 1 = post-crisis. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. ZSCORE is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

Nominal interest rate (INTER)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV*CRISIS	-0.233***	-11.90	-0.064***	-4.55
SIZE	-0.008***	-15.43	-0.009***	-16.39
COL	-0.014***	-3.79	-0.009**	-2.43
MTB	-0.001	-1.34	-0.001	-1.19
INTA	-0.018**	-2.01	-0.011	-1.02
CF	0.050***	4.76	0.056***	4.18
DPAY	0.118***	3.86	0.169***	4.14
INF	-0.072**	-2.35	-0.134***	-4.05
POP	-0.011***	-6.71	-0.034***	-12.46
KKM	0.086***	15.83	-0.002	-0.25
ZSCORE	-	-	0.001**	2.34
Cov(e.1,e.2)	0.000***	9.50	-	-
# Obs	7,235		7,235	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.2179	

Notes: significance levels of 1% (***) and 5% (**).

3.6.6 Leverage negatively affects MNCs´ interest rates in nonfinancial crisis moments (H2b)

Table 3.6 presents the SEM and OLS results for the H2b hypothesis. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC × LEV) is statistically significant. It has a negative relationship with interest expenses (-0.104). Thus, for every 1% increase in the MNC´s leverage, there is a 0.104% reduction in its interest expenses. These results confirm the H2b hypothesis: *leverage negatively affects MNCs´ interest rates in nonfinancial crisis moments*, similar to those obtained in the crisis period (H2a).

The negative coefficients of size (-0.007) and collateral (-0.013) stand out among the control variables. It appears for every 1% increase in these variables, there is a reduction of 0.007% and 0.013% in the total interest paid by companies. Larger companies and those with more assets that can be pledged as collateral have lower credit risk, making contracting debt feasible at lower interest rates (Agustini, 2016).

Table 3.6 – Interest rates and leverage in nonfinancial crisis moments

The dependent variable INTER is the real interest rate, calculated as the ratio of interest expenses to total debt. The independent variable (MNC*LEV) is the interaction variable between the MNC and LEV. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. ZSCORE is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

Nominal interest rate (INTER)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV	-0.104***	-14.20	-0.059***	-9.39
SIZE	-0.006***	-21.01	-0.007***	-20.96
COL	-0.014***	-6.23	-0.009***	-3.59
MTB	-0.001	1.43	-0.002***	-3.42
INTA	-0.010**	-2.21	-0.020***	-3.75
CF	0.030***	4.39	0.041***	4.50
DPAY	0.147***	7.08	0.171***	5.98
INF	-0.009	-0.39	-0.055**	-2.10
POP	-0.013***	-13.09	-0.024***	-17.81
KKM	0.042***	11.68	0.011***	2.62
ZSCORE	-	-	0.001***	3.27
Cov(e.1,e.2)	0.000***	9.66	-	-
# Obs	14,590		14,590	
# Firms	977		977	
LR test	0.0000		-	

Table 3.6 – Interest rates and leverage in nonfinancial crisis moments (continuation)

Nominal interest rate (INTER)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.1782	

Notes: significance levels of 1% (***) and 5% (**).

3.6.7 Leverage negatively affects MNCs´ tax benefit in financial crisis moments (H3a)

Table 3.7 presents the SEM and OLS results obtained for H3a. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC × LEV × CRISIS) was statistically significant. It has a negative relationship with tax benefits (-0.923). This means that for every 1% increase in the MNC's leverage during the crisis, there is a 0.923% reduction in the company's tax benefits. Becker and Riedel (2012) suggest that the parent-subsidiary relationship in the capital stock generates a negative fiscal externality in corporate taxation. Such results confirm the H3a hypothesis that *leverage negatively affects MNCs´ tax benefits in financial crisis moments*, in agreement with the results of similar studies (Gupta & Newberry, 1997; Richardson & Lanis, 2007).

For the control variables, the negative coefficients of collaterals (-0.093) and intangibles (-0.233) stand out. Note that for every 1% increase in these variables, there are tax benefit reductions of 0.093% and 0.233%, respectively. The tax sensitivity of the capital structure is most impacted by mitigating factors unrelated to debt depreciation, amortization, and carried losses (Buettner *et al.*, 2011).

Table 3.7 – Tax benefit and leverage in financial crisis moments

The dependent variable LDMTR is the last dollar marginal tax rate, calculated as the ratio between the change in a company´s tax liability resulting from the deleveraging company and an increase in its debt by 2% of total assets to that amount. The independent variable MNC*LEV*CRISIS is the interaction variable among the MNC, LEV, and CRISIS. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. CRISIS is the global financial crisis calculated by proxy, where 0 = pre-crisis and 1 = post-crisis. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and

Mastruzzi indices. Zscore is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

Last-dollar marginal tax rate (LDMTR)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV*CRISIS	-0.923***	-17.21	-0.271***	-10.09
SIZE	0.007***	5.53	0.007***	5.92
COL	-0.093***	-10.40	-0.073***	-8.41
MTB	-0.010***	-4.73	-0.022***	-10.54
INTA	-0.233***	-10.68	-0.182***	-8.80
CF	0.860***	33.18	0.763***	31.06
DPAY	0.724***	9.60	0.613***	9.48
INF	-0.003	-0.04	0.140**	2.02
POP	0.043***	11.01	0.064***	13.69
KKM	0.124***	9.22	0.180***	11.87
ZSCORE	-	-	0.007***	11.74
Cov(e.1,e.2)	0.002***	14.64	-	-
# Obs	7,238		7,478	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.2356	

Notes: significance levels of 1% (***) and 5% (**).

3.6.8 Leverage negatively affects MNCs´ tax benefit in moments of nonfinancial crisis (H3b)

Table 3.8 presents the SEM and OLS results for the H3b hypothesis. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC*LEV) is statistically significant and shows a negative relationship with the tax benefit (-0.203). It appears that there is a 0.203% reduction in its tax benefit for every 1% increase in the MNC's leverage. These results corroborate the H3b hypothesis that *leverage negatively affects MNCs´ tax benefit in nonfinancial crisis moments*, similar to the results found in the crisis period (H3a).

For the control variables, the positive coefficients of cash flow (0.921) and dividend distribution (0.850) stand out. The result of the free cash flow is expected due to taxation on the company's profit. Desai *et al.* (2007) claim that repatriations help headquarters meet their financing needs. There is a positive relationship between dividends distributed to external shareholders and their internal repatriation. Shared ownership of foreign subsidiaries contributes to the routine dividend repatriation policy.

Table 3.8 – Tax benefit and leverage in nonfinancial crisis moments

The dependent variable LDMTR is the last dollar marginal tax rate, calculated as the ratio between the change in a company´s tax liability resulting from the deleveraging company and an increase in its debt by 2% of total

assets to that amount. The independent variable (MNC*LEV) is the interaction variable between the MNC and LEV. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. ZSCORE is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

Last-dollar marginal tax rate (LDMTR)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV	-0.203***	-11.33	-0.225***	-13.78
SIZE	0.001	1.03	0.003***	3.77
COL	-0.114***	-20.81	-0.085***	-15.30
MTB	-0.007***	-5.41	-0.016***	-11.93
INTA	-0.089**	-7.76	-0.075***	-6.87
CF	0.921***	55.18	0.821***	49.85
DPAY	0.850***	16.97	0.773***	16.51
INF	0.122**	2.12	0.066	1.13
POP	0.015***	6.21	0.006**	2.39
KKM	0.001	0.01	-0.016*	-1.81
ZSCORE	-	-	0.006***	15.90
Cov(e.1,e.2)	0.000***	4.65	-	-
# Obs	14,596		14,962	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.2513	

Notes: significance levels of 1% (***), 5% (**), and 10% (*).

3.6.9 Leverage positively affects MNCs´market value in financial crisis moments (H4a)

Table 3.9 presents the SEM and OLS results obtained for H4a. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent variable (MNC × LEV × CRISIS) was statistically significant. It has a positive relationship with the company's value (2.124). This means for every 1% increase in MNCs´ leverage during the crisis, there is a 2.124 % improve in market value. The positive effects of MNCs' leverage on the company's value during the crisis period result from a greater possibility of diversification and leverage (Chkir & Cosset, 2001; Yung *et al.*, 2015). These results confirm the H4a hypothesis: *leverage positively affects MNCs´ market value in financial crisis moments,* in line with other empirical studies (Graham, 2000; Van Binsbergen *et al.*, 2010; Huynh *et al.*, 2020).

Concerning the control variables, the positive coefficients for dividend payouts (3.291) and inflation (6.051) stand out. Kim *et al.* (2021) found a positive relationship between dividend payouts and company value. Multinational companies increase their leverage in scenarios of high inflation – a more frequent scenario in emerging countries. For Olalere *et al.* (2021), the inflation rate also positively and significantly impacts the company's value.

Table 3.9 – Market value and leverage in financial crisis moments

The dependent variable MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. The independent variable (MNC*LEV*CRISIS) is the interaction variable among the MNC, LEV, and CRISIS. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. CRISIS is the global financial crisis calculated by proxy, where 0 = pre-crisis and 1 = post-crisis. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. ZSCORE is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

Market-to-book (MTB)				
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV*CRISIS	2.124***	8.17	0.461**	2.39
SIZE	-0.047***	-7.04	-0.038***	-5.24
COL	-0.024	-0.48	-0.033	-0.63
INTA	0.766***	6.33	0.703***	6.15
CF	0.054	0.36	0.995	0.57
DPAY	3.291***	8.14	3.759***	9.83
INF	6.051***	14.70	6.774***	15.91
POP	0.587***	29.69	0.685***	20.96
KKM	1.066***	15.07	1.473***	12.82
ZSCORE	0.104***	31.68	0.099***	25.66
Cov(e.1,e.2)	-0.005**	-8.22	-	-
# Obs	6,720		7,478	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.3562	

Notes: significance levels of 1% (***) and 5% (**).

3.6.10 Leverage positively affects MNCs´ market value in moments of nonfinancial crises (H4b)

Table 3.10 presents the SEM and OLS results for H4b. Both models are valid according to the likelihood ratio (0.000) and probability F (0.000) tests. The independent

variable (MNC × LEV) is statistically significant. It has a positive relationship with the company's value (0.657). For every 1% increase in the MNC's leverage, there is a 0.657% increase in the company's value. These results confirm the H4b hypothesis: *Leverage positively affects MNCs´ market value at nonfinancial crisis moments,* similar to the results obtained for the crisis period (H4a).

Among the control variables, the positive coefficients of two macroeconomic variables stand out: population size (0.524) and regulatory environment (0.597), the latter being measured by the Kaufmann Kraay and Mastruzzi index (KKM). In the case of population size, a larger local consumer market improves its results, impacting its value (Fang *et al.*, 2018). According to Cumming *et al.* (2017), countries with better business environments generate more value for companies.

Table 3.10 – Market value and leverage in nonfinancial crisis moments

MTB is the Neperian logarithm of the market-to-book value, calculated as the ratio of the market value of equity to the book value of equity. The independent variable (MNC*LEV) is the interaction variable between the MNC and LEV. The MNC is the multinational company index, calculated as the ratio of total international revenue to total global revenue. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. SIZE is calculated as the Neperian logarithm of the total assets. COL is the collateral calculated as the ratio of inventory and net fixed assets to total assets. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. CF is the cash flow, calculated as the ratio of earnings before extraordinary items and depreciation minus the dividend distribution to total assets. DPAY is the dividend payout, calculated as the ratio of dividend distribution to total assets. INF is the inflation calculated as the ratio of the consumer price index in year 1 minus the consumer price index in year 0 to the consumer price index in year 0. The POP is the Neperian logarithm of the population. KKM is the Kaufmann–Kraay and Mastruzzi indices. ZSCORE is the Z-score obtained by Altman. Cov (e.1,e.2) is the covariance between the two equation error terms.

	Market-to-book (MTB)			
Variables	**Structural Equation Model**		**Ordinary Least Squares**	
	Coef	**Z**	**Coef**	**T**
MNC*LEV	0.657***	5.40	0.633***	6.32
SIZE	-0.161***	-34.74	-0.088***	-18.06
COL	-0.418***	-11.81	0.040	1.09
INTA	0.552***	7.48	0.769***	10.48
CF	1.212***	11.33	-0.407***	-3.21
DPAY	7.417***	24.01	5.411***	18.50
INF	2.001***	5.37	2.306***	6.42
POP	0.524***	34.49	0.375***	21.79
KKM	0.597***	10.78	0.327***	5.66
ZSCORE	-	-	0.103***	44.28
Cov(e.1,e.2)	-0.031	-0.97	-	-
# Obs	14,970		14,962	
# Firms	977		977	
LR test	0.0000		-	
Industry-year effect	-		Yes	
Country-year effect	-		Yes	
Prob > F	-		0.0000	
R^2	-		0.3166	

Notes: significance level of 1% (***).

3.7 Conclusion

This study contributes to investigating the impact of the leverage level of MNCs and DCs on bankruptcy costs, interest rates, tax benefits, and market value in periods of crisis and financial stability. This analysis occurs on companies whose parents are located in emerging countries that are among OECD's key partners. The sample is composed of 977 publicly traded companies, excluding those developing financial and real estate activities. The hypotheses are tested through structural equations, including more efficient tax proxies for capturing short- and long-term decisions. The analysed period considers pre- (2004–2007) and postcrisis (2008–2011), as well as a long-run period (2004–2019).

For 2004 to 2019 period, the results indicate an increase in the leverage level of MNCs (0.59%), in contrast to the reduction presented by DCs (-0.04%). It is worth noting that this distinction takes place when central banks apply countercyclical policies - reduction of interest rates, an incentive to credit offer, and reduction of official tax rates. Notwithstanding this fact, it appears that DCs have higher interest rates (7.20%). On the other hand, MNCs have greater financing capacity through transfers among affiliates, which reduces their debt costs (6.51%).

According to the World Bank (2021a), between 2004 and 2019, the deposit interest rate shows a negative variation of -35.82% for the countries in the sample. This reduction is also seen in the companies´ nominal interest rates; however, to a lesser extent, MNCs (-4.23%) and DCs (-2.47%). Another impact associated with the fall in interest rates is the retraction of the bankruptcy costs of MNCs (-0.13%) and DCs (-0.09%), even though countries point to their increase, ranging from 7.04% (1994) to 8.46% (2019). In macroeconomic terms, the official tax rate decreased from 34.74% (2004) to 27.80% (2019) (World Bank, 2021a). This fact is reflected in the reductions in the tax benefits of MNCs (-5.53%) and DCs (-2.87%).

The results of the econometric model tests indicate that the MNC´s leverage increases their bankruptcy costs, both during the crisis (H1a) and in the financial stability period (H1b). The existence of a negative relationship between MNCs´ leverage and nominal interest rates in times of crisis (H2a) or not (H2b), as well as in relation to the tax benefit, in both scenarios (H3a and H3b) is found. Finally, an MNC's higher leverage increases its market value in crisis scenarios (H4a) or not (H4b).

These results indicate that MNCs take advantage of the liquidity excesses and reductions in their interest rates, promoted by governments´ countercyclical policies, and

increase their financing, even in scenarios where bankruptcy costs rise. This suggests that MNCs have more favourable characteristics for the absorption of these stimuli. Furthermore, they create value during this period, starting a virtuous cycle. This process continues until 2014, when more modest economic growth is observed; the annual variation in the gross domestic product (GDP) was reduced from 4.44% (2014) to 3.29% (2019) (World Bank, 2021a).

Another contribution of this study is tax proxies that are rarely used in the literature because of the complexity of their procurement. The marginal benefit area (AREA) variable projects how much debt the company can contract to reach its optimal leverage point, consistent with trade-off theory. The marginal tax rate (MTR) variable projects future taxable income, with which the company makes debt decisions, as well as its respective tax benefit. The last-dollar marginal tax rate (LDMTR) variable estimates the corporate tax rate, considering debt contracting. The non-debt tax shields (NDTS) variable considers tax benefits not arising from leverage, depreciation, and amortization. Additionally, this study analyses leverage relationships while considering both micro-and macroeconomic variables.

A limitation of this research is the collection of data from only six years (1998–2003) to calculate the MTR. The emerging countries in the sample are in the process of maturing their economies, restricting the availability of long-term historical series. Finally, for future research, analyses of different financial position aspects that may influence MNCs´ tax benefits, such as transfer prices, thin capitalization rules, and production verticalization, are suggested.

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4. TAX HAVENS AND TRANSFER PRICING: TAX EFFECTS FOR MULTINATIONAL COMPANIES OF EMERGING COUNTRIES

Abstract

This study analyses specific multinational companies' characteristics that cause them to establish in tax havens. Taxation has always been a problem both for governments and companies. Globalization and last crises led to an increase in the asset intangibility level, allowing its greater fluidity among the companies' subsidiaries. Such a fact facilitates the transfer of results to jurisdictions with a lower tax burden. This reality is challenging for government authorities, as it offers room for corporate strategies of tax elision and tax evasion. The sample, for the period from 2010 to 2020, is composed of companies headquartered in emerging countries: Brazil, China, India, Indonesia, and South Africa. Among the contributions of this study, the use of the unconventional tax proxy - AREA - stands out, which controls the ground for taking advantage of the tax benefit from short-and long-term debts. In addition, this research innovates by considering in its models, the transfer pricing intensity variable - obtained through manually collected data. The result is that multinational companies make use of different instruments – even combined – with the objective of reducing the conglomerate´s total taxation. Therefore, despite the efforts of governments and international organizations – proposing the adoption of a minimum tax rate among countries, these companies have been efficient in reducing or even in not paying taxes.

Keywords: Tax haven, transfer pricing, taxation, intangible assets, emerging markets.

4.1 Introduction

At present, developed countries, led by US President Joe Biden, defend the need to establish a minimum global tax rate. Large worldwide corporations, such as Amazon, Google and Starbucks, are accused of tax evasion through the transfer of intangible assets to countries with low taxation, imperiling government collection (Jones & Temouri, 2016). The Organization for Economic Co-operation and Development (OECD), in turn, is seeking an agreement ensure that foreign subsidiaries of multinational companies (MNCs) pay taxes in the places where they operate and not only in the country of their headquarters. Additionally, governments and international entities propose the adoption of a statutory minimum rate among countries (Poux, 2021). Such practices, enhanced by globalization and economy dematerialization, grow and become a challenge to supranational organizations, tax authorities and markets (Merle *et al.*, 2019).

MNCs adopt different techniques to reduce their tax base. Among them, manipulation the price of commercial transfers and the registering intangible assets such as patents and trademarks in countries with reduced taxation stand out (Beer *et al.* 2020). Transfer price manipulation occurs when a company negotiates with its affiliates and values of goods and services below or above market values (Brock & Pogge, 2014). Thus, profits are artificially deflated (inflated) in jurisdictions with high (low) tax rates. This strategy offers MNCs the opportunity to reallocate their profits to countries with lower tax rate.

The OECD estimates that these intra-group flows represent more than 70% of the total world trade (Merle et al., 2019). Because intangible assets have unique values that are difficult to assess, they become instruments for profit shifting. This occurs through the payment of royalties and licenses by the subsidiary, which operates in a country with high tax rates, to another located in a country with lower tax rates (Jones & Temouri, 2016). According to Thum-Thysen et al. (2019), American and European companies already have more investments in intangible than in tangible assets.

Merle et al. (2019) analyze the impact of traditional determinants of French MNCs on their decisions to shift profits to tax havens. In turn, Bratta et al. (2021) compared effective income tax rates in different jurisdictions, with high and low taxations. Dharmapala (2020) observed that choosing to locate MNCs in tax havens go beyond the reduced rate to include governance, legal, and financial environments. Finally, Devereux et al. (2018) emphasized the importance of using more complex tax proxies, which capture long-term

decisions, in research on the MNCs´ capital structures. These studies were carried out in developed countries, considering traditional tax rate proxies.

Monitoring of MNCs' operations is often challenging because their activities may include foreign countries with different business cultures, currencies, socio-economic risks, disclosure rules and weaker legal institutions (Akamah, 2017). Greater business flows stimulate tax rate reductions among countries. Tax competition becomes the consequence of the budget deficits acceleration in developed countries. Besides that, the 2007 financial crisis and the Covid-19 pandemic also affected the world economy, making official tax rates decrease. Capital structure of MNCs from emerging countries has specific features due their huge growing capability.

This essay - as mentioned at the Introduction of this thesis (item 1.3) - has the objective of studying the impact of taxation on MNC´s investment decisions. Moreover, this study differs from the others for verifying specific characteristics of MNCs - from emerging countries – that induces their decision to open subsidiaries in tax havens, as a tax planning strategy. Among these determinants are direct taxe's rate, intangible assets and transfer pricing mechanisms. The hypotheses arising from this analysis are: H1 – Direct corporate taxes are positively related to the use of tax havens, H2 - Intangible assets are positively related to the use of tax havens, and H3 - Transfer pricing policies by companies are positively related to the use of tax havens.

Such verifications use logistic and panel data regression models for publicly traded non-financial MNCs from the OECD´s five key partner countries – Brazil, China, India, Indonesia, and South Africa (OECD, 2021a). Companies and countries´ data were obtained from Capital IQ Pro (CIQ Pro) and World Bank databases, respectively, for the period from 2010 to 2020. The final sample comprised 448 MNCs, totaling 4,928 observations.

Among the differentials of this study, the use of the unconventional tax proxy, AREA, stands out, which controls the opportunity to take advantage of the tax benefit from short-and long-term debts (see appendix 1). In addition, the transfer pricing intensity variable was obtained from manually collected data. The result is that multinational companies make use of different instruments, even combined, with the objective of reducing the conglomerate´s total taxation. Therefore, despite the efforts of governments and international organizations, these companies have been efficient in reducing or not paying taxes.

4.2 Literature review

Tax evasion practices have been criticized by politicians, regulators and, society (Akamah *et al.*, 2017). Tax legislation mechanisms, which allow countries to neutralize the impact of the use of tax havens by MNCs, have been in existence for a long time. These mechanisms include controlled foreign company (CFC) rules, under which the countries of residence (where the MNCs are headquartered) tax certain types of income earned by affiliates, in tax haven, from their resident MNCs using thin capitalization rules (TCRs), which limit the interest deductibility, under certain circumstances. The use of these tax law provisions has grown in recent decades. However, more attention it still needed from governments (Dharmapala, 2020). The Base Erosion and Profit Shifting Action Plan (OECD, 2020) aims to help governments reduce MNC´s abilities to shift profits to low-tax jurisdictions. Transactions with related parties must be based on the arm´s length principle which means that transacions between related parties must be carried out as if these parties are not related.

MNCs derive several benefits from international trade, such as expanding trade; employment opportunities, technology transfer, information flow from the international market, technical research and development, economic growth, and tax increase. MNCs seek to obtain these benefits, with the ultimate goal of maximizing the conglomerate's profit and minimizing their global taxes by locating their affiliates in countries with very a low or zero tax rates (Lin & Chang, 2010). A tax haven is, in essence, a mechanism to impose a low or zero tax rate on certain types of income earned by MNCs (Dharmapala, 2020). MNCs use tax havens for profit shifting, intangible payments, and financing among subsidiaries (Brock & Pogge, 2014). The hypotheses arising from these arguments are as follow:

H1a: Direct taxes are positively related to the use of tax havens

H1b: Direct taxes - of companies with investment in intangible assets - have a positive relationship with the use of tax havens

H1c: Direct taxes - of companies with transfer pricing practices - have a positive relationship with the use of tax havens

The advancement of economies towards innovation-intensive knowledge models make taxing the returns on capital challenging, as intangible assets may be placed in low-tax jurisdictions. For developing countries, evasion is an even bigger problem than for developed countries because of failures in tax administration and corruption. This issue is

amplified by the COVID-19 pandemic, which accelerates digital transformation, increasing the problems faced by governments in managing public finances deficits (Ciuriak & Akinyi, 2021). The arm´s length principle requires companies to find comparable transactions to set their own prices. However, in the case of intangible assets, examining their comparisons a challenge for tax authorities because of the lack of similar operations in the market (Gravelle, 2010). As these assessments are subject to the companies' own analysis, they allow managers to take advantage of discrepancies in tax rates among jurisdictions, moving these assets to subsidiaries in tax havens (Atwood & Lewellen, 2019). These arguments support the following hypothesis:

H2: Intangible assets are positively related to the use of tax havens

Another aspect related to the intensification of international trade refers to the fact that nation-states are concerned with the malleability of transfer pricing and with their role in avoiding revenue leakage, which harms the citizens´ well-being (Sikka & Willmott, 2010). Transfer pricing manipulation is the main evasion mechanism used by companies, in an attempt of maximize their global profit via tax reduction (Jansky & Kokeš, 2015). According to Richardson (2008), tax evasion is a deliberate illegal behavior or activity that involves direct violation of tax legislation. Tax evasion theory is when tax paid is less than what is effectively due.

This procedure may be harmful to the public purse. However, there are risks for companies that engage in aggressive practices. If one or more tax authorities reject the transfer pricing, the defaulting company undergoes a tax adjustment, which, in the absence of bilateral correction measures, may result in a double taxation (Merle et al., 2019). Additionally, tax evasion mechanisms provide space for opportunistic managers to pursue selfish goals and to manage results to benefits themselves and not the shareholders (Desai & Dharmapala, 2009).

Given the differences in corporate income tax rates across jurisdictions, some companies manipulate the prices of goods and services to reduce their total tax burden. A strategic possibility is the under- or over-invoicing of sales prices made between subsidiaries abroad, with the objective of minimizing net payments to the government (Ciuriak & Akinyi, 2021). The intensity of this transfer pricing practice can be verified by the proportions of a subsidiaries´ sales revenue and assets in relation to the company's totals. From these points the following hypotheses were formed:

H3a: Transfer pricing policies are positively related to the use of tax havens

H3b: Transfer pricing policies - by companies with investment in intangible assets - are positively related to the use of tax havens

4.3 Data description and sample selection

The initial sample consists of 2,333 publicly traded companies with parents in OECD´s key partner countries: Brazil, China, India, Indonesia, and South Africa. Companies and countries´ data were obtained from the Capital IQ Pro and World Bank databases, respectively, for the period from 2010 to 2020. It is noteworthy that the implementation of transfer pricing guidelines began in 2010 (OECD, 2010). From the initial sample are the MNCs that: had insufficient data (1,773), negative equity (8), negative operational result (21), belong to the financial sector (62), which is subject to specific regulation, and (e) employ less than 250 employees (21) to meet the large company criterion (OECD, 2021b).

Thus, the final sample included 448 non-financial publicly traded companies, totaling 4,928 observations. All of them had total assets valued at more than US$2 million during the sample period. The variables were winsorized between the 1st and the 99th percentiles, to reduce the effects of outliers. Creating a sample for studies on investments in tax havens, mainly for MNCs headquartered in non-developed countries, is challenging for researchers (Merle *et al.*, 2019; Taylor *et al.*, 2015). In general, companies do not make their records available in formats that can be organized in databases. Therefore, the information related to the sales and assets of all the MNCs´ subsidiaries and tax haven were processed manually from the information available in the CIQ Pro database.

4.4 Methodology

In this section, the econometric models and their variables are described for each hypothesis.

4.4.1 Variables description

Table 4.1 presents the description of the variables used in the econometric models of this study (see equations 2 to 7), including the references to substantiate the use of the variables.

Table 4.1 – Variables of econometric models

Initials	Name	Formula	Components	References
Dependent variables				
DTH	Dummy for tax haven	Dummy for tax haven	1= If the firm has at least one subsidiary in a tax haven country 0 = Otherwise	Gumpert *et al.* (2016); Akamah *et al.* (2017); Hebous & Johannesen, 2021; Vineeth & Nidheesh (2021)
Independent variables				
ETR (H1)	Effective tax rate	ETR= TP / EBT	TP = Tax paid EBT = Earnings before tax	Al-Hadi *et al.* (2021); Christensen *et al.* (2021)
INTA (H2)	Intangible assets	INTA = NIA / TA	NIA = Net intangible assets TA = Total assets	Akamah *et al.* (2017); Merle *et al.* (2019); Vineeth & Nidheesh (2021)
Initials	**Name**	**Formula**	**Components**	**References**
TPI (H3)	Transfer pricing intensity	TPI = [(TSR / TCR) - (TSA / TCA)]	TSR = Total subsidiaries revenue TCR = Total company revenue TSA = Total subsidiaries assets TCA = Total company assets	Proxy defined like the variables considered by Solikhah *et al.* (2021)
Companies´control variables				
AREA	Marginal benefit area	$A = \int_0^{\infty} f_{it}(x_{it})dx_{it}$	A = Integral of the x function x = Marginal benefit on an additional dollar (US$) of income	Van Binsbergen *et al.* (2010)
ETR (a)	Effective tax rate	ETR= TP / EBT	TP = Tax paid EBT = Earnings before tax	Jones & Temouri (2016); Merle *et al.* (2019)
SIZE	Size	SIZE = nl TA	nl=Napierian logarithm TA = Total assets	Gumpert *et al.* (2016); Akamah *et al.* (2017); Vineeth & Nidheesh (2021)
LEV	Leverage	LEV = TD / TA	TD = Total debt (short and long terms) TA = Total assets	Akamah *et al.* (2017); Merle *et al.* (2019)
RD	Research and development	RD = RDE / TS	RD = Research and development expenses TS = Total sales	Atwood & Lewellen, (2019); Alnori (2021)
Countries´control variables				
STR	Statutory tax rate	Statutory corporate income tax rate at the beginning of the year	n/a	Egger & Merlo (2011); Jones & Temouri (2016)
GDP	Gross domestic product per capita´s growth	Gross domestic product per capita´s growth (annual %)	n/a	Ayanwale (2007); Kurtović *et al.* (2020)

Notes: (a) The ETR variable is considered as being of control in equations 5, 6 and 7 - which test the hypotheses 2a, 2b and 3 - respectively; (b) n/a means not applicable.

4.5 Econometric models

Econometric models consist of logistic regressions (logit), applied both the total and sub-sample of countries, ensuring the robustness of the results. Mescall and Klassen (2018) applied logit to investigate transfer pricing in mergers and acquisitions. Akamah *et al.* (2018) and Vineeth and Nidheesh (2021) applied logit to analyze MNCs´ investment decision in tax havens. In addition, another robustness test was performed by applying regression models to panel data, considering country, industry and year fixed effects (Gumpert *et al.*, 2016; Hebous & Johannesen, 2021; Merle *et al.*, 2019). For all models, presented in equations 2 to 7, the dependent variable was a dummy, equal to one if the company had at least one subsidiary in a tax haven (see Appendix 2) and equal to zero otherwise. The interpretation of logit results was performed using the odds ratio (Hailpern & Visintainer, 2003). Industry and country fixed effects are used to control for time-varying characteristics across sector and macroeconomic types (Schmal *et al.*, 2021).

The potential for the increase in the total tax benefit derived from MNCs´ debts were controlled by the variable $AREA_{it}$ (see equation 1) (Van Binsbergen *et al.*, 2010). MNCs adopt strategies for their tax burden reduction through their subsidiaries´ leverage by recording debts in locations where the use of the tax benefit are the highest and statutory rates are the lowest possible (Amidu *et al.*, 2019). Appendix 1 presents an example of variable AREA calculus.

$$AREA_{it} = \int_{0}^{\infty} f_{it}\,(x_{it})dx_{it} \tag{1}$$

4.5.1 Hypothesis 1

To verify hypothesis H1a (see equation 2), the independent variable, direct taxes, was represented by ETR_{it}. This tax proxy captures the effect of the payment of MNCs´ taxes on their operating result. To test hypothesis H1b (see equation 3), the interaction variable ETR_{it}*$INTA_{it}$ was added, to analyze the impact of direct taxes on companies` decisions to maintain subsidiaries in tax haven, where these companies invest in intangible assets. To verify hypothesis H1c (see equation 4), the interaction variable $ETR_{it}$* $TPI_{it}$ was included to analyze the impact of direct taxes on companies` decisions to establish subsidiaries in tax havens where they use transfer pricing.

H1a: Direct taxes are positively related to the use of tax havens

$$DTH_{it} = \beta_0 + \beta_1 ETR_{it} + \beta_2 AREA_{it} + \beta_3 SIZE_{it} + \beta_4 LEV_{it} + \beta_5 RD_{it} + \beta_6 STR_{kt} + \beta_7 GDP_{kt} + \lambda_k + \nu_j + \theta_t + \mu_{ijkt} \quad (2)$$

Where, dependent variable = DTH; independent variable = ETR; company´s controls = AREA, SIZE, LEV, and RD; country´s controls = STR and GDP; i = company; j = industry; k = country; t = year; λ_k = country effect; ν_j = industry-effect; θ_t = year effect; μ_{ijkt} = residuals.

H1b: Direct taxes - of companies with investment in intangible assets - have a positive relationship with the use of tax havens

$$DTH_{it} = \beta_0 + \beta_1 ETR_{it} + \beta_2 (ETR_{it} * INTA_{it}) + \beta_3 AREA_{it} + \beta_4 SIZE_{it} + \beta_5 LEV_{it} + \beta_6 RD_{it} + \beta_7 STR_{kt} + \beta_8 GDP_{kt} + \lambda_k + \nu_j + \theta_t + \mu_{ijkt} \quad (3)$$

Where, dependent variable = DTH; independent variables = ETR and (ETR*INTA); company´s controls = AREA, SIZE, LEV, and RD; country´s controls = STR and GDP; i = company; j = industry; k = country; t = year; λ_k = country effect; ν_j = industry-effect; θ_t = year effect; μ_{ijkt} = residuals.

H1c: Direct taxes - of companies with transfer pricing practices - have a positive relationship with the use of tax havens

$$DTH_{it} = \beta_0 + \beta_1 ETR_{it} + \beta_2 (ETR_{it} * TPI_{it}) + \beta_3 AREA_{it} + \beta_4 SIZE_{it} + \beta_5 LEV_{it} + \beta_6 RD_{it} + \beta_7 STR_{kt} + \beta_8 GDP_{kt} + \lambda_k + \nu_j + \theta_t + \mu_{ijkt} \quad (4)$$

Where, dependent variable = DTH; independent variables = ETR and (ETR*TPI); company´s controls = AREA, SIZE, LEV, and RD; country´s controls = STR and GDP; i = company; j= industry; k = country; t = year; λ_k = country effect; ν_j = industry-effect; θ_t = year effect; μ_{ijkt} = residuals.

4.5.2 Hypothesis 2

For the investigation of the hypothesis H2 (see equation 5), the ratio between net intangible assets and total assets is used as independent variable (Akamah *et al.*, 2017; Vineeth & Nidheesh, 2021). MNCs may eventually move intangible assets to jurisdictions with lower tax rates. Their objective are to obtain gains from these assets, in these locations, through the usufruct of items such as trademarks, patents, industrial secrets, and projects. (Merle *et al.*, 2019).

H2: Intangible assets are positively related to the use of tax havens

$$DTH_{it} = \beta_0 + \beta_1 INTA_{it} + \beta_2 ETR_{it} + \beta_3 AREA_{it} + \beta_4 SIZE_{it} + \beta_5 LEV_{it} + \beta_6 RD_{it} + \beta_7 STR_{kt} + \beta_8 GDP_{kt} + \lambda_k + \nu_j + \theta_t + \mu_{ijkt} \quad (5)$$

Where, dependent variable = DTH; independent variable = INTA; company´s controls = ETR, AREA, SIZE, LEV, and RD; country´s controls = STR and GDP; i = company; j = industry; k = country; t = year; λ_k = country effect; ν_j = industry-effect; θ_t = year effect; μ_{ijkt} = residuals.

4.5.3 Hypothesis 3

For the analysis of the hypothesis H3a (see equation 6), the ratio between the proportions of subsidiaries´ revenue and assets, in relation to the MNC, was considered as a proxy for the independent variable, transfer pricing. The higher this ratio is, the greater is the organization's transfer pricing policy. Transfers of revenues and assets among associated parties play a relevant role in MNCs' tax planning (Amidu *et al.*, 2019; Solikhah *et al.*, 2021). The examination of hypothesis H3b (see equation 7) – occurs via inclusion of the interaction variable $TPI_{it}*INTA_{it}$. To analyze the influence of transfer pricing policies on the opening of subsidiaries in tax havens by companies with investment in intangible assets.

H3a: Transfer pricing policies are positively related to the use of tax havens.

$$DTH_{it} = \beta_0 + \beta_1 TPI_{it} + \beta_2 ETR_{it} + \beta_3 AREA_{it} + \beta_4 SIZE_{it} + \beta_5 LEV_{it} + \beta_6 RD_{it} + \beta_7 STR_{kt} + \beta_8 GDP_{kt} + \lambda_k + \nu_j + \theta_t + \mu_{ijkt} \quad (6)$$

Where, dependent variable = DTH; independent variable = TPI_{it}; company´s controls = ETR, AREA, SIZE, LEV, and RD; country´s controls = STR and GDP; i = company; j = industry; k = country; t = year; λ_k = country effect; ν_j = industry-effect; θ_t = year effect; μ_{ijkt} = residuals.

H3b: Transfer pricing policies - by companies with investment in intangible assets - are positively related to the use of tax havens

$$DTH_{it} = \beta_0 + \beta_1 TPI_{it} + \beta_2 (TPI_{it}*INTA_{it}) + \beta_3 ETR_{it} + \beta_4 AREA_{it} + \beta_5 SIZE_{it} + \beta_6 LEV_{it} + \beta_7 RD_{it} + \beta_8 STR_{kt} + \beta_9 GDP_{kt} + \lambda_k + \nu_j + \theta_t + \mu_{ijkt} \quad (7)$$

Where, dependent variable = DTH; independent variables = TPI_{it} and $(TPI_{it}*INTA_{it})$; company´s controls = ETR, AREA, SIZE, LEV, and RD; country´s controls = STR and GDP; i = company; j = industry; k = country; t = year; λ_k = country effect; ν_j = industry-effect; θ_t = year effect; μ_{ijkt} = residuals.

4.6 Analysis of results

Each hypothesis was verified using descriptive statistics, correlation analysis and econometric outputs.

4.6.1 Descriptive analysis

Table 4.2 describes the main characteristics of the variables in the models of equations 2 to 7 for the total sample and sub-samples of MNCs with and without subsidiaries in tax havens. The effective tax payment rate´s (ETR) average was 0.22. This means that MNCs commit about 22% of their operating profit before taxes (EBT) on tax obligations, in a scenario with an average statutory rates (STR) of 28%. The negative minimum value of ETR (-0.53) indicates that, even with a negative operating result, MNC stills direct 53% of its EBT to the payment of taxes. The tax legislation of some countries indicates that companies must make adjustments based on the results presented in the financial statements to determine the income tax to be effectively paid.

In turn, intangible assets (INTA), on average, represent approximately 7% of companies´ total assets. The analysis of sub-samples confirms the relevance of this instrument for transferring profits among units. MNCs with subsidiaries in tax havens have a higher average of intangible assets (8%) than those that do not register businesses in these locations (6%).

The transfer pricing intensity (TPI) has an average of 0.16. However, MNCs with subsidiaries in tax havens have an average TPI of 0.22 and earn 22% more revenue than the value of their total assets held. In contrast, MNCs without subsidiaries in tax havens had an average TPI of 0.10. Thus, the former present a higher billing to their subsidiaries for a given volume of assets than the latter, characterizing a behavior expected by companies operating in tax havens. TPI variable shows a negative sign for companies whose subsidiaries' assets in relation to the company's total outweigh the respective proportions of revenue.

AREA is a proxy for the opportunity of reducing tax based on companies' leverage levels. Thus, the greater its value, the greater is the possibility of MNCs using its benefit. It is verified that, on an average, the marginal area of this benefit was 3.94. In the case of MNCs with (without) subsidiaries in tax havens, the average was 3.89 (4.00). Thus, companies with less ground to take advantage of the tax benefit are more likely to set up subsidiaries in tax havens.

The size (SIZE) of the MNCs, indicate total assets on average of US$ 5.74 billion. For the sub-samples, those that (do not) have subsidiaries in tax havens, their mean total assets were US$ (4.60) 6.75 billion, respectively. This fact corroborates the results of Davies et al. (2018) that large MNCs are the main users of tax havens. Leverage`s (LEV) median (0.21) confirms that MNCs, with units in tax havens, are more indebted than those without (0.18). It is noteworthy that leverage is a way to transfer profits among subsidiaries via tax benefit arising from the payment of interest.

The proportion of research and development expenses to total sales (RD) have similar means (0.02) for both MNCs located in tax havens and those not located there. However, when analyzing just only the numerator of this ratio, the first group of MNCs spent an average of US$ 64.4 million on research and development. This amount corresponds to almost double the expenses incurred by MNCs that do not have subsidiaries in tax havens (US$ 31.71 million). This type of expense is considered as a proxy for intangible assets related to the ownership of trademarks and patents.

Jone and Temouri (2016) found that MNCs transfer intangible assets to subsidiaries located in countries with low or no corporate taxation, high levels of secrecy, and significant institutional strength, which favors the protection of private ownership rights. Finally, as for GDP per capita growth, on an average, the annual variation was 4.59%. The countries contributing the most to this high average were China (6.68%), India (4.14%), and Indonesia (3.44%). In contrast, Brazil has a mean close to zero (0.12%), while South Africa`s GDP declined (-0.57%).

In summary, MNCs consider several variables in their decisions for tax planning considers to implement tax elision policies. Among them, benefits arise from debt interest deductibility. Another alternative is setting up subsidiaries in tax havens, which have low statutory tax rates. An even greater stimulus for these corporate decisions by MNCs is that they have higher levels of intangible assets adopt transfer pricing policies.

Table 4.2 – Descriptive statistics

This table presents the descriptive statistics for the dependent, independent, and control variables: ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. INTA is intangible assets, calculated as the ratio of net intangible assets to total assets. TPI is the transfer pricing intensity, which is the ratio between participation of subsidiaries´ revenue and total assets in the total of company. AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Napierian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is research and development, which is the ratio of research and development expenses to total sales. STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

Variables	Observations	Mean	Median	Stand. Dev.	Minimum	Maximum
Total MNCs sample						
ETR	4,919	0.22	0.22	0.22	-0.53	1.40
INTA	4,928	0.07	0.02	0.11	0.00	0.52
TPI	4,928	0.16	0.09	0.23	-0.34	0.81
AREA	4,928	3.94	4.33	1.60	0.00	6.53
SIZE (a)	4,928	5.74	0.54	24.52	0.002	392.49
LEV	4,917	0.22	0.20	0.16	0.01	0.70
RD	4,924	0.02	0.01	0.04	0.00	0.56
STR	4,928	0.28	0.25	0.04	0.25	0.35
GDP	4,928	4.59	5.67	3.85	-8.87	10.10

Sub-samples of MNCs with subsidiaries in tax haven (TH) and non-tax haven (NTH) countries												
Variables	**TH**	**NTH**	**TH**	**NTH**	**TH**	**NTH**	**TH**	**NTH**	**TH**	**NTH**	**TH**	**NTH**
ETR	2,603	2,316	0.23	0.21	0.23	0.21	0.25	0.19	-0.53	-0.53	1.40	1.40
INTA	2,609	2,319	0.08	0.06	0.25	0.02	0.11	0.10	0.00	0.00	0.52	0.52
TPI	2,609	2,319	0.22	0.10	0.15	0.03	0.21	0.23	-0.34	-0.34	0.81	0.81
AREA	2,609	2,319	3.89	4.00	4.32	4.37	1.65	1.54	0.00	0.00	6.53	6.53
SIZE (a)	2,609	2,319	6.75	4.60	0.72	0.36	29.14	17.90	0.002	0.002	392.49	335.78
LEV	2,599	2,318	0.22	0.22	0.21	0.18	0.16	0.18	0.01	0.01	0.70	0.70
RD	2,605	2,319	0.02	0.02	0.01	0.01	0.05	0.04	0.00	0.00	0.56	0.41
STR	2,609	2,319	0.29	0.28	0.25	0.25	0.04	0.04	0.25	0.25	0.35	0.35
GDP	2,609	2,319	4.49	4.71	5.67	6.19	3.85	3.84	-8.87	-8.87	10.10	10.10

Note: The size variable is measured by the MNC´s total assets, in US$ billion

4.6.2 Correlation analysis

Table 4.3 presents the correlation analysis between the variables mentioned in equations 2 to 7. According to finance literature, there is a positive relationship between ETR and DTH (Al-Hadi et al., 2021; Christensen et al., 2021). The higher the tax paid, the greater the possibility that MNCs will opt for subsidiaries in tax havens. Furthermore, there is a negative relationship between AREA and DTH. This means that, as expected, the smaller the opportunity for taking advantage of tax benefit, arising from debts, the greater the possibility for MNCs to install subsidiaries in tax havens.

In addition, there are positive relationships between INTA and TPI with DTH ratifying hypotheses H2 and H3. In fact, a greater proportion of intangible assets and transfer pricing practices point to a greater possibility for MNCs to operate in tax havens. These results corroborate those by Desai and Dharmapala (2009), Amidu et al. (2019), Merle et al. (2019) and Vineeth and Nidheesh (2021). According to Richardson and Taylor (2015), MNCs transfer profits, intangible assets, and consequently, taxes among their subsidiaries through transfer pricing. Alternatively, they seek tax reduction through the use of tax havens.

Regarding size (SIZE), as foreseen there is a positive relationship with DTH. Large MNCs are the main users of tax havens. As expected, leverage`s (LEV), negative correlation (40%) with AREA stands out. In fact, the more third-party capital is used, the less is the

opportunity for taking advantage of the tax benefits arising from debts. Another aspect to be highlighted is the positive correlation between countries´ STR and the presence of MNCs´ subsidiaries in tax havens (DTH). Thus, the greater an MNC´s consolidated tax burden, the greater the possibility for it to have subsidiaries in tax havens to mitigate this expense.

However, AREA`s positive correlation with STR (30%) stands out. This result is equally expected. MNCs in countries with high STR are more likely to take advantage of the tax benefit arising from debts. Finally, a problem of multicollinearity is not identified among the explanatory variables, as there are no positive correlations above 80% (Gujarati et al., 2012).

Table 4.3 – Correlation Analysis

VAR	DTH	ETR	INTA	TPI	AREA	SIZE	LEV	RD	STR	GDP
DTH	1.00									
ETR	0.04 **[0.01]**	1.00								
INTA	0.04 **[0.00]**	-0.04 **[0.01]**	1.00							
TPI	0.25 **[0.00]**	-0.04 [0.84]	-0.13 **[0.00]**	1.00						
AREA	-0.04 **[0.01]**	0.14 **[0.00]**	-0.04 **[0.00]**	-0.04 **[0.00]**	1.00					
SIZE	0.14 **[0.00]**	0.01 [0.52]	0.14 **[0.00]**	-0.12 **[0.00]**	-0.10 **[0.00]**	1.00				
LEV	0.03 [0.07]	0.01 [0.64]	0.07 **[0.00]**	-0.11 **[0.00]**	-0.40 **[0.00]**	0.21 **[0.00]**	1.00			
RD	0.02 [0.25]	-0.12 **[0.00]**	0.08 **[0.00]**	0.01 [0.42]	-0.09 **[0.00]**	-0.08 **[0.00]**	-0.09 **[0.00]**	1.00		
STR	0.03 **[0.02]**	0.13 **[0.00]**	-0.05 **[0.00]**	0.04 **[0.00]**	0.30 **[0.00]**	-0.16 **[0.00]**	0.08 **[0.00]**	-0.17 **[0.00]**	1.00	
GDP	-0.03 **[0.05]**	0.03 [0.06]	-0.09 **[0.00]**	0.05 **[0.00]**	0.03 [0.01]	-0.04 **[0.00]**	-0.10 **[0.00]**	0.14 **[0.00]**	-0.22 **[0.00]**	1.00

Notes: Upper values correspond to the correlation coefficient, while lower values [in brackets] correspond to its significance level. Values highlighted in bold are those that have statistical significance levels of 1% or 5%. VAR means variables; DTH: tax haven dummy; ETR: effective tax rate; INTA: intangible assets; TPI: transfer pricing intensity; AREA is the marginal benefit area; SIZE: Napierian logarithm of the total assets; LEV: leverage ratio; RD: research and development; STR: statutory tax rate at the beginning of the year; GPD: gross domestic product per capita growth (annual %).

4.6.3 Logit model (LM) and ordinary least squares (OLS) results

Tables 4.4 to 4.9 present the LM and OLS results, respectively. LM was verified by the likelihood ratio (LR), receiver operating characteristic curve (ROC curve), model classification, and Hosmer-Lemeshow goodness-of-fit post estimation test. The OLS assumption tests applied were Breusch–Pagan, Cameron-Trivedi, Shapiro–Francia, regression specification error test (RESET), variance inflation factor (VIF), and White. It is important to point out that the OLS results were consistent with the LM results.

4.6.4 H1a: Direct taxes are positively related to the use of tax havens

Table 4.4 presents the LM and OLS results for the hypothesis H1a. Both models are valid according to the Chi-squared (0.000) and F-probability (0.000) tests. The ETR variable had a positive relationship with the possibility of the MNC installing subsidiaries in a tax haven. The economic interpretation of the odds ratio is that, for each 1% increase in ETR, there is a 98.80% [100*(1.988–1)] increase in the chances of establishing MNC´s subsidiaries in tax havens. In other words, MNCs respond positively to the opening of subsidiaries in tax havens, based on the increase in tax payable. Therefore, hypothesis H1a is confirmed - *Direct taxes are positively related to the use of tax havens.*

In contrast, the AREA variable had a negative relationship with the possibility of the MNC installing subsidiaries in a tax haven. The economic interpretation of the odds ratio is that, for each 1% increase in AREA, there was a 7.80% [100*(0.922–1)] reduction in the chances of establishing MNC´s subsidiaries in tax havens. This reduction was because MNCs have less opportunity to use the tax benefit arising from debts. This negative relationship is in line with the result obtained in Table 4.3 – Correlation analysis.

Furthermore, a country's STR has a positive relationship with DTH, with an odds ratio of 25.933 indicating that, for each 1% increase in STR, an increase of 2,493.30% is expected [100*(25.933– 1)] in the probability of MNCs having subsidiaries installed in tax havens. These results confirmed tax avoidance theory and were in agreement with the studies of Clausing (2009), Gumpert et al. (2016), Hebous and Johannesen (2021), and Lin and Chang, (2010).

Table 4.4 – Direct corporate taxes and tax havens

The dependent variable DTH is a tax haven dummy, equal to 1 if the firm has at least one subsidiary in a tax haven country, and 0 otherwise. The independent variable ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. The company´s control variables: AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Napierian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is the research and development, which is the ratio of research and development expenses to total sales. The countries´ control variables: STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

	Tax haven dummy (DTH)				
Variables	**Logit Model**			**Ordinary Least Squares**	
	Coef	**Odds Ratio**	**Z**	**Coef**	**T**
ETR	0.687***	1.988	2.87	0.165***	3.22
AREA	-0.081***	0.922	-3.63	-0.020***	-6.44
SIZE	0.174***	1.189	10.24	0.048***	12.40
LEV	-0.431**	0.650	-1.96	-0.104**	-3.63
RD	2.216***	9.169	3.25	0.529***	2.96
STR	3.256***	25.933	4.00	0.780***	5.29
GDP	-0.023**	0.978	-2.53	-0.005**	-2.17

Table 4.4 – Direct corporate taxes and tax havens (continuation)

	Tax haven dummy (DTH)	
	Logit Model	**Ordinary Least Squares**
# Obs	4,904	4,904
Industry effect	Yes	Yes
Year effect	Yes	Yes
# Firms	448	448
Prob > chi2	0.0000	n/a
Prob > F	n/a	0.0000
R^2	n/a	0.0329
Hosmer-Lemeshow	0.4398	n/a

Notes: n/a means not applicable, while the significance levels are 1% (***) and 5% (**)

4.6.5 H1b: Direct taxes - of companies with investment in intangible assets - have a positive relationship with the use of tax havens

Table 4.5 presents the LM and OLS results for hypothesis H1b. Both models are valid according to the Chi-squared (0.000) and F-probability (0.000) tests. The ETR variable showed a positive relationship with DTH. There was a 73.8% [100*(1.738–1)] increase in the probability that MNC establish a subsidiary in a tax haven. The interaction variable (ETR* INTA) also had a positive relationship with DTH. Despite its significance level being lower than ETR´s, MNCs with intangible assets increase the chance of operating in tax haven by 380.8% [100 (4.808–1)]. MNCs impacted by tax burden and with intangible assets opt to transfer assets to subsidiaries in tax havens. Taxation on royalty income is reduced or eliminated in this type subsidiary and deductible in companies located in jurisdictions with a higher corporate tax rate.

According to Dharmapala (2020), tax haven jurisdictions reduce, or even exempt, corporate tax to attract mobile capital - tax competition theory. These results were confirmed by regression tests with panel data, which are in line with the studies of Amidu et al. (2019), Pavel et al. (2021) and Vineeth and Nidheesh (2021), there by, confirming hypothesis H1b - Direct taxes - of companies with investment in intangible assets - have a positive relationship with the use of tax havens.

The AREA variable controls the possibility of using the debt tax benefit. For each 1% increase in AREA, there was a 7.70% [100*(0.923–1)] reduction in the chances of establishing MNC´s subsidiaries in tax havens. With the control variables, the positive relationship between research and development (RD) expenditure is highlighted. For each 1% increase in RD, an increase of 741.10% [100*(8.411–1)] is expected in the probability for MNCs to have a subsidiary installed in tax havens. Investment in RD confirms the shift of intangible assets to units in tax havens.

Table 4.5 – Direct taxes of companies with intangible assets and tax havens

The dependent variable DTH is a tax haven dummy, equal to 1 if the firm has at least one subsidiary in a tax haven country, and 0 otherwise. The independent variables: ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. ETR*INTA is the interaction variable between the ETR and INTA (INTA is intangible assets, calculated as the ratio of net intangible assets to total assets). The company´s control variables: AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Napierian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is the research and development, which is the ratio of research and development expenses to total sales. The countries´ control variables: STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

Tax haven dummy (DTH)					
Variables	**Logit Model**			**Ordinary Least Squares**	
	Coef	**Odds Ratio**	**Z**	**Coef**	**T**
ETR	0.552**	1.738	2.19	0.142**	2.41
ETR*INTA	1.570*	4.808	1.71	0.256**	2.11
AREA	-0.081***	0.923	-3.62	-0.194***	-3.65
SIZE	0.172***	1.187	10.07	0.041***	10.62
LEV	-0.452**	0.636	-2.06	-0.108**	-2.04
RD	2.130***	8.411	3.12	0.515***	3.24
STR	3.333***	28.099	4.08	0.791***	4.05
GDP	-0.217**	0.979	-2.34	-0.005**	-2.39
# Obs		4,904		4,904	
Industry effect		Yes		Yes	
Year effect		Yes		Yes	
# Firms		448		448	
Prob > chi2		0.0000		n/a	
Prob > F		n/a		0.0000	
R^2		n/a		0.0333	
Hosmer-Lemeshow		0.4465		n/a	

Notes: n/a means not applicable, while the significance levels are 1% (***), 5% (**), and 10% (*).

4.6.6 H1c: Direct taxes - of companies with transfer pricing practices - have a positive relationship with the use of tax havens

Table 4.6 presents the LM and OLS results for the hypothesis H1c. Both models are valid according to the Chi-squared (0.000) and F-probability (0.000) tests. The interaction variable (ETR*TPI) has a positive relationship with the possibility of the MNC installing a subsidiary in a tax haven. The economic interpretation of the odds ratio is that for each 1% increase in ETR*TPI there is an increase of 3,151.10% [100 (32.511-1)] in the chances for the MNC establishing a subsidiary in tax havens. In contrast, The ETR variable had a negative relationship with DTH. These results suggest that the practice of transfer pricing, combined with the tax actually paid, predominates in the MNCs´ tax planning. Profit is transferred through sales of goods and services to jurisdictions with lower or no taxation - tax evasion theory. These results are supported by regression tests with panel data, which are in agreement with the studies by Brock and Pogge (2014), Merle *et al.* (2019) and

Solikhah *et al.* (2021). Thus, hypothesis H1c is confirmed: *Direct taxes - of companies with transfer pricing practices - have a positive relationship with the use of tax havens.*

The variable AREA had a negative relationship with the possibility of the MNC installing a subsidiary in a tax haven. The economic interpretation of the odds ratio is that for each 1% increase in AREA, there is a 9.30% [100*(0.907–1)] reduction in the chance of establishing MNC´s subsidiaries in tax havens. This is because MNCs have less opportunity to use the tax benefit arising from debts. Regarding control variables, the negative relationship between leverage (LEV) and tax haven stands out. That is to say, for each 1% increase in LEV, MNCs reduce by 34.80% [100*(0.652–1)] their possibility of operating with subsidiaries in tax havens. The MNCs with more tax benefit from debts - trade-off theory - are less prone to tax planning via subsidiary´s establishment in a tax haven.

Table 4.6 – Direct taxes of companies with transfer pricing practices and tax havens

The dependent variable DTH is a tax haven dummy, equal to 1 if the firm has at least one subsidiary in a tax haven country, and 0 otherwise. The independent variables: ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. ETR*TPI is the interaction variable between the ETR and TPI (TPI is the transfer pricing intensity, which is the ratio between participation of subsidiaries´ revenue and total assets in the total of company). The company´s control variables: AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Napierian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is the research and development, which is the ratio of research and development expenses to total sales. The countries´ control variables: STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

	Tax haven dummy (DTH)				
Variables	**Logit Model**			**Ordinary Least Squares**	
	Coef	**Odds Ratio**	**Z**	**Coef**	**T**
ETR	-0.732***	0.481	-2.81	-0.176***	-2.96
ETR*TPI	10.389***	32.511	13.06	2.209***	15.57
AREA	-0.098***	0.907	-4.26	-0.022***	-4.23
SIZE	0.221***	1.248	12.73	0.050***	13.59
LEV	-0.427**	0.652	-2.07	-0.097**	-2.07
RD	2.569***	13.051	3.74	0.568***	3.68
STR	3.202***	24.576	3.75	0.745***	3.86
GDP	-0.033***	0.968	-3.34	-0.006***	-3.12
# Obs		4,904		4,904	
Industry effect		Yes		Yes	
Year effect		Yes		Yes	
# Firms		448		448	
Prob > chi2		0.0000		n/a	
Prob > F		n/a		0.0000	
R^2		n/a		0.0812	
Hosmer-Lemeshow		0.1565		n/a	

Notes: n/a means not applicable, while the significance levels are 1% (***) and 5% (**).

4.6.7 H2: Intangible assets are positively related to the use of tax havens

Table 4.7 presents the LM and OLS results for the hypothesis H2. Both models are valid according to the Chi-squared (0.000) and F-probability (0.000) tests. The INTA

variable presents a positive relationship with the possibility of installing a subsidiary in a tax haven, by the MNC. The odds ratio is interpreted economically as for each 1% increase in INTA, the chances of establishing MNC´s subsidiaries in tax havens increase by 81.80% [100*(1.818–1)]. For Ciuriak and Akinyi (2021), the expansion of intangible assets increases tax erosion. These results are confirmed by regression tests with panel data, which are in accordance with the studies of Beer *et al.* (2020). Thus, hypothesis H2 is confirmed - *Intangible assets are positively related to the use of tax havens.*

In contrast, the control variable of the tax actually paid (ETR) maintains a positive relationship with DTH. AREA which controls the opportunity for using the debt tax benefit, has a negative relationship with DTH. For each 1% increase in AREA, there was a 7.90% [100*(0.921–1)] reduction in the chances of establishing MNC´s subsidiaries in tax havens. Furthermore, for each 1% increase in RD, MNCs increase by 699.80% [100 (7.998–1)] the possibility of operating with subsidiaries in tax havens. Investment in RD confirms the transfer of intangible assets to companies in tax havens.

Table 4.7 – Intangible assets and tax havens

The dependent variable DTH is a tax haven dummy, equal to 1 if the firm has at least one subsidiary in a tax haven country, and 0 otherwise. The independent variable INTA is the intangible assets, calculated as the ratio of net intangible assets to total assets. The company´s control variables: ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Napierian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is the research and development, which is the ratio of research and development expenses to total sales. The countries´ control variables: STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

	Tax haven dummy (DTH)				
Variables	**Logit Model**			**Ordinary Least Squares**	
	Coef	**Odds Ratio**	**Z**	**Coef**	**T**
INTA	0.598*	1.818	1.70	0.144*	1.66
ETR	0.701***	2.015	2.92	0.169***	2.66
AREA	-0.083***	0.921	-3.71	-0.020***	
SIZE	0.170***	1.185	9.87	0.041***	11.07
LEV	-0.459**	0.632	-2.08	-0.111**	-0.82
RD	2.079***	7.998	3.04	0.496***	3.11
STR	3.364***	28.907	4.11	0.804***	3.04
GDP	-0.022**	0.979	-2.32	-0.005**	-2.48
# Obs		4,904		4,904	
Industry effect		Yes		Yes	
Year effect		Yes		Yes	
# Firms		448		448	
Prob > chi2		0.0000		n/a	
Prob > F		n/a		0.0000	
R^2		n/a		0.0335	
Hosmer-Lemeshow		0.4418		n/a	

Notes: n/a means not applicable, while the significance levels are 1% (***), 5% (**), and 10% (*).

4.6.8 H3a: Transfer pricing policies are positively related to the use of tax havens.

Table 4.8 presents the LM and OLS results for the hypothesis H3a. Both models are valid according to the Chi-squared (0.000) and F-probability (0.000) tests. The TPI variable is positively related to the possibility of the MNC installing a subsidiary in a tax haven. The economic interpretation of the odds ratio is that, for each 1% increase in TPI, there is an increase of 1,383.0% [100*(14.830-1)] in the chances of establishing MNC´s subsidiaries in tax havens. MNCs´ productive process of verticalization, for example, allows greater transfer intensity of goods among their subsidiaries (Cen et al., 2017). These results are confirmed by regression tests with panel data, in accordance with the studies of Dharmapala (2008), Solikhah et al. (2021), and Taylor and Richardson (2012). Therefore hypothesis H3a is confirmed: *Transfer pricing policies are positively related to the use of tax havens.*

The AREA variable continues to show a negative relationship with the MNC´s possibility of establishing a subsidiary in a tax haven. For each 1% increase in AREA, there was a 7.4% [100*(0.926–1)] reduction in the chances of establishing MNC´s subsidiaries in tax havens. Regarding the other control variables, the negative relationship between GDP and DTH in all models stands out (equations 2 to 7). In the specific case of equation 7, for each 1% reduction in GDP, MNCs increase by 3.4% [100*(0.966–1)] the possibility of operating with subsidiaries in tax havens. This fact stems from MNC's desire to maintain its profitability through tax planning.

Table 4.8 – Transfer pricing and tax havens

The dependent variable DTH is a tax haven dummy, equal to 1 if the firm has at least one subsidiary in a tax haven country, and 0 otherwise. The independent variable TPI is the transfer pricing intensity, which is the ratio between participation of subsidiaries´ revenue and total assets in the total of company. The company´s control variables: ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Napierian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is the research and development, which is the ratio of research and development expenses to total sales. The countries´ control variables: STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

	Tax haven dummy (DTH)				
Variables	**Logit Model**			**Ordinary Least Squares**	
	Coef	**Odds Ratio**	**Z**	**Coef**	**T**
TPI	2.697***	14.830	15.96	0.586***	18.92
ETR	0.529***	1.697	3.78	0.120***	4.00
AREA	-0.077***	0.926	-3.36	-0.017***	-3.39
SIZE	0.222***	1.248	12.65	0.050***	13.48
LEV	-0.214	0.808	-1.00	-0.044	-0.93
RD	2.517***	12.390	3.60	0.524***	3.49
STR	2.936***	18.847	3.42	0.645***	3.37
GDP	-0.034***	0.966	-3.45	-0.007***	-3.25

Table 4.8 – Transfer pricing and tax havens (continuation)

	Tax haven dummy (DTH)	
Variables	**Logit Model**	**Ordinary Least Squares**
# Obs	4,904	4,904
Industry effect	Yes	Yes
Year effect	Yes	Yes
# Firms	448	448
Prob > chi2	0.0000	n/a
Prob > F	n/a	0.0000
R^2	n/a	0.1033
Hosmer-Lemeshow	0.1559	n/a

Notes: n/a means not applicable, while the significance level is 1% (***).

4.6.9 H3b: Transfer pricing policies - by companies with investment in intangible assets - are positively related to the use of tax havens

Table 4.9 presents the LM and OLS results for the hypothesis H3b. Both models are valid according to the Chi-squared (0.000) and F-probability (0.000) tests. Similar to the results in Table 4.8, TPI had a positive relationship with DTH. In addition, the interaction variable (TPI*INTA) was also positively related to DTH, however, it had a significantly higher coefficient than TPI. The economic interpretation of the odds ratio is that for each 1% increase in TPI*INTA, there is a 912.20% [100 (10.122–1)] increase in the chances of establishing MNC´s subsidiaries in tax havens. The transfer pricing policy is increased by revenues from intangible assets, such as trademark licenses, patents, and industrial projects. These results are in agreement with those of Alnori (2021), Merle *et al.* (2019), and Richardson and Taylor (2015). Therefore, hypothesis H3b is confirmed: *Transfer pricing policies - by companies with investment in intangible assets - are positively related to the use of tax havens.*

Transactions among a conglomerate´s subsidiaries must comply with the arm´s length principle. This means that the prices of products or services in these transactions must be defined by the MNCs, based on the amounts involved in transactions comparable to those of other companies. Nevertheless, tax authorities find it difficult to establish a reference price for high-value intangible assets for taxation purposes (Merle et al., 2019). Thus, MNCs register these intangible assets in tax havens and create relationships among the companies to transfer profits to jurisdictions with low or no taxation.

Regarding the control variables, AREA continued to show a negative relationship with DTH. For each 1% increase in AREA, there was a 7.5% reduction [100*(0.925–1)] in the possibility of opening MNC´s subsidiaries in tax havens. In contrast, size (SIZE) had a positive relationship with DTH, in all models in this study (equations 2 to 7). In the specific

case of equation 7, for each 1% increment in SIZE, the probability MNCs of establishing subsidiaries in tax havens increase by 23.70% [100* (1.237–1)].

Table 4.9 – Intangible assets with transfer pricing and tax havens

The dependent variable DTH is a tax haven dummy, equal to 1 if the firm has at least one subsidiary in a tax haven country, and 0 otherwise. The independent variable TPI is the transfer pricing intensity, which is the ratio between participation of subsidiaries´ revenue and total assets and the total of company. INTA is the intangible assets, calculated as the ratio of net intangible assets to total assets. TPI*INTA is the interaction variable between TPI and INTA. The company´s control variables: ETR is the effective tax rate, which is the ratio between tax paid and the earnings before tax. AREA is the marginal benefit area that simulates the marginal tax benefit function through the marginal tax rate. SIZE was calculated as the Neperian logarithm of the total assets. LEV is the leverage ratio, calculated as the ratio of total debt to total assets. RD is the research and development, which is the ratio of research and development expenses to total sales. The countries´ control variables: STR is the statutory tax rate at the beginning of the year. GPD is the gross domestic product per capita growth (annual %).

	Tax haven dummy (DTH)				
Variables	**Logit Model**			**Ordinary Least Squares**	
	Coef	**Odds Ratio**	**Z**	**Coef**	**T**
TPI	2.252***	9.506	11.87	0.513**	14.27
TPI*INTA	9.223***	10.122	4.08	1.259***	4.84
ETR	0.550***	1.733	3.94	0.123***	4.10
AREA	-0.078***	0.925	-3.40	-0.017***	-3.43
SIZE	0.213***	1.237	11.99	0.047***	12.94
LEV	-0.250	0.779	-1.17	-0.052	-1.09
RD	2.072***	7.941	2.93	0.461***	3.07
STR	3.208***	24.741	3.74	0.671***	3.51
GDP	-0.032***	0.968	-3.29	-0.007***	-3.15
# Obs		4,904		4,904	
Industry effect		Yes		Yes	
Year effect		Yes		Yes	
# Firms		448		448	
Prob > chi2		0.0000		n/a	
Prob > F		n/a		0.0000	
R^2		n/a		0.1069	
Hosmer-Lemeshow		0.1616		n/a	

Notes: n/a means not applicable, while the significance levels are 1% (***) and 5% (**).

Table 4.10 shows the results of classification of the models, considering the measures of sensitivity, specificity, and general assertiveness. Sensitivity assesses the total number of correct answers that the model obtains in relation to the event, namely the existence of a subsidiary in a tax haven. From the total of 4,904 observations in the sample, 2,589 point to this situation. In the case of the hypothesis H1a, for example, the model of equation 2 had a hit percentage of 65.78% (1,703/2,589).

Specificity assesses the total number of correct answers as opposed in the event of interest. Thus, from the total of 4,904 observations in the sample, 2,315 point to the non-existence of subsidiaries in a tax haven. In the case of the hypothesis H1a, the model in equation 2 had a percentage of correctness of 50.28% (1,164/2,315) for MNCs with no subsidiaries in tax havens. The total percentage of correctly classified observations in the

model that tests the hypothesis H1a was 58.46% (2,867/4,904). According to Duppati et al. (2021), assertiveness represents the overall quality of the model.

In summary, on average, the models of equations 2 to 7 had an assertiveness index of 61.37%, with the minimum (maximum) percentage shown in the model testing the hypothesis H1b (H3b). In addition, it appears that the average prediction abilities of the models are greater where the dummy is equal to one, when there are subsidiaries in tax havens (66.69%), compared to cases where the dummy is equal to zero, when are no subsidiaries in a tax haven (55.41%).

Table 4.10 – Model´s hit level classification

Hypotheses	**H1a**	**H1b**	**H1c**	**H2**	**H3a**	**H3b**
Equations	**2**	**3**	**4**	**5**	**6**	**7**
Independent variables	**ETR**	**ETR*INTA**	**ETR*TPI**	**INTA**	**TPI**	**TPI*INTA**
Positive classification (+)	1,703	1,700	1,731	1,712	1,763	1,751
Negative classification (-)	1,164	1,164	1,363	1,163	1,411	1,431
Total	2,867	2,864	3,094	2,875	3,174	3,182
Total dummy (+) = 1				2,589		
Total dummy (-) = 0				2,315		
Total model				4,904		
Sensitivity (dummy = 1)	65.78%	65.66%	66.86%	66.13%	68.10%	67.63%
Specificity (dummy = 0)	50.28%	50.28%	58.88%	50.24%	60.95%	61.81%
Correctly classified	58.46%	58.40%	63.09%	58,63%	64.72%	64.89%

Finally, Table 4.11 summarizes the confirmation of each hypothesis. All hypotheses have their classifications predicted as being equal to those observed, according to the Hosmer Lemeshow post-estimate test (goodness-of-fit). In addition, the independent variables had statistically significant coefficients and sign, as expected, confirming the financial theories supporting them.

Table 4.11 – Result synthesis of the hypotheses

Hypotheses	**H1a**	**H1b**	**H1c**	**H2**	**H3a**	**H3b**
Equations	**2**	**3**	**4**	**5**	**6**	**7**
Independent variables	**ETR**	**ETR*INTA**	**ETR*TPI**	**INTA**	**TPI**	**TPI*INTA**
Expected relationship	**+**	**+**	**+**	**+**	**+**	**+**
Found relationship	**+**	**+**	**+**	**+**	**+**	**+**
Hosmer-Lemeshow	Yes	Yes	Yes	Yes	Yes	Yes
Confirmed hypotheses	Yes	Yes	Yes	Yes	Yes	Yes

4.6.10 Robustess test

Robustness tests are applied for correcting possible endogeneity problems, between tax payment and the decision of the MNC operating in tax havens. On the first alternative regression model, the independent variable ETR is lagged by one year (ETR_{t-1}) (Harris &

Feeny, 2003). On the second alternative regression model, the independent variable long run effective tax rate (LRETR) is calculated as the sum of the tax paid for the last three years, divided by the accounting profit before taxes (Taylor & Richardson, 2012). The LRETR seeks to capture MNC's decision for opening a subsidiary at a tax haven country. This resolution takes time to be done, since it involves asset´s migration, business plant´s tranfer and other long-term processes. The results are not presented, as they have been surpassed by the use of the independent ETR variable in level.

4.7 Conclusion

The main objective of this study is verifying specific characteristics of MNCs - from emerging countries – that induces their decision to open subsidiaries in tax havens, as a tax planning strategy. Among these determinants are direct taxe's rate, intangible assets and transfer pricing mechanisms.

The sample is composed of 448 non-financial publicly traded companies, whose headquarters are located in the five OECD´s key partner countries: Brazil, China, India, Indonesia, and South Africa. Data were obtained from the CIQ Pro and the World Bank databases for the period from 2010 to 2020. The hypotheses were tested using logistic regression models – see equations 2 to 7. In addition, a robustness test was applied through econometric models with panel data, considering the fixed effects of industry, country, and year.

This study takes an innovative approach for capture transfer pricing intensity (TPI) with manually collected data. In addition, a tax proxy (AREA) was used to control the tax benefits of debt, which is more efficient in capturing short-and long-term decisions. In summary, all hypotheses were confirmed. Additionally, the post-estimate tests determined the goodness of fit for all the models.

Regarding the hypothesis *H1a - Direct taxes are positively related to the use of tax havens*. It is understood that, for each 1% increase in the tax actually paid (ETR), there is a 98.80% increase in the probability of opening MNC´s subsidiaries in tax havens. This result is in line consistent with those of Christensen *et al.* (2021) and Garcia-Bernardo *et al.* (2021). The confirmation of the hypothesis *H1b - Direct taxes - of companies with investment in intangible assets - have a positive relationship with the use of tax havens,* corroborates the studies of Dharmapala (2020) where the author states that tax haven jurisdictions reduce, or even exempt, corporate tax, to attract mobile capital - tax competition theory.

As for the hypothesis *H1c - Direct taxes - of companies with transfer pricing practices - have a positive relationship with the use of tax havens,* for each 1% increase in the interaction variable (ETR*TPI), there is an increase of 3,151.10% in the chances of MNC establishing a subsidiary in tax havens. Profit is transferred through the sales of goods and services to jurisdictions with less or no taxation - tax evasion theory. These results are in agreement with those of Brock and Pogge (2014), Merle *et al.* (2019) and Solikhah *et al.* (2021).

Hypothesis *H2 - Intangible assets are positively related to the use of tax havens*, which is in accordance with the studies by Beer *et al.* (2020) and Ciuriak and Akinyi (2021) who found that the expansion of intangible assets increases tax erosion, contributing to transfer to tax havens. As for the hypothesis *H3a - Transfer pricing policies are positively related to the use of tax havens*, for each 1% increase in TPI, an increase of 1,383.0% was found in the possibility of the MNC establishing subsidiaries in tax havens. Finally, with regard to hypothesis *H3b - Transfer pricing policies - by companies with investment in intangible assets - are positively related to the use of tax havens*, for each 1% increase in the interaction variable (TPI*INTA), an increase of 912.20% was verified in the probability of the MNC establishing subsidiaries in tax havens. This confirms the findings of Solikhah *et al.* (2021).

Concerning the control variables, for all models, those presenting a positive relationship with the possibility of the MNCs establishing subsidiaries in tax havens (DTH) are: a country's statutory rate (STR), which ratifies the tax avoidance theory, spending on research and development (RD), which confirms the shift of intangible assets to units in tax havens, and size (SIZE) because large MNCs are the main users of tax havens.

The following variables consistently have negative relationship with DTH: AREA when an MNC has less opportunity to use the tax benefit arising from debts, leverage (LEV) MNCs with greater debt tax benefit according to trade-off theory are less likely to establish a subsidiary in a tax haven, and gross domestic product (GDP) where the reduction in GDP encourages MNCs to maintain their profitability through tax planning and to consider alternatives to tax havens.

These results indicate that MNCs make use of different instruments and even combine them to reduce a conglomerate´s total taxation. Despite the efforts of governments and international organizations, these companies have been efficient in reducing or not paying taxes. Currently, global entities such as OECD, Group of Seven (G7), Group of

Twenty (G20) propose the application of a minimum tax rate at a global level (Valenduc, 2021).

Thus, this study suggests that the policies proposed by such institutions may not achieve the desired effects if jurisdictions with low-tax options remain, such as tax havens. It is important to note that the use of the official rate sometimes hides deductions that allow effective taxation below the eventual rate defined as minimum. Furthermore, in a more intangible world, administrators find greater opportunities to implement tax elision or evasion practices, which represent a major challenge for government authorities.

For managers and shareholders, this study presents alternatives for tax solutions that increase the net margins of MNCs. It was found that almost half of the companies do not adopt these tax planning options. This fact may explain why MNCs that have subsidiaries in tax havens have higher operating income (11.57%) than those that do not (10.56%).

Among the limitations of this study, the difficulty in data collection is notable. Companies typically do not provide intercompany transactions or related information in detail for strategic and investment reasons and for tax planning reasons so that governments cannot accurately assess their taxes (Schmal *et al.*, 2021). Finally, there are opportunities for future studies, such as the marginal tax rate, the marginal tax rate of the last dollar, the use of multilevel econometric models, application of artificial intelligence modelling, the analysis of non-tax advantages in invested countries, the consideration of bilateral tax agreements, and the examination of the role of tax audits.

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Appendix 1

The AREA is obtained through the relationship between the current interest deduction rate or increase in interest expenses (%) - Line A of Table 12 (Axis X) and the marginal tax rate (MTR) - Line E of Table 12 (Axis Y). The purpose of this measure was to verify the marginal reduction in the companies´ tax rate, for each percentage increase in the level of interest expenses, which is deductible from income tax. In other words, AREA allows companies to identify an optimal level of leverage to obtain the maximum reduction in tax payments - trade-off theory (Ko & Yoon, 2011; Van Binsbergen *et al.*, 2010).

For purposes of exemplification, published data from two companies in the sample (I and II) were considered as the base or zero scenarios for the year 2016. The data obtained were from interest expenses and earnings before taxes (EBT), which are US$ 217.82 and US$ 377.55 million, respectively – Lines B and C of Table 12. The increase in interest expenses percentages varied from 0% to 700%, in intervals of 20% (0%-200%) and 100 % (200%-700%). These percentage ranges are merely illustrative, having been equally adopted by Van Binsbergen *et al.* (2010). Such percentage increases with interest expenses are based on the zero scenario (year 2016). The total AREA consisted of the sum of the AREAS by scenario - Lines F of Table 12, being 2.9292 (Company I) and 7.3427 (Company II) - lines G of Table 12. Finally, the AREA of company II (7.3427) is highlighted as exceeding the maximum value shown in Table 2 - descriptive statistics (6.5307). However, it is worth remembering that sample data are winsorized between the 1st and the 99th percentiles – to reduce the effect of outliers.

Figure 4.1 – AREA

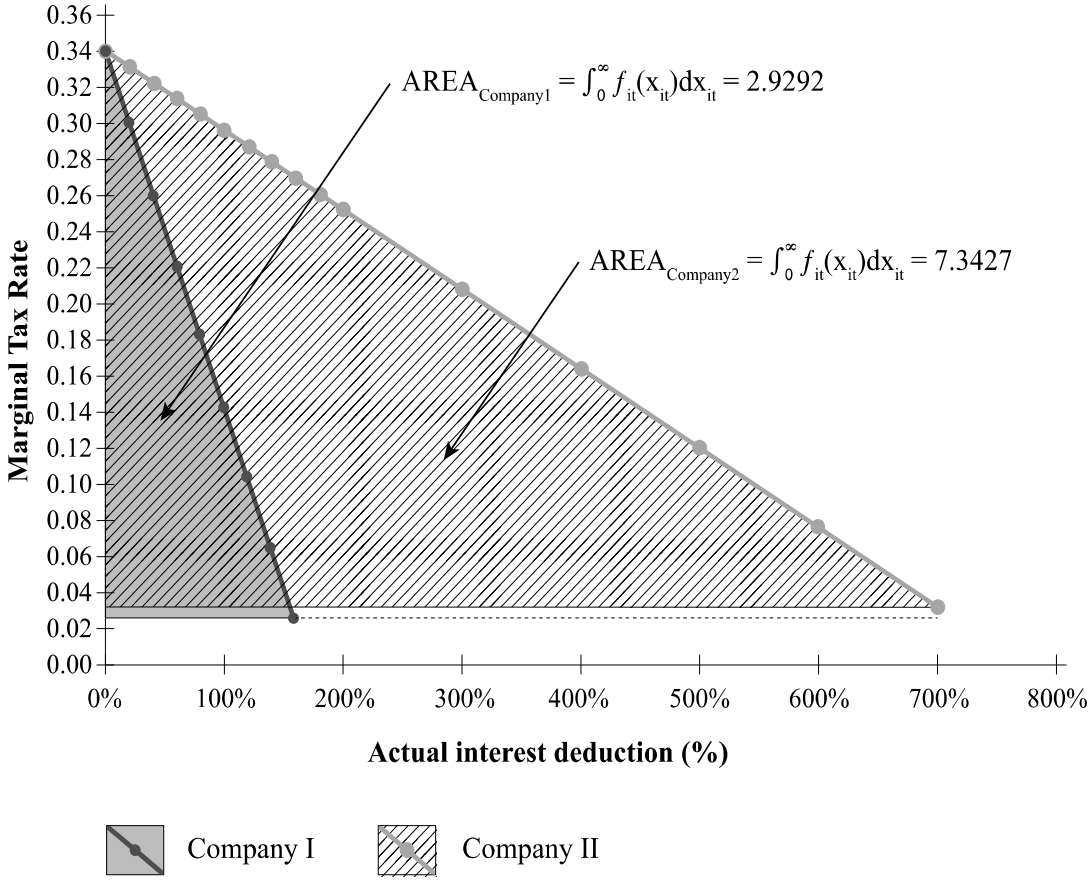


Table 4.12 – Data for calculating the AREA

enarios	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
Increase in interest expenses)	**0**	**20**	**40**	**60**	**80**	**100**	**120**	**140**	**160**	**180**	**200**	**300**	**400**	**500**	**600**	**700**
COMPANY I																
Interest expenses = Interest nario 0 x Increase % new interest	**217.8**	43.5 (i)	87.1	130.6	174.2	**217.8 (vi)**	261.3	304.9	348.5							
EBT = EBT scenario 0 – New rest	**377.5**	333.9 (ii)	290.4	246.8	203.2	159.7	116.1	72.6	29.0							
Income Tax = EBT x 34%	128.3	113.5 (iii)	98.7	83.9	69.1	54.1	39.0	24.6	9.8							
MTR = new IT / EBT scenario 0	**0.340**	**0.300 (iv)**	**0.261**	**0.222**	**0.183**	**0.143**	**0.104**	**0.065**	**0.026**							
AREA per scenario = MTR vious scenario + new MTR		0.640 (v)	0.562	0.483	0.405	0.326	0.248	0.170	0.091							
Total AREA = Sum of MTRs of EAs per scenario	**2.929**															
COMPANY II																
Interest expenses = Interest nario 0 x Increase % new interest	**19.7**	3.9	7.9	11.8	15.7	**19.7 (vi)**	23.6	27.6	31.5	35.5	39.4	59.2	78.9	98.7	118.4	138.1
EBT = EBT scenario 0 – New rest	152.6	148.7	144.7	140.8	136.8	132.9	128.9	125.0	121.1	117.1	113.2	93.4	73.7	53.9	34.2	14.5
Income Tax = EBT x 34%	51.9	50.5	49.2	47.8	46.5	45.2	43.8	42.5	41.1	39.8	38.4	31.7	25.0	18.3	11.6	4.9
MTR = new IT / EBT scenario 0	**0.340**	**0.331**	**0.322**	**0.313**	**0.304**	**0.296**	**0.287**	**0.278**	**0.269**	**0.260**	**0.252**	**0.208**	**0.164**	**0.120**	**0.076**	**0.032**
AREA per scenario = MTR vious scenario + new MTR		0.671	0.653	0.636	0.618	0.600	0.583	0.565	0.548	0.530	0.513	0.460	0.372	0.284	0.196	0.108
Total AREA = Sum of MTRs of EAs per scenario	**7.342**															

s: (i) = 217.82 x 0,20 = 43.56; (ii) = 377.55 - 43.56 = 333.99; (iii) = 333.99 x 0.34 = 113.56; (iv) = 113.56 / 377.55 = 0.3008; (v) = 0.3400 + 0.3008 = 0.6408; (vi) interest expenses the 100% increase level – are those actually paid by companies in the zero scenario, that is, in 2016.

Appendix 2

In this study, those mentioned by, at least, three organizations were considered as tax havens. Botswana was included as it is a relevant tax haven to one of the countries in the sample - South Africa. This list of organizations work to combat tax evasion and money laundering and they identify 28 jurisdictions as locations with low or no taxation. This survey was conducted in August, 2021.

Table 4.13 – Tax Haven jurisdiction

Jurisdiction/ Sources	OECD	CBI	EU	OXFAM	FATF	IMF	CTH	EX21	FSI	Total	FSI Rank
Cayman Islands	✓		✓	✓		✓	✓	✓	✓	7	1
Bahamas	✓	✓		✓			✓	✓	✓	6	22
Bermuda	✓			✓		✓	✓	✓	✓	6	40
British Virgin Islands	✓			✓		✓	✓	✓	✓	6	9
Malta	✓	✓		✓			✓	✓	✓	6	18
Mauritius	✓			✓	✓		✓	✓	✓	6	51
Panama	✓		✓		✓		✓	✓	✓	6	15
Antigua and Barbuda	✓	✓		✓				✓	✓	5	121
Bahrain	✓	✓		✓				✓	✓	5	81
Seychelles	✓	✓	✓					✓	✓	5	95
Anguilla	✓			✓			✓	✓	✓	5	62
Cyprus	✓	✓					✓	✓	✓	5	27
Hong Kong				✓		✓	✓	✓	✓	5	4
Ireland				✓		✓	✓	✓	✓	5	29
Luxembourg				✓		✓	✓	✓	✓	5	6
Singapore				✓		✓	✓	✓	✓	5	5
Netherlands				✓		✓	✓		✓	4	8
United Arab Emirates		✓		✓			✓		✓	4	10
Aruba	✓			✓				✓	✓	4	112
Barbados		✓			✓			✓	✓	4	63
Marshall Islands	✓			✓				✓	✓	4	49
Switzerland				✓			✓	✓	✓	4	3
Trinidad and Tobago			✓	✓					✓	3	126
Virgin Islands			✓	✓					✓	3	89
Malaysia		✓						✓	✓	3	32
Isle of Man	✓						✓	✓		3	-
Liechtenstein	✓						✓	✓		3	-
Botswana				✓					✓	2	113

Notes:
OECD - Organization for Economic Cooperation and Development. https://www.oecd.org/general/oecdremovesliberiafromitslistofunco-operativetaxhavens.htm
CBI - Citizenship by Investment. https://www.oecd.org/tax/automatic-exchange/crs-implementation-and-assistance/residence-citizenship-by-investment/
EU - Europe Union. https://www.consilium.europa.eu/en/policies/eu-list-of-non-cooperative-jurisdictions/#
OXFAM - Oxford Committee for Famine Relief. https://www.oxfamamerica.org/explore/research-publications/
FATF - Financial Action Task Force. https://www.fatf-gafi.org/media/fatf/documents/brochuresannualreports/FATF-annual-report-2019-2020.pdf
IMF - International Monetary Fund blacklist. https://www.imf.org/external/NP/ofca/OFCA.aspx
CTH - Corporate Tax Haven. https://cthi.taxjustice.net/cthi2021/country-list.pdf
EX21 - Exhibit 21 of form 10K. https://sites.google.com/site/scottdyreng/Home/data-and-code/EX21-Dataset
FSI - Financial Secrecy Index. https://fsi.taxjustice.net/en/introduction/fsi-results

5. CONCLUSION

This research addresses the influence of taxes on MNCs´ investment and financing decisions. The theme is increasingly relevant due to the advance of globalization, digitalization of information and token usage for assets. Currently, countries and supranational organizations are seeking to coordinate the application of a minimum global taxation. The set of actions in movement aims at mitigating the implementation of corporate strategies which have the purpose of tax payment reduction – such as tax havens. For the academy, the topic is equally challenging, given the countries´ complexity of tax systems.

This study presents three related essays. All of them have the objective of analyzing the influence of MNCs´ taxes in their corporate decisions. The first one performs a systematic review and bibliometric analysis about the theme, providing a research agenda for future research. Among the pointed gaps, are the issues verified by essays two and three of this research.

The second essay addresses how MNCs structure their capital, in order to obtain tax / financial benefits. This analysis is performed in times of crisis and economic stability. There is evidence that the allocation of resources among branches generates gains for these corporations. The third essay investigates the MNC´s tax planning for reduction of their tax burden. Among the analyzed aspects are the adoption of transfer pricing among subsidiaries, the intensive use of intangible assets and the establishment of subsidiaries in tax havens.

According to the first essay, the papers have estimation problems of models, due to the tax proxy calculation difficulty. Because of this fact, they point to opposite results with the same information bases and research problems, depending on how the variable is calculated.

Thus, the essay proposes the use of tax proxies allowing for an analysis of short and long term decisions, such as MTR, AREA, NTDS – considered in the second and third essays. Another issue highlighted is the verification that FDI are driven by the possibility of tax payment reduction through the use of tax haven.

This way, in the second essay, these tax proxies are defined and inserted in their econometric models. The variable AREA estimates the amount of debt that the company may still contract to reach its optimal leverage point. The MTR variable projects the future taxable profit as well as company ´s tax benefit expectation – due to the increase of its financial leverage. As for the LDMTR variable, it foresees MTR, based on the contracting of new debts.

At last, NDTS variable measures the tax benefits not arising from indebtedness - such as: depreciation and amortization.

Among the results of the second essay, the existence of a positive relationship between MNCs´ leverage level on their bankruptcy costs and value is verified. On the other hand, interest rates and tax benefits have a negative relationship with MNCs´ indebtedness. When analyzing MNCs and DCs – comparatively – an increase (decrease) in MNCs (DC)´s leverage level is observed in the considered period. In addition, MNCs have greater negotiating power and may obtain better interest rates than DCs. These results are fundamental for understanding the financial behavior of MNCs in emerging countries, in periods of crisis and financial stability.

Finally, the third essay investigates the investments of MNCs in countries with low or no taxation, as a strategy to reduce the conglomerate's total taxes. MNCs seek to obtain benefits in these jurisdictions classified as tax havens, with the ultimate purpose of maximizing the group's profit by reducing their taxes. Among the contributions of the study, the use of the unconventional tax proxy - AREA - stands out. In addition, TPI variable is obtained through manually collected data.

The result is that MNCs make use of different instruments – even combined – with the objective of reducing the conglomerate´s total taxation. Therefore, despite the efforts of governments and international organizations – proposing the adoption of a minimum tax rate among countries – these companies have been efficient in reducing or even in not paying taxes.

Among the limitations of this study - in the first essay – just papers presenting econometric models are considered. Thus, other equally relevant papers, which apply different methodologies from this one, are excluded. On the second essay, MTR´s calculation demands long-term historical series. Nonetheless, since sample consists of emerging countries - that are in the process of maturing their economies – there are restrictions of data availability for the desired period. This occurs for strategic reasons and/or to avoid assessment of tax planning by governments and investors.

Finally, as an opportunity for the evolution of this study, the use of different methodologies, such as multilevel econometric models and structural equations is suggested. In addition, an analysis is proposed to other researchers, about the impact of the following aspects in MNCs´ investment and financing decisions: a) non-tax advantages b) bilateral tax agreements, c) local companies´ agglomeration around productive chain´s central processes d) sub-capitalization rules, and e) production verticalization.